

Identifiability of the correction term: Worst-Case Leakage Is Governed by the Principal Angles of the Dictionary

Xiangrui Meng*

DeepSeek V4.1 Flash[†]

Qwen 3.8 Flash[‡]

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Abstract

A scientific-ML model is written as a prior operator plus a learned correction, $A^* = \sum_i \theta_i^* E_i + \sum_j \psi_j^* F_j + \Delta_{\text{hidden}}$. Fitting the coefficients by least squares from n input–output pairs answers a question that is not about accuracy but about *attribution*: how much of an unmodeled operator Δ_{hidden} is forced into the correction block? We show this is a question in matrix geometry with an exact answer. Four identities hold for the operator (Frobenius) Gram $\mathcal{G}$ of the dictionary: its block form, an isometry, a square identity, and $\det \mathcal{G} = \det(\Lambda_E) \sin^{2q} \alpha$. They give a two-sided bound on $\lambda_{\min}(\mathcal{G})$ whose intrinsic quantity is the frame invariant $\|\Lambda_E^{-1} H\|_2^2$, and a sampling bound whose remainder is *independent of the principal angle*. The resulting certificate

$$\|\hat{C} - C^*\|_F \leq \frac{\|\Delta\|_F + \sigma\sqrt{d}}{\sin \alpha_{\min}} \sqrt{1 + 2 \cos^2 \alpha_{\min} \|\Lambda_E^{-1} H\|_2^2} \left(1 - \frac{c}{2n} - \frac{0.88}{\sqrt{n}}\right)^{-1}$$

contains no fitted constant, is attained within 1% in the worst sampled case, and holds with the sample-size factor set to 1 in 4320/4320 adversarial trials spanning $n = 30 \dots 1800$ and $\alpha = 2^\circ \dots 85^\circ$. We also measure the identity err/bound $= q_4 q_1 q_2 q_3$ and show that the sample-size remainder *cancels structurally* ($q_1 q_2 \approx 1$) whenever the perturbing operator is realized through the same data matrix whose Gram it perturbs: the tightness of the certificate is a property of $\mathcal{G}$ alone. Since $\lambda_{\min}(\mathcal{G})$ attains the upper branch sample by sample, the dictionary factor q_4 is a closed-form quantity, and with the block-mass factor q_3 the observed looseness collapses to a single dictionary scalar, $(1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2)^{-1/2}$, which reproduces the fitted ratios of the adversarial table to within 1% and needs no calibration at any angle. The upper side is then not a bound but a reduction: $\lambda_{\min}(\mathcal{G})$ is the unique root of a scalar equation in a $q \times q$ matrix pencil, and that root is the closed form above *exactly* when $p = q$. Section 7 states what we do not claim, including the real-data leg of this project, which was retired by its own measurements rather than by argument.

Keywords. saddle-point matrix; arrowhead secular equation; principal angles between subspaces; identifiability; attribution error; scientific machine learning; certified a posteriori bound; Gauss quadrature. **AMS 2020 subject classifications.** 15A18, 65F15, 62J05, 68T07.

*School of Mathematics and Physics, Shanghai Normal University, Shanghai, China. Lead (human) author, ORCID 0009-0006-8902-0519.

[†]AI co-author (Agent A). Role: experiments, certificate construction, falsification of the manuscript’s own claims.

[‡]AI co-author (Agent B). Role: derivation, theorems, numerics, literature verification.

Introduction

Hybrid models write a dynamical operator as something known plus something learned, $A^* = \sum_i \theta_i^* E_i + \sum_j \psi_j^* F_j + \Delta_{\text{hidden}}$, fit both blocks from the same n input–output pairs, and are then validated by predictive accuracy. Accuracy answers a different question from the one the model is asked to address. When the correction block is reported as a *finding* — a closure term, a missing forcing, a diagnosed coupling — the claim is about attribution: what fraction of an unmodeled operator Δ_{hidden} was *forced into* $\hat{\psi}$ by the fit itself. A model can forecast perfectly and attribute nothing.

This note shows that attribution is a question in matrix geometry with an answer that is computable *before any data are collected*. For a dictionary whose correction directions make principal angle α with the span of the prior block, four identities hold for the operator (Frobenius) Gram $\mathcal{G}$: a block form, an isometry, a square identity, and $\det \mathcal{G} = \det(\Lambda_E) \sin^{2q} \alpha$. They pin $\lambda_{\min}(\mathcal{G})$ between two branches whose intrinsic quantity is the frame invariant $\|\Lambda_E^{-1} H\|_2^2$, and the resulting leakage factor is $1/\sin \alpha_{\min}$ times a correction that never exceeds $\sqrt{1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2}$. Nothing in that chain depends on n , on the noise level σ , or on a fitted constant.

The second point is methodological, and it is why we insist on the constant-free form. The alternative in the literature is an empirically calibrated envelope $c_1 \|\Delta\| + c_2 \sigma$. The two constants do not carry the same dependence on anything: Corollary 5.2 makes the model-form one a $\sqrt{q}/d$ -type dilution factor (the average case is a dimensional artifact, not physics; the $\sqrt{q}$ is block aggregation), while the noise one is the Gaussian quantile $z_{1-\delta/2}/\sqrt{n}$, with no d at all and no mechanism in between. Fitting a single pair (c_1, c_2) averages two regimes whose scalings disagree and tells a reader nothing about which regime their experiment sits in. The disagreement is not a technicality: in the two-arm least-squares experiment behind this note (design by Agent A, audit **b40**, **b41**, **b40b**, $q = 3$), the measured model-form constant is the random-projection fraction $\mathbb{E}\|P_{\mathcal{F}}\Delta\|/\|\Delta\|$ — a Beta mean $B(q/2 + \frac{1}{2}, (d^2 - q)/2)/B(q/2, (d^2 - q)/2)$, leading term $\kappa(q)\sqrt{q}/d$ with κ the χ_q mean factor — matched to within 8% over $d = 6 \dots 20$ and 6% over $q = 1 \dots 6$, and flat in n to within 5% over $n = 150 \dots 2400$; the noise constant is instead $\sqrt{q}/n$ to within 13%. Their ratio is $d/\sqrt{n}$, measured from 0.90 at $n = 150$ to 0.21 at $n = 2400$. One fitted constant cannot describe both, and a table that reports one is reporting whichever regime its sample size happened to sit in.

Contributions, tagged by status.

- Exact determinant law $\det \mathcal{G} = \det(\Lambda_E) \sin^{2q} \alpha$, and three companion identities *[proved; residuals $\leq 10^{-12}$]*
- Two-sided bound on $\lambda_{\min}(\mathcal{G})$, with the direction of the inequality chosen so that a wrong dictionary always *enlarges* the certificate *[proved; 6000 draws over three scripts, 0 violations on either side]*
- The upper side sharpens to a closed form $\text{cf}(t, \alpha)$, realised as the Rayleigh minimum on an explicit plane; and $\lambda_{\min}(\mathcal{G})$ reduces *exactly* to a one-dimensional root find over a $q \times q$ matrix pencil *[both proved; the exact formula to $1.8 \cdot 10^{-11}$ on 6480 dictionaries, the closed form identically $\lambda_{\min}(\mathcal{G})$ when $p = q$; the root find survives an arbitrary principal-cosine spectrum, the bound direction of the closed form does not]*
- The conjecture that the plane direction is the top singular direction is *false* for $q < p$, and the counterexample was found only by replacing random sampling with an exhaustive circle search *[disproved; defect $\leq 10^{-6}$ relative]*

- Sampling remainder $O(\sqrt{(p+q)/n})$ that does not know about α , and a refined one-sided $1/n$ bias term computable from the dictionary *[proved in shape; the constants are calibrated, not derived]*
- Worst-case leakage bound with constant *exactly* 1, plus a construction attaining it *[proved, subject to $n > d$]*
- Certificate $(\star)$ and its 4320-trial adversarial test; the analytic form of its remaining slack *[measured; the closed form matches the exact dictionary factor over 4400 dictionaries and eleven angles, worst per-sample deviation 0.004]*
- The achieved ratio factorizes as $q_4 q_1 q_2 q_3$ (dictionary, sampling, realization, block mass), and with q_3 included the observed looseness of $(\star)$ collapses to the single dictionary scalar $(1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2)^{-1/2}$ *[identity verified per trial; the collapse is measured, not proved]*
- Three structural negative results: the sample-size margin cancels and must not be tabulated; the α -uniformity of the bias is *not* proved although it is flat over the scanned grid; and the $q_1 q_2$ cancellation is only partial under a correlated (trajectory) design, where it reaches 0.975 at $\rho = 0.95$ — always in the safe, looser direction *[measured, with the grid stated]*

Sections 1–2 hold the geometry, 3–4 the two-sided bound and the sampling bridge, 5–6 the worst case and the certificate, 6.2 the measured factorization, and 7 the limitations, including seventeen claims we retracted ourselves and two open items. Section 8 states what is not new here.

1 Setting, notation, and the one geometric quantity

Let $A^\star \in \mathbb{R}^{d \times d}$ be the true linear operator and let a *dictionary* be a pair of finite families of operators,

$$A^\star = \underbrace{\sum_{i=1}^p \theta_i^\star E_i}_{\text{prior block } P} + \underbrace{\sum_{j=1}^q \psi_j^\star F_j}_{\text{correction block } C^\star} + \Delta, \quad (1)$$

where Δ is the *model-form error*: the part of the dynamics that the chosen dictionary does not span. We observe

$$Y = A^\star U + \Sigma, \quad U \in \mathbb{R}^{d \times n} \text{ i.i.d. standard Gaussian}, \quad \Sigma \in \mathbb{R}^{d \times n} \text{ i.i.d. } \mathcal{N}(0, \sigma^2), \quad (2)$$

and fit $(\hat{\theta}, \hat{\psi})$ by least squares over the design matrix

$$G = [\text{vec}(E_1 U) \cdots \text{vec}(E_p U) \mid \text{vec}(F_1 U) \cdots \text{vec}(F_q U)] \in \mathbb{R}^{dn \times (p+q)}. \quad (3)$$

All geometry below lives in the vectorized operator space $\mathbb{R}^{d^2}$ with the Frobenius inner product. Define the *operator Gram*

$$\mathcal{G}_{ab} = \langle M_a, M_b \rangle_F, \quad \{M_a\}_{a=1}^{p+q} = \{E_i\} \cup \{F_j\}, \quad (4)$$

the prior Gram $\Lambda_E = (\langle E_i, E_j \rangle_F) \in \mathbb{R}^{p \times p}$, and the cross-Gram $H = (\langle E_i, a_j \rangle_F) \in \mathbb{R}^{p \times q}$, where a_j is the component of F_j inside $\text{span}\{E_i\}$ (made precise in (5)). Two abbreviations recur: $M := \Lambda_E^{-1/2} H$ and $C := \Lambda_E^{-1} H$.

Definition 1.1 (dictionary angle). *The construction used throughout (and in the experiments of Agent A) is*

$$F_j = \cos \alpha a_j + \sin \alpha b_j, \quad \{a_j\}_{j=1}^q \text{ orthonormal} \subset \text{span}\{E_i\}, \quad \{b_j\}_{j=1}^q \text{ orthonormal} \subset \text{span}\{E_i\}^\perp, \quad (5)$$

with all orthonormality in the Frobenius norm of $\mathbb{R}^{d^2}$. Then $\|F_j\|_F = 1$, $\langle F_i, F_j \rangle_F = \delta_{ij}$, and $\langle E_i, F_j \rangle_F = \cos \alpha \langle E_i, a_j \rangle_F$; α is the principal angle between the two blocks, and $\alpha \rightarrow 0$ means the correction is aligned with the prior.

Remark 1.1 (where exactness lives). (5) is imposed in the *vectorized* space, hence $\|F_j\|_F = 1$ and $\langle F_i, F_j \rangle_F = \delta_{ij}$ hold sample by sample, not in expectation. What is *not* exact is Λ_E : the prior operators are random and mutually correlated, with off-diagonal entries of size $O(\sqrt{d})$ relative to the diagonal. Every looseness reported in this chapter traces back to $\kappa(\Lambda_E)$, not to the correction block. This is a correction of an earlier draft of ours, which assumed the opposite.

The single scalar that controls the certificate is

$$\|\Lambda_E^{-1} H\|_2^2 = \|\Lambda_E^{-1} H\|_2^2 = \|M\|_2^2 : \text{the frame condition functional of the dictionary.} \quad (6)$$

It is computable from the dictionary alone — no data, and (Theorem 3.1 remarks) no α . Replacing it by the quantity $\rho_E^{-2} = \|E_i\|_F^{-2}$ that an earlier version of this chapter used is wrong except when $\Lambda_E \approx \rho_E^2 I$; the measured gap is $\|\Lambda_E^{-1} H\|_2^2 \cdot d = 1.005\text{--}1.073$ for $d \geq 16$ but $1.5\text{--}3.0$ at $d = 4$. A computation pitfall worth recording: by Proposition 2.2, $\|\Lambda_E^{-1/2} H\|_2^2 \equiv 1$ identically, so the invariant must be formed as a *solve*, $\Lambda_E^{-1} H = \text{solve}(\text{Lambda_E}, H)$, never as a Cholesky half-inverse times H . The latter measures $\|R^{-1} H\|_2^2$ and returns $1.05\text{--}1.07$ at $d = 8$, where the true value is 0.17 .

2 Four exact identities

Proposition 2.1 (block Gram). *With (5), $\mathcal{G} = \begin{pmatrix} \Lambda_E & \cos \alpha H \\ \cos \alpha H^\top & I_q \end{pmatrix}$.*

Proposition 2.2 (isometry). *$H^\top \Lambda_E^{-1} H = M^\top M = I_q$; consequently $K := MM^\top$ is the orthogonal projection of rank q onto $\text{ran } \Lambda_E^{-1/2} H$.*

Proposition 2.3 (square identity). *For every coefficient pair (θ, w) ,*

$$\theta^\top \Lambda_E \theta + 2 \cos \alpha \theta^\top H w + \|w\|^2 = \|\Lambda_E^{1/2} \theta + \cos \alpha M w\|_2^2 + \sin^2 \alpha \|w\|_2^2. \quad (7)$$

The left-hand side is $\|\sum_i \theta_i E_i + \sum_j w_j F_j\|_F^2$.

Proof. Expand the right-hand side: $\theta^\top \Lambda_E \theta + 2 \cos \alpha \theta^\top \Lambda_E^{1/2} M w + \cos^2 \alpha \|M w\|^2 + \sin^2 \alpha \|w\|^2$. Now $\Lambda_E^{1/2} M = H$ and, by Proposition 2.2, $\|M w\|^2 = w^\top M^\top M w = \|w\|^2$. $\square$

Theorem 2.4 (determinant law). $\det \mathcal{G} = \det(\Lambda_E) \sin^{2q} \alpha$.

Proof. Take the Schur complement of the I_q block in Proposition 2.1: $I_q - \cos^2 \alpha H^\top \Lambda_E^{-1} H = I_q - \cos^2 \alpha M^\top M = \sin^2 \alpha I_q$ by Proposition 2.2. Hence $\det \mathcal{G} = \det(\Lambda_E) \det(\sin^2 \alpha I_q) = \det(\Lambda_E) \sin^{2q} \alpha$. $\square$

Corollary 2.5. *Each correction direction contributes one factor $\sin^2 \alpha$ of identifiability loss; the generalized volume of a q -dimensional correction block collapses as $\sin^{2q} \alpha$.*

Numerical status. Measured against a directly computed Frobenius Gram of $\{E_i\} \cup \{F_j\}$, the block form (4) agrees to 4.0×10^{-16} , 5.8×10^{-16} , 7.4×10^{-16} at $d = 8, 16, 32$; the isometry residual is $\leq 4.8 \times 10^{-16}$; and the determinant law satisfies $|\det \mathcal{G}/(\det \Lambda_E \sin^{2q} \alpha) - 1| \leq 1.6 \times 10^{-12}$ over $d = 3 \dots 20$, $\alpha = 1^\circ \dots 85^\circ$. The four identities are exact, not asymptotic; the numbers above are round-off. (Script: `b11_theorem_remainder.py`.)

3 The dictionary side: a two-sided bound for $\lambda_{\min}(\mathcal{G})$

Parametrize the quadratic form of (7) by the free variable $z = \Lambda_E^{1/2} \theta + \cos \alpha M w$, so that $\theta = \Lambda_E^{-1/2} z - \cos \alpha \Lambda_E^{-1/2} M w = \Lambda_E^{-1/2} z - \cos \alpha C w$. Then $\lambda_{\min}(\mathcal{G})$ is the Rayleigh quotient

$$\lambda_{\min}(\mathcal{G}) = \min_{(z,w) \neq 0} \frac{\|z\|^2 + \sin^2 \alpha \|w\|^2}{\|\Lambda_E^{-1/2} z - \cos \alpha C w\|^2 + \|w\|^2}. \quad (8)$$

Theorem 3.1 (two-sided bound). *For every d, p, q and every $\alpha \in (0, \pi/2]$,*

$$\begin{aligned} \min\left(\frac{\lambda_{\min}(\Lambda_E)}{2}, \frac{\sin^2 \alpha}{1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2}\right) &\leq \lambda_{\min}(\mathcal{G}) \leq \\ &\leq \frac{\sin^2 \alpha}{1 + \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2}. \end{aligned} \quad (9)$$

Proof. Upper bound. In (8) restrict to the feasible choice $z = 0$ and any unit w : the quotient becomes $\sin^2 \alpha / (\cos^2 \alpha \|C w\|^2 + 1)$, whose minimum over $\|w\| = 1$ is the right-hand side of (9) at w aligned with the top singular direction of C . *Lower bound.* For the denominator, Young's inequality with $\varepsilon = 1$ gives $\|\Lambda_E^{-1/2} z - \cos \alpha C w\|^2 \leq 2\|\Lambda_E^{-1/2} z\|^2 + 2\cos^2 \alpha \|C w\|^2 \leq 2\lambda_{\min}(\Lambda_E)^{-1} \|z\|^2 + 2\cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2 \|w\|^2$, hence with $c_1 := 2/\lambda_{\min}(\Lambda_E)$ and $d_2 := 1 + 2\cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2$ the quotient in (8) is at least $(a + \sin^2 \alpha b)/(c_1 a + d_2 b)$ where $a = \|z\|^2$, $b = \|w\|^2$. Writing this as $(c_1 a) \cdot (1/c_1) + (d_2 b) \cdot (\sin^2 \alpha / d_2)$ over $c_1 a + d_2 b$ exhibits it as a weighted average of $1/c_1$ and $\sin^2 \alpha / d_2$, so it is bounded below by $\min(1/c_1, \sin^2 \alpha / d_2)$, which is the left-hand side of (9). $\square$

Remark 3.1 (the inequality direction is the safety-critical part). An earlier version of this chapter stated $\sigma_{\min}(\mathcal{G}^{1/2}) = \sin \alpha / \sqrt{1 + \cos^2 \alpha \|C\|_2^2}$ as an *equality*. The right-hand side of Theorem 3.1 is an **upper** bound on $\lambda_{\min}(\mathcal{G})$, i.e. an upper bound on $\sigma_{\min}(\mathcal{G}^{1/2})$. Substituting an upper bound on a smallest singular value into $\|\hat{C} - C^*\| \leq \|r\|/\sigma_{\min}(G)$ *underestimates* the error — the unsafe direction. Quantitatively, the ratio of true to this surrogate is 0.112 at the corner ($d = 3, \alpha = 85^\circ$), i.e. a factor-of-nine underestimation, while in the working region $\alpha \leq 20^\circ$, $d \geq 6$ it is ≥ 0.97 . Certificates must use the *lower*-bound multiplier $\sqrt{1 + 2\cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2} / \sin \alpha$, which at $d = 10$ costs a factor 1.08: essentially free.

Remark 3.2 (scaling invariance). $E_i \mapsto sE_i$ gives $H \mapsto sH$, $\Lambda_E \mapsto s^2 \Lambda_E$, hence $\|\Lambda_E^{-1} H\|_2^2 \mapsto \|\Lambda_E^{-1} H\|_2^2 / s^2$ and the multiplier $\rightarrow 1$. Enlarging the prior dictionary can therefore manufacture identifiability *in coefficient space* while leaving the operator-space error untouched. This is the analytic explanation of the reparameterization invariance observed numerically (error invariant for $s = 0.01 \dots 10$).

Remark 3.3 (case distinction in the small- α regime). The branch $\lambda_{\min}(\Lambda_E)/2$ does not shrink with α , so for small α the $\sin^2 \alpha$ branch controls. The lower bound is tight in the regime that matters ($\lambda_{\min}(\Lambda_E)/2$ -versus-true ratio 0.84–1.00 at $d = 8$, 0.94–1.00 at $d = 20$) and loose only at $d = 3$ (0.50–0.58 at $\alpha = 85^\circ$), consistent with Remark 1.1: looseness is carried by $\kappa(\Lambda_E)$.

Numerical status. $d \in \{3, 4, 6, 8, 12, 20\} \times \alpha \in \{1, 5, 20, 45, 70, 85\}^\circ \times 25 \text{ samples} = 900 \text{ draws}$: zero violations of either side. (Script: `b13_two_sided_bound.py`.)

The upper side is attained, sample by sample. A sharper test (2400 further draws, $d = 8$, $p = 4$, $q = 3$, $\alpha \in \{1, 2, 5, 10, 20, 45, 70, 85\}^\circ$) compares each draw's $\lambda_{\min}(\mathcal{G})$ with *its own* upper branch: zero violations, and the ratio $\lambda_{\min}(\mathcal{G})/\text{UB}$ has mean 0.99278–0.99999 and minimum 0.98489 over the eight cells (script `b29_upper_bound_check.py`). The $\sin^2 \alpha$ branch is therefore not merely a bound: it is the right-hand side of an identity up to 1.5%, with the worst case at $\alpha = 45^\circ$. Since $\lambda_{\min}(\mathcal{G})$ and $\|\Lambda_E^{-1}H\|_2^2$ are functions of the same draw, the mean-versus-mean comparison is not a test of the bound; only the per-sample ratio above is, and it is what makes the slack of $(\star)$ analytic (§6.2).

Proposition 3.2 (the upper side, restricted to a plane). *Write $t := \|\Lambda_E^{-1}H\|_2^2$ and, for $t > 0$ and $\alpha \in (0, \pi/2)$,*

$$\text{cf}(t, \alpha) := \frac{(1+t) - \sqrt{(1+t)^2 - 4t \sin^2 \alpha}}{2t}, \quad (10)$$

the smaller root of $Q(\lambda) := t\lambda^2 - (1+t)\lambda + \sin^2 \alpha$; by continuity $\text{cf}(0, \alpha) = \sin^2 \alpha$. Then

$$\lambda_{\min}(\mathcal{G}) \leq \text{cf}(t, \alpha) < \frac{\sin^2 \alpha}{1 + \cos^2 \alpha t}, \quad (11)$$

so cf is a closed-form upper bound for $\lambda_{\min}(\mathcal{G})$, depending on the dictionary through t alone, that is strictly sharper than the upper branch of Theorem 3.1.

Proof. In the coordinates $x = \Lambda_E^{-1/2}z$ of (8), with $C := \Lambda_E^{-1}H$, $\lambda_{\min}(\mathcal{G}) = \min_{(x,w) \neq 0} (x^\top \Lambda_E x + \sin^2 \alpha \|w\|^2) / (\|x - \cos \alpha Cw\|^2 + \|w\|^2)$. Let w_0 be a unit top right-singular vector of C , put $u := \cos \alpha Cw_0$, and restrict the minimiser to the two-dimensional family $x = cu$, $w = w_0$: the minimum over a subset is at least the global one, so it suffices to evaluate it. The isometry $C^\top \Lambda_E C = I_q$ gives $u^\top \Lambda_E u = \cos^2 \alpha$ and $\|u\|^2 = \cos^2 \alpha t$, whence the quotient on the family is $\varphi(c) = (c^2 \cos^2 \alpha + \sin^2 \alpha) / (c^2 \|u\|^2 - 2c\|u\|^2 + \|u\|^2 + 1)$, whose sublevel condition $\varphi(c) \geq \lambda$ for all c reads $Q(\lambda) \geq 0$ after division by $\cos^2 \alpha$ (the leading coefficient of $\varphi - \lambda$ is positive exactly when $\lambda < 1/t$, and $Q(1/t) = \sin^2 \alpha - 1 < 0$ places $1/t$ between the roots). The largest admissible λ is therefore the smaller root of Q , which is $\text{cf}(t, \alpha)$. For the second inequality of (11), substituting the upper branch gives $Q(\sin^2 \alpha / (1 + \cos^2 \alpha t)) = -\cos^2 \alpha \sin^2 \alpha t^2 / (1 + \cos^2 \alpha t)^2 < 0$, and a quadratic which is negative at a point puts that point strictly between its roots. $\square$

Remark 3.4 (what cf buys, and what it does not). The gain over Theorem 3.1's branch is small in α -units but systematic: over 42 cells $\times$ 150 dictionaries ($d = 3 \dots 16$, $\alpha = 1^\circ \dots 89^\circ$) the ratio cf/UB is ≤ 1 on every draw, with worst-case slack $|\lambda_{\min}(\mathcal{G})/\text{cf} - 1| \leq 2.2 \times 10^{-3}$ against 1.5×10^{-2} for UB, both at $\alpha = 45^\circ$ (scripts `b33_lambda_min_sharp.py`, `b36_cf_chain.py`). The identity of cf with the plane minimum was checked against brute-force one-dimensional minimisation on all 6300 draws, agreeing to $\leq 4.5 \times 10^{-12}$: the reduction is exact, not fitted. Like the branch it replaces, (10) bounds $\lambda_{\min}(\mathcal{G})$ from *above*, so it is an *unsafe* direction for a certificate (Remark 3.1); no downstream estimate in this chapter uses cf , which appears only as the comparison point of Corollary 3.4. Its role is comparative: with Theorem 3.3 available, it says how much of Theorem 3.1's upper branch is an artifact of the restriction $z = 0$ rather than of the geometry.

Theorem 3.3 (exact scalar formula for $\lambda_{\min}(\mathcal{G})$). *Put $C := \Lambda_E^{-1}H$ and, for $\lambda \in (0, \lambda_{\min}(\Lambda_E))$, $B(\lambda) := C^\top \Lambda_E (\Lambda_E - \lambda I)^{-1} C \in \mathbb{R}^{q \times q}$. Then $\lambda_{\min}(\mathcal{G}) = \min(\lambda_{\min}(\Lambda_E), \lambda^*)$, where λ^* is the unique root in $(0, \lambda_{\min}(\Lambda_E))$ of*

$$\lambda + \cos^2 \alpha \cdot \lambda \cdot \lambda_{\max}(B(\lambda)) = \sin^2 \alpha, \quad (12)$$

taken as $\lambda_{\min}(\Lambda_E)$ when no root exists in that interval. Thus $\lambda_{\min}(\mathcal{G})$ is a property of the dictionary, obtainable to absolute round-off by a monotone one-dimensional root find on a $q \times q$ eigenvalue, with no eigenvalue of the $(p+q)$ -dimensional Gram needed; Remark 3.9(iii) quantifies what the word round-off can and cannot mean here.

Proof. In the coordinates $x = \Lambda_E^{-1/2} z$ the quotient (8) is invariant under $(x, w) \mapsto r(x, w)$, so the unit sphere splits into the face $w = 0$, where it equals $x^\top \Lambda_E x / \|x\|^2$ with minimum $\lambda_{\min}(\Lambda_E)$, and the slice $\|w\| = 1$, on which we may minimise over x first. Fix unit w , write $y = Cw$ and $u = \cos \alpha y$, and let $\mu(w) := \min_x (x^\top \Lambda_E x + \sin^2 \alpha) / (\|x - u\|^2 + 1)$. A number λ satisfies $\lambda \leq \mu(w)$ iff the form $x^\top \Lambda_E x + \sin^2 \alpha - \lambda(\|x - u\|^2 + 1)$ is nonnegative for all x ; for $\lambda < \lambda_{\min}(\Lambda_E)$ that form is a strictly convex quadratic, minimised at $x = -\lambda(\Lambda_E - \lambda I)^{-1} u$ with minimal value $-\Phi_w(\lambda)$, where, using $I + \lambda(\Lambda_E - \lambda I)^{-1} = \Lambda_E(\Lambda_E - \lambda I)^{-1}$ to collapse the two u -terms,

$$\Phi_w(\lambda) := \lambda + \cos^2 \alpha \cdot \lambda \cdot y^\top \Lambda_E (\Lambda_E - \lambda I)^{-1} y - \sin^2 \alpha. \quad (13)$$

So $\mu(w)$ is the largest $\lambda < \lambda_{\min}(\Lambda_E)$ with $\Phi_w(\lambda) \leq 0$. Because $\Lambda_E(\Lambda_E - \lambda I)^{-1} = (I - \lambda \Lambda_E^{-1})^{-1}$ is Loewner-increasing in λ , each Φ_w is strictly increasing, and its root is unique; for $\lambda \geq \lambda_{\min}(\Lambda_E)$ the form is unbounded below, so $\mu(w) \leq \lambda_{\min}(\Lambda_E)$ and $\lambda_{\min}(\mathcal{G}) = \min(\lambda_{\min}(\Lambda_E), \min_{\|w\|=1} \mu(w))$. It remains to perform the w -minimisation. By (13), Φ_w depends on w only through $y^\top \Lambda_E (\Lambda_E - \lambda I)^{-1} y$, so $\Psi(\lambda) := \max_{\|w\|=1} \Phi_w(\lambda) = \lambda + \cos^2 \alpha \cdot \lambda \cdot \lambda_{\max}(B(\lambda)) - \sin^2 \alpha$, with equality at a top eigenvector $w^*(\lambda)$ of $B(\lambda)$. Ψ is strictly increasing by the same argument and $\Psi(0^+) = -\sin^2 \alpha < 0$. If Ψ has no root in the interval, set $\lambda^* := \lambda_{\min}(\Lambda_E)$ and every $\mu(w)$ equals $\lambda_{\min}(\Lambda_E)$, which is the claimed value; otherwise let λ^* be its root. Then $\Phi_w(\lambda^*) \leq \Psi(\lambda^*) = 0$ gives $\mu(w) \geq \lambda^*$ for every unit w , while $w^* := w^*(\lambda^*)$ satisfies $\Phi_{w^*}(\lambda^*) = \Psi(\lambda^*) = 0$, i.e. $\mu(w^*) = \lambda^*$. Hence $\min_{\|w\|=1} \mu(w) = \lambda^*$. $\square$

Corollary 3.4 (the square case: cf is an identity, not a bound). *If $p = q$ and C is invertible, then $\lambda_{\min}(\mathcal{G}) = \text{cf}(\|\Lambda_E^{-1} H\|_2^2, \alpha)$ exactly.*

Proof. $C^\top \Lambda_E C = I_q$ with C square and invertible gives $\Lambda_E = C^{-\top} C^{-1}$, so $\Lambda_E^{-1} = C C^\top$, $\lambda_{\min}(\Lambda_E) = 1/\sigma_{\max}^2(C) = 1/t$, and with $S := C^\top C$, $B(\lambda) = C^\top (I - \lambda C C^\top)^{-1} C = (I - \lambda S)^{-1} S = f(S)$, $f(x) = x/(1 - \lambda x)$, using the intertwining $C^\top g(C C^\top) = g(C^\top C) C^\top$. On the admissible interval $\lambda < 1/t$ every eigenvalue of S is $< 1/\lambda$ and f is increasing, so $\lambda_{\max}(B(\lambda)) = f(\lambda_{\max}(S)) = t/(1 - \lambda t)$, attained at w_0 ; and (12) is $\lambda + \cos^2 \alpha \cdot \lambda t/(1 - \lambda t) = \sin^2 \alpha$, i.e. after multiplying by $1 - \lambda t$, exactly $t\lambda^2 - (1+t)\lambda + \sin^2 \alpha = 0$. Its larger root is at least $1/t$, because $\Delta := (1+t)^2 - 4t \sin^2 \alpha \geq (1-t)^2$ (as $4t \cos^2 \alpha \geq 0$) gives $(1+t + \sqrt{\Delta})/(2t) \geq (1+t + |1-t|)/(2t) \geq 1/t$; since $\lambda < \lambda_{\min}(\Lambda_E) = 1/t$, the only admissible root is the smaller one, cf(t, α). $\square$

Remark 3.5 (w_0 is not optimal when $q < p$; what the closed form then costs). The direction selected by Theorem 3.3 is a top eigenvector of $B(\lambda^*)$, not of $C^\top C$; for $q < p$ the two differ, and w_0 -optimality is *false*. A dense search over S^1 (4000 points) beats $\mu(w_0)$ in 157 of the 840 dictionaries tested in **b37**, and the breakdown is the point: of the 180 draws in the two non-square $q = 2$ cells with $\alpha \geq 20^\circ$ — where a circle search is *exhaustive*, so the count is a measurement and not a sampling rate — 143 admit a direction with $\mu(w) < \mu(w_0)$, i.e. the failure is the norm (79%), not an exception, with angular separation from w_0 up to 0.30 rad. The remaining 14 come from random sampling in the $q = 3$ cells and are therefore a lower bound on what a sphere search would find. An earlier version of this remark reported that “no random unit w beats w_0 ” in 3200 draws; that was an artifact of sampling — 4000 random points do not resolve S^2 , and the claim was only ever testable at $q = 2$, where a circle search is exhaustive. The same script contains the exception that identifies the true theorem: its two square cells, $(d, p, q) = (5, 2, 2)$ and $(6, 3, 3)$, report 0 violations and $\mu(w_0)/\lambda_{\min}(\mathcal{G}) - 1$ at the 10^{-13} level. Corollary 3.4 is what explains them

— those two rows had been logged as an unexplained anomaly in the round that produced them. The cost is nevertheless small: the gap between $\mu(w_0)$ and the sphere minimum is $\leq 9.6 \cdot 10^{-7}$ in $\lambda_{\min}(\mathcal{G})$ units (the $\mu(w_0)$ restriction itself sits $\leq 1.3 \cdot 10^{-5}$ above $\lambda_{\min}(\mathcal{G})$), it grows with α (the w -dependence beyond $\|Cw\|^2$ enters with weights λ^j), and (10) stays a valid upper bound, short of the exact value by $\leq 1.9 \times 10^{-3}$ at $(d, p, q) = (8, 4, 3)$ and $\leq 6.9 \times 10^{-3}$ at $(6, 4, 3)$, both at $\alpha = 45^\circ$ (script `b39_square_case_cf_exact.py`, 200 draws per cell, which also confirms $\lambda_{\min}(\mathcal{G})/\text{cf} - 1 = 0$ to 1.4×10^{-11} on all seven square configurations $\times$ five angles). Theorem 3.3 itself was checked against the exact Gram eigenvalue on 6480 dictionaries, nine (d, p, q) configurations and $\alpha = 1^\circ \dots 89^\circ$: worst relative deviation 1.8×10^{-11} , i.e. round-off, no exception (script `b38_exact_scalar_formula.py`).

Proposition 3.5 (the equal-angle hypothesis is an artifact of the construction). *Let $\Lambda_E \in \mathbb{R}^{p \times p}$ be positive definite, let $K \in \mathbb{R}^{p \times q}$, and let $\mathcal{G} = \begin{pmatrix} \Lambda_E & K \\ K^\top & I_q \end{pmatrix}$ be positive semidefinite, so that the singular values $c_1 \geq \dots \geq c_q \geq 0$ of $\Lambda_E^{-1/2}K$ are principal cosines. On $(0, \bar{\lambda})$ with $\bar{\lambda} := \min(\lambda_{\min}(\Lambda_E), 1)$, put*

$$h(\lambda) := (1 - \lambda) - \lambda_{\max}(K^\top(\Lambda_E - \lambda I)^{-1}K). \quad (14)$$

Then h is strictly decreasing and $\lambda_{\min}(\mathcal{G}) = \min(\bar{\lambda}, \lambda^)$, where λ^* is the root of h if one lies in the interval, $\bar{\lambda}$ if $h > 0$ throughout, and 0 if $h(0^+) \leq 0$.*

Proof. For $\lambda < \bar{\lambda}$ the block $\Lambda_E - \lambda I$ is positive definite, so $\mathcal{G} - \lambda I$ is positive semidefinite iff its Schur complement in that block, $(1 - \lambda)I_q - K^\top(\Lambda_E - \lambda I)^{-1}K$, is. That complement is strictly decreasing in the Loewner order on the interval: $\lambda \mapsto 1 - \lambda$ decreases and $\lambda \mapsto (\Lambda_E - \lambda I)^{-1}$ increases. Hence its positivity set is an interval $[0, \lambda^*]$ and $\lambda_{\min}(\mathcal{G})$, its supremum, is as claimed. At $\lambda = 0$ the complement is $I_q - K^\top \Lambda_E^{-1}K$, positive semidefinite because $\mathcal{G}$ is, and singular exactly when $c_1 = 1$; that is the third case, the two subspaces meeting. $\square$

Corollary 3.6 (the isotropic locus: cf is exact with no angle hypothesis). *Let $\Lambda_E = \gamma I_p + Z$ with $\gamma > 0$, $Z \in \mathbb{R}^{p \times p}$ symmetric, $ZK = 0$, and let $K \in \mathbb{R}^{p \times q}$ be arbitrary, with $\mathcal{G}$ positive semidefinite; write $c^2 := \lambda_{\max}(K^\top \Lambda_E^{-1}K) \in [0, 1]$. Then, with $\bar{\lambda} := \min(\lambda_{\min}(\Lambda_E), 1)$,*

$$\lambda_{\min}(\mathcal{G}) = \min(\bar{\lambda}, \lambda_-), \quad \lambda_- = \frac{(\gamma + 1) - \sqrt{(\gamma - 1)^2 + 4\gamma c^2}}{2} = \text{cf}(\gamma^{-1}, \alpha), \quad \cos^2 \alpha := c^2. \quad (15)$$

In particular $\lambda_{\min}(\mathcal{G}) = \lambda_-$ and cf is exact whenever $\lambda_{\min}(\Lambda_E) \geq \lambda_-$, which holds automatically for $Z \succeq 0$; for $\gamma = 1$ and $Z \preceq 0$ the condition reads $c \geq 1 - \lambda_{\min}(\Lambda_E)$. Under the chapter's normalisation $\text{diag } \Lambda_E = 1$ isotropy forces $\gamma = 1$, i.e. orthonormal atoms, and (15) reads $\lambda_{\min}(\mathcal{G}) = \min(\bar{\lambda}, 1 - c)$. The hypothesis $ZK = 0$ is non-vacuous exactly when K has a non-trivial left kernel, i.e. $p > q$ or a rank-deficient dictionary: over-completeness is what makes a whole direction of Λ_E invisible to $\lambda_{\min}(\mathcal{G})$.

Proof. $\Lambda_E - \lambda I = (\gamma - \lambda)I + Z$ leaves $\text{ran } Z$ and its orthogonal complement invariant, is $(\gamma - \lambda)I$ on that complement, and K maps into it because $ZK = 0$; hence $(\Lambda_E - \lambda I)^{-1}K = (\gamma - \lambda)^{-1}K$ whether or not Z is zero. So $\lambda_{\max}(K^\top(\Lambda_E - \lambda I)^{-1}K) = \gamma c^2/(\gamma - \lambda)$ on $(0, \bar{\lambda})$ and (14) becomes $h(\lambda) = (1 - \lambda) - \gamma c^2/(\gamma - \lambda)$, whose zero set is the quadratic $\lambda^2 - (\gamma + 1)\lambda + \gamma(1 - c^2) = 0$. Its smaller root is the λ_- displayed, and $\lambda_- \leq \gamma$ and $\lambda_- \leq 1$ are each equivalent to $|\gamma - 1| \leq \sqrt{(\gamma - 1)^2 + 4\gamma c^2}$, so $\lambda_- \leq \min(\gamma, 1)$; for $Z = 0$ that is $\lambda_- \leq \bar{\lambda}$ and Proposition 3.5 gives $\lambda_{\min}(\mathcal{G}) = \lambda_-$ (the quadratic is non-negative at 0 and equals $-\gamma c^2 \leq 0$ at $\bar{\lambda}$; at $c = 0$ it returns $\bar{\lambda}$, which is $\lambda_{\min}$ of $\gamma I_p \oplus I_q$). With $Z \neq 0$ the same argument runs on $(0, \bar{\lambda})$ and gives the announced $\min(\bar{\lambda}, \lambda_-)$, the cap being taken at $\lambda_{\min}(\Lambda_E) = \min(\gamma, \gamma + \lambda_{\min}(Z))$ rather than at γ . At $\gamma = 1$ with $Z \preceq 0$ the criterion $\lambda_- \leq \lambda_{\min}(\Lambda_E) = 1 - |\lambda_{\min}(Z)|$ is $h(\lambda_{\min}(\Lambda_E)) \leq 0$, i.e. $c^2 \geq |\lambda_{\min}(Z)|^2$, the stated $c \geq 1 - \lambda_{\min}(\Lambda_E)$. On

the other hand $\Lambda_E^{-1}K = \gamma^{-1}K$ again by $ZK = 0$, so $t = \|\Lambda_E^{-1}K\|_2^2/c^2 = \gamma^{-1}$ whatever Z , and clearing the factor $1/\gamma$ in $\text{cf}(t, \alpha)$ of (10) turns its discriminant into $(\gamma + 1)^2 - 4\gamma(1 - c^2) = (\gamma - 1)^2 + 4\gamma c^2$, which is the second equality. Script **b44** checks (15) on 18 cells ($\leq 1.7 \cdot 10^{-15}$ absolute): K a plain Gaussian block rescaled to prescribed top cosine 0.3/0.9/0.999, full and rank-1, (p, q) equal to (4, 3), (3, 3) and (3, 5), and $\gamma \in \{1, 1.5, 0.6\}$. Script **b45** checks the $Z \neq 0$ clause: $\Lambda_E = I_4 - \delta w w^\top$ with $w \in \ker K^\top$ ($p = 4 > q = 3$), $\delta = 10^{-3}, 10^{-6}$, one spread and one equal-angle spectrum, 400 draws — $|t - 1| \leq 1.3 \cdot 10^{-15}$, median $|\lambda_{\min}(\mathcal{G}) - \lambda_-| \leq 3.0 \cdot 10^{-16}$ and $\max |D| \leq 1.3 \cdot 10^{-15}$ absolute. $\square$

Remark 3.6 (the two exactness loci are two fibres of one mechanism). Corollary 3.4 ($p = q$, isometry) and Corollary 3.6 (Λ_E scalar, any K) look alike because in both cases the pencil $\lambda \mapsto B(\lambda) = C^\top \Lambda_E (\Lambda_E - \lambda I)^{-1} C$ is an *increasing scalar function applied to one fixed Hermitian matrix* — $B(\lambda) = (I - \lambda C^\top C)^{-1} C^\top C$ in the square case, $B(\lambda) = (\gamma/(\gamma - \lambda)) C^\top C$ here — so its top eigenvalue is determined by one number and (12) collapses to a quadratic. Neither hypothesis is the other: the square-isometric locus has non-scalar Λ_E ($\Lambda_E = C^{-\top} C^{-1}$), and the isotropic locus needs no isometry at all. What they share is the conclusion, and it is the honest reading of cf : the scalar reduction is exact when Λ_E carries no directional information that t could lose. At scalar Λ_E the resolvent is a scalar multiple of the identity, so (14) decouples into q quadratics, one per principal cosine — the classical arrowhead case rather than a new formula. What it does for this chapter is name the hypothesis that cf was really using.

Proposition 3.7 (the residual is a moment defect of one scalar measure, and its sign is decidable). Write $\Lambda_E = \sum_{i=1}^p \ell_i u_i u_i^\top$, $k_i := K^\top u_i$, and for a unit $v \in \mathbb{R}^q$ let $\nu_v := \sum_i (k_i \cdot v)^2 \delta_{\ell_i}$, so that $\int x^{-j} d\nu_v = v^\top K^\top \Lambda_E^{-j} K v$ for every $j \geq 1$. Let v maximise $K^\top \Lambda_E^{-1} K$, so that $\int x^{-1} d\nu_v = c^2$, and on $[0, \bar{\lambda})$ put

$$f(\lambda) := \lambda_{\max}(K^\top (\Lambda_E - \lambda I)^{-1} K), \quad g(\lambda) := \frac{c^2}{1 - t\lambda}, \quad D(\lambda) := f(\lambda) - g(\lambda).$$

Then g is $\int d\nu/(x - \lambda)$ for the single atom $\hat{\nu} := (c^2/t) \delta_{1/t}$, which matches ν_v in $\int x^{-1}$ and over-states it in $\int x^{-2}$, and (i) in the uncapped case $\lambda_{\min}(\mathcal{G}) < \bar{\lambda}$, with $\text{cf} \in (0, \bar{\lambda})$,

$$\begin{aligned} D(\text{cf}) &= (\text{cf} - \lambda_{\min}(\mathcal{G})) + (f(\text{cf}) - f(\lambda_{\min}(\mathcal{G}))), \\ \text{sgn } D(\text{cf}) &= \text{sgn}(\text{cf} - \lambda_{\min}(\mathcal{G})), \quad |\text{cf} - \lambda_{\min}(\mathcal{G})| \leq |D(\text{cf})|; \end{aligned} \tag{16}$$

if f is differentiable between the two points, $\text{cf} - \lambda_{\min}(\mathcal{G}) = D(\text{cf})/(1 + f'(\xi))$ for some ξ . (ii) As $\lambda \downarrow 0$,

$$D(\lambda) = -\lambda \text{ slack} + o(\lambda), \quad \text{slack} := c^2 t - \lambda_{\max}(E_0^\top K^\top \Lambda_E^{-2} K E_0) \geq 0, \tag{17}$$

where E_0 is the top eigenspace of $K^\top \Lambda_E^{-1} K$: so the leading term of D is always on the flip side ($\text{cf} < \lambda_{\min}(\mathcal{G})$), it vanishes exactly when some maximiser of $K^\top \Lambda_E^{-1} K$ also maximises $K^\top \Lambda_E^{-2} K$, and where it vanishes the sign is decided by the remainder. (iii) Three named loci have $\text{slack} = 0$: $q = 1$ (scalars), equal angles ($K^\top \Lambda_E^{-1} K = c^2 I_q$, so $E_0 = \mathbb{R}^q$), and Corollary 3.6 (there $D \equiv 0$, not merely to first order).

Proof. $\hat{\nu}$ gives $\int d\hat{\nu}/(x - \lambda) = (c^2/t)/(1/t - \lambda) = c^2/(1 - t\lambda) = g(\lambda)$, $\int x^{-1} d\hat{\nu} = c^2 = \int x^{-1} d\nu_v$, and $\int x^{-2} d\hat{\nu} = c^2 t = \|\Lambda_E^{-1} K\|_2^2 \geq v^\top K^\top \Lambda_E^{-2} K v = \int x^{-2} d\nu_v$. Both f and g are increasing on $[0, \bar{\lambda})$ ($\Lambda_E - \lambda I$ decreases in the Loewner order, and $t > 0$). At $\lambda_{\min}(\mathcal{G})$, which is the root λ^* of (14) in the uncapped case, $(1 - \lambda_{\min}) = f(\lambda_{\min})$; at cf , clearing q in (10) gives $(1 - \text{cf}) = g(\text{cf})$, and $\text{cf} < 1/t$ is strict because $q(1/t) = -c^2 \leq 0$ while cf is the smaller root, so cf is inside the interval where g is finite. Subtracting the two scalar identities yields the first display of (i); its

second factor has the sign of $\text{cf} - \lambda_{\min}(\mathcal{G})$ because f is increasing, giving the sign equality and, since the two summands then share a sign, the bound $|\text{cf} - \lambda_{\min}(\mathcal{G})| \leq |D|$. The mean-value reading is the same identity with $f(\text{cf}) - f(\lambda_{\min}) = f'(\xi)(\text{cf} - \lambda_{\min})$ and $1 + f'(\xi) \geq 1$. For (ii), $g(\lambda) = c^2 + \lambda c^2 t + O(\lambda^2)$, while the right derivative at 0 of $\lambda \mapsto \lambda_{\max}(A + \lambda B)$ at a multiple top eigenvalue of A is $\lambda_{\max}(E_0^\top B E_0)$ (Lidskii–Danskin), here with $A = K^\top \Lambda_E^{-1} K$, $B = K^\top \Lambda_E^{-2} K$; the displayed ≥ 0 is $\lambda_{\max}(E_0^\top B E_0) \leq \lambda_{\max}(B) = c^2 t$, with equality iff a maximiser of A attains the top of B . Hence $D(\lambda) = -\lambda \text{slack} + o(\lambda)$, and if $\text{slack} > 0$ then $D < 0$ on some $(0, \delta)$: a cf landing there sits below $\lambda_{\min}(\mathcal{G})$. (iii) is immediate: $q = 1$ and equal angles make $E_0^\top B E_0 = B$ a scalar or give $E_0 = \mathbb{R}^q$, and at scalar Λ_E the resolvent identity used in Corollary 3.6 gives $f = g$ on the whole interval. $\square$

Remark 3.7 (what the side-bound test measured, and what it did not). Proposition 3.7 exists because the size of the residual had been attributed by **b44** but its *direction* unexplained. Script **b45** tests (i)–(iii) on ~ 6650 draws, eigvalsh-based floors included, and the pre-registered reading is the measured one: (i) gives $\text{sgn } D(\text{cf}) \neq \text{sgn}(\text{cf} - \lambda_{\min})$ in 0 of 500 resolved draws, and the magnitude reading is a *quantitative* oracle, not merely a sign oracle — $\text{gap} / (D(\text{cf}) / (1 + f'(\lambda_{\min})))$ has median deviation $\leq 1.1 \cdot 10^{-4}$ from 1 in every cell (best $2.6 \cdot 10^{-8}$). (ii) holds where it was claimed: $\text{slack} \geq -1.8 \cdot 10^{-15}$ in all cells, and in every flipping cell $|\text{gap} / (\text{cf} \cdot \text{slack})| = 0.92\text{--}1.02$, while the two families that do *not* flip sit at 13 and 126–394: the linear term is the whole story exactly when it wins, which is the content of the $o(\lambda)$ and not of a fitted law. At $\Lambda_E = \gamma I_p$ the diagnostic collapses as (iii) demands ($\max |D| \leq 1.3 \cdot 10^{-15}$ over 15 cells, $\gamma \in \{1, 1.5, 0.6\}$). What the script does *not* buy: a criterion for flipping in terms of the cosine spectrum alone — see retraction (m) of Section 7 — and any statement about $\lambda_{\min}(\mathcal{G})$ at capped $\text{cf} \geq \bar{\lambda}$, which the draws record as skipped rather than read (43 of the 500 $q = 1$ draws, all at the smallest cosine $\cos \alpha = 0.2$).

Proposition 3.8 (the chord surrogate: an upper bound with no angle hypothesis). *Keep the notation of Proposition 3.7 ($\Lambda_E \succ 0$, $K \in \mathbb{R}^{p \times q}$ with $\mathcal{G}$ positive semidefinite, $c^2 := \lambda_{\max}(K^\top \Lambda_E^{-1} K) \in (0, 1]$, E_0 the top eigenspace of $K^\top \Lambda_E^{-1} K$) and put*

$$t_1 := \frac{\lambda_{\max}(E_0^\top K^\top \Lambda_E^{-2} K E_0)}{c^2} \leq t, \quad \text{cf}_1 := \text{cf}(t_1, \alpha), \quad \cos^2 \alpha := c^2. \quad (18)$$

Then, with no hypothesis on the angles, on Λ_E , and no case split at $\bar{\lambda}$,

$$\lambda_{\min}(\mathcal{G}) \leq \text{cf}_1, \quad \text{cf} \leq \text{cf}_1, \quad (19)$$

the second being equality exactly where $\text{slack} = 0$ in (17). More generally, for every unit $v \in \mathbb{R}^q$ with $B(v) := v^\top K^\top \Lambda_E^{-1} K v$ and $C(v) := v^\top K^\top \Lambda_E^{-2} K v$, the smaller root $\lambda(v)$ of

$$C(v) \lambda^2 - (B(v) + C(v)) \lambda + B(v)(1 - B(v)) = 0 \quad (20)$$

satisfies $\lambda_{\min}(\mathcal{G}) \leq \lambda(v)$, and cf_1 is the member of that family with the largest intercept ($B = c^2$, which forces $v \in E_0$) and, among those, the largest slope.

Proof. Fix a unit v with $B(v) > 0$, write $a_i := (k_i \cdot v)^2$, and let $\rho_v := \sum_i (a_i / \ell_i) \delta_{1/\ell_i}$, a positive measure of mass $B(v)$ and first moment $C(v)$. For $\lambda \in [0, \bar{\lambda})$ every atom has $\lambda / \ell_i < 1$ and $y \mapsto (1 - \lambda y)^{-1}$ is convex there, so Jensen's inequality $\int h d\rho_v \geq \rho_v(\mathbb{R}) h(\int y d\rho_v / \rho_v(\mathbb{R}))$ in the variable $y = 1/x$ turns the measure ν_v of Proposition 3.7 into the *chord*

$$\phi_v(\lambda) := \sum_i \frac{a_i}{\ell_i - \lambda} \geq \frac{B(v)}{1 - \lambda C(v) / B(v)} =: s_v(\lambda), \quad \text{hence} \quad f \geq s_v \text{ on } [0, \bar{\lambda})$$

for every v , because $f(\lambda) = \max_{\|v\|=1} \phi_v(\lambda)$. Let $F(\lambda) := (1 - \lambda)(B(v) - \lambda C(v)) - B(v)^2$, so that $s_v = 1 - \lambda$ reads $F = 0$, i.e. (20): $F(0) = B(v)(1 - B(v)) \geq 0$, F opens upward in λ , and $F(B(v)/C(v)) = -B(v)^2 < 0$ where s_v blows up, so the smaller root $\lambda(v)$ is the first contact and obeys $\lambda(v) \leq B(v)/C(v)$, where s_v is finite and increasing. If $\lambda(v) < \lambda_{\min}(\Lambda_E)$, then $f(\lambda(v)) \geq s_v(\lambda(v)) = 1 - \lambda(v)$, and as $\lambda \mapsto f(\lambda) - (1 - \lambda)$ is strictly increasing on $[0, \bar{\lambda}]$ with unique zero λ^* (Proposition 3.5), this gives $\lambda(v) \geq \lambda^* \geq \min(\bar{\lambda}, \lambda^*) = \lambda_{\min}(\mathcal{G})$. If instead $\lambda(v) \geq \lambda_{\min}(\Lambda_E)$, then $\lambda(v) \geq \lambda_{\min}(\Lambda_E) \geq \bar{\lambda} \geq \lambda_{\min}(\mathcal{G})$, the last step being the second clause of Proposition 3.5. That is (19) for the family. Take now $v_1 \in E_0$ maximising C over the unit sphere of E_0 : $K^\top \Lambda_E^{-1} K = c^2 I$ on E_0 gives $B(v_1) = c^2$, and $C(v_1) = c^2 t_1$ by the definition of t_1 , so $s_{v_1}(\lambda) = c^2/(1 - t_1 \lambda)$ and $F(\lambda) = 0$ is exactly $(1 - \lambda)(1 - t_1 \lambda) = c^2$, i.e. $Q(\lambda, t_1) = 0$ of (10), whose smaller root is cf_1 . Compressing cannot raise a maximum, so $t_1 \leq t$; differentiating $Q(\lambda, t) = 0$ at the smaller root gives $\partial \lambda / \partial t = \lambda(1 - \lambda)/(2t\lambda - 1 - t) < 0$ because $2t\lambda < 1 + t$ there, so cf decreases in t , whence $\text{cf} \leq \text{cf}_1$ with equality iff $t_1 = t$, which is slack = 0. $\square$

Remark 3.8 (what the chord-surrogate tests measured). cf_1 was derived *before* it was run, as the surrogate that replaces g by the chord through the two moments it should have used; the pre-registered tests were four, and a fifth test (**b48**) searches rather than samples. *One*: the flips found by the two-cosine scan (**b45**) must go. On the same cells cf lands below $\lambda_{\min}(\mathcal{G})$ in 100/100, 100/100 and 98/100 draws while cf_1 does so in 0/100 in every cell, uncapped and capped alike (1600 further draws in the capped rescan **b47**: 0 resolved sign flips, $\text{min gap}_1 = 1.1 \cdot 10^{-11}$ over the thirteen cells whose gap clears the floor; the three isotropy cells report 0/0 because there the gap is identically zero). *Two*: the correction should cancel the first-order term and leave the remainder. The ratio $\text{med}|\text{gap}_1|/\text{med}|\text{gap}|$ is 0.001–0.31 in every cell where slack dominates, and 1.00–1.10 in the ten where the slope inflation is absent or negligible — the five $q = 1$ controls, the two equal-angle cells, and the three cells (the 30-88-89 family of that scan) that run left unexplained; among them the cell of retraction (m), which cf_1 leaves untouched, as it must: the correction removes only what the slope inflation put there. Along that scan’s top-cosine ladder the log-log slope of $|\text{gap}_1|$ against cf_1 is 2.081, the predicted value being 2 and the falsifier 1. *Three*: the $q = 1$ and isotropy nulls must be unchanged, for $t_1 = t$ there: $\max|\text{cf}_1 - \text{cf}| = 1.3 \cdot 10^{-15}$ over five $q = 1$ cells, and at $\Lambda_E = \gamma I_p$, $\gamma \in \{1, 1.5, 0.6\}$, $|t_1 - 1/\gamma| \leq 2.0 \cdot 10^{-15}$, $|\text{cf}_1 - \text{cf}| \leq 2.5 \cdot 10^{-16}$, $\max|D_1| \leq 1.3 \cdot 10^{-15}$ — which independently re-verifies that the Lidskii–Danskin slope of Proposition 3.7(ii) really is the right derivative at 0. *Four*: the inequality itself, attacked directly: the Jensen residual $\phi_{v_1} - s_{v_1}$ is ≥ 0 at $\lambda = 0.3, 0.7, 0.95$ of cf_1 in every cell (worst $-5.6 \cdot 10^{-16}$, in the isotropy cells where the gap is identically 0), and the whole family $\lambda(v)$ of (20) stayed above $\lambda_{\min}(\mathcal{G})$ for all 200 random directions of all 1600 draws — 320,000 tests, 0 violations. That scan also seemed to show nothing beats cf_1 , and the polar-net search (**b48**, 96×240 directions) instead found that it does: the first variation of $\lambda(v)$ at v_1 is $-2v_1^\top K^\top \Lambda_E^{-2} K u \cdot \lambda(1 - \lambda)/|\partial_\lambda F|$ along $v(t) = \cos t v_1 + \sin t u$, $u \perp v_1$, generally nonzero because t_1 maximises C over E_0 alone — so cf_1 is a saddle of the family, stationary iff $K^\top \Lambda_E^{-2} K v_1 \parallel v_1$, a condition that holds exactly on the slack = 0 loci ($\dim E_0 = q$; there the net finds 0 competitors) and fails generically (84/84 draws). The propagation is limited: the best ray removes 0.07–0.50 of cf_1 ’s over-estimate at the median (0.58 worst), every competitor buys slope with intercept $(C/(c^2 t_1) = 1.000005 \dots 1.000499$ against $B/c^2 \leq 1 - 10^{-6}$), and no family member drops below $\lambda_{\min}(\mathcal{G})$ on the net or on 144,000 ray points, so the proposition stands (retraction (n) records the two artifacts of that run, a mis-chosen v_1 and a normalised noise direction). The closed-form sizing of that descent (**b49**) needs no optimiser. On the steepest ray, $u \parallel P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1$, the two jets are $B(t) = c^2 + b_2 t^2 + O(t^4)$ — no linear term, since B is an exact trigonometric quadratic on a great circle when $A_0 v_1 = c^2 v_1$ — and $C(t) = C(v_1) + c_1 t + c_2 t^2 + O(t^3)$ with $c_1 = 2\|P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1\|$, and implicit differentiation of (20) at the smaller root gives $\lambda(t) = \text{cf}_1 + L_1 t + L_2 t^2$, $L_1 = -F_C c_1 / \partial_\lambda F < 0$

and $L_2 = -(\partial_{\lambda\lambda} F L_1^2 + 2\partial_{\lambda C} F L_1 c_1 + 2F_B b_2 + 2F_C c_2)/(2\partial_{\lambda} F)$, where all coefficients of $F(\lambda; B, C) = C\lambda^2 - (B + C)\lambda + B - B^2$ are evaluated at $(cf_1, c^2, C(v_1))$: $F_C = \lambda(\lambda - 1)$, $F_B = 1 - \lambda - 2B$, $\partial_{\lambda} F = 2C\lambda - B - C < 0$, $\partial_{\lambda\lambda} F = 2C$, $\partial_{\lambda C} F = 2\lambda - 1$, so the ray has an interior minimum at $t^* = -L_1/(2L_2)$ of depth $\text{dip} = -L_1^2/(4L_2)$: *quadratic* in the off- E_0 gradient although the descent itself is linear in t . Measured over 280 saddle draws, $\text{dip}_{\text{meas}}/\text{dip} = 1$ to within $6 \cdot 10^{-6} \dots 1.0 \cdot 10^{-3}$ per cell and $t_{\text{meas}}/t^* \in [0.993, 1.008]$, with $L_2 > 0$ in 280/280; the single ray step removes 0.0004–0.50 of cf_1 's over-estimate at the median (0.76 worst), reproducing the net's range on a different seed by a different method. $cf_1 + \text{dip}$ *is not offered as a bound* — one-sidedness would need a signed third jet, and the measured drift with $|c_1|$ has no consistent sign, which is exactly the audit I ask of other people's closed forms. What is a bound is the family member itself, $cf_2 := \lambda(\cos t^* v_1 + \sin t^* u)$: Proposition 3.8 applied to that explicit unit vector gives $\lambda_{\min}(\mathcal{G}) \leq cf_2$ with no new proof, at the price of one 2×2 root, and the two-jet supplies only where to step and how much is there. A 64×64 net on the same draws gets at most $\sim 10\%$ deeper than the ray (median ray / net = 0.906–1.009, and that net is not a superset of the ray at 5.6° resolution, so the statement is “sampled, none found”, not exhaustive); the absolute size is $|\text{dip}| \leq 1.2 \cdot 10^{-7}$, so this leg closes here — no closed form for the family minimum, and no marginal left worth chasing. Two caveats are recorded in the statement rather than hidden: the proposition uses $\lambda_{\max}$ over E_0 , not “any top eigenvector”, because at a multiple top eigenvalue the vector an eigensolver returns need not maximise C — the equal-angle cells have $\dim E_0 = 3$ and $|C(v_1) - c^2 t_1|$ up to $6.6 \cdot 10^{-1}$ — and (19) is an upper bound, the *unsafe* side for a certificate (Remark 3.1), which no downstream estimate in this chapter uses. What changes is the status of the closed form: an upper estimate of $\lambda_{\min}(\mathcal{G})$ now exists with no equal-angle, no isometry and no simplicity hypothesis, at the price of one extra $q \times q$ compression; and the flip of Proposition 3.7 is explained as the one unlicensed step in (10) — pairing the intercept c^2 with a slope measured in a direction where that intercept was never attained.

Remark 3.9 (what Proposition 3.5 gives, and what it takes away). Put $K = \cos \alpha H$, $C := \Lambda_E^{-1} H$, and assume the isometry $C^\top \Lambda_E C = I_q$ of Theorem 3.3. Then $\Lambda_E(\Lambda_E - \lambda I)^{-1} = I + \lambda(\Lambda_E - \lambda I)^{-1}$ gives $K^\top (\Lambda_E - \lambda I)^{-1} K = \cos^2 \alpha (I_q + \lambda B(\lambda))$, so $h(\lambda) = 0$ reads (12): Theorem 3.3 is the *equal-angle fibre* of Proposition 3.5, and the cost statement transfers verbatim — a monotone one-root find on a $q \times q$ eigenvalue, no eigenvalue of the $(p + q)$ -dimensional Gram. Three consequences, the first two of them against my own interest. (i) *The general statement is not new, and (14) is not offered as a contribution.* $h(\lambda) = 0$ is the secular equation of an arrowhead (bordered diagonal) matrix: the eigenvalues of a diagonal-plus-rank-one matrix are the roots of one monotone scalar equation, each computable separately, which is the basis of the rank-one modification and divide-and-conquer solvers of numerical linear algebra [1, 2, 3] and of the shift-and-invert arrowhead solvers [4], whose rank-one secular equation is the case $q = 1$, K a single column, of (14). Proposition 3.5 is the rank- q reading of that device, with the scalar equation replaced by the top eigenvalue of a $q \times q$ pencil; we state it because the chapter's own theorem is the $q \geq 2$ case of it and not because the reduction is new. What this note owns is the identification: the dictionary factor of $(\star)$ is an arrowhead secular equation. (The classical records above are verified bibliographically — DOI, authors, title, venue — and the connection was drawn at abstract level, not against those papers' theorems; if the rank- q case is already spelled out somewhere in that literature, Proposition 3.5 should be replaced by a citation to it.) (ii) *Relaxing the angle buys almost nothing for $\lambda_{\min}$, and the residual cf leaves behind is not an angle effect at all.* The prescribed-spectrum run **b42** prescribes cosine spectra in place of the constructed single angle ($p = 4$, $q = 3$, $q = 2$ cells, 80–120 draws per cell). Proposition 3.5 reproduces $\lambda_{\min}(\mathcal{G})$ to $\leq 1.6 \cdot 10^{-12}$ relative over spread spectra, and to $\leq 1.3 \cdot 10^{-15}$ against the independent closed answer $\lambda_{\min}(\mathcal{G}) = 1 - c_1$ available at $\Lambda_E = I_p$; on the equal-angle cells it agrees with (12) to $\leq 4.8 \cdot 10^{-13}$, so it is a generalisation and not a correction. Yet $cf(t, \alpha)$, given the

most generous angle available (the top cosine) and the rescaled t , is still within 3.4% of $\lambda_{\min}(\mathcal{G})$ — and the 2×5 design (b44) settles *which* hypothesis that residual is paying for. It is not the angle distribution, as a first reading of these numbers said (retraction (l) of Section 7): with the cosine spectrum fixed, replacing a generic Gram by $\Lambda_E = \gamma I_p$ cuts the median relative gap by eight to twelve orders of magnitude ($1.3 \cdot 10^{-4} \rightarrow 5.5 \cdot 10^{-15}$ at 20° , $1.8 \cdot 10^{-3} \rightarrow 4.0 \cdot 10^{-13}$ at $2\text{--}45\text{--}88^\circ$), exactly the collapse Corollary 3.6 asserts; with Λ_E fixed, replacing the equal angle by a spread spectrum raises it by only thirteen to three hundred. The anisotropy of Λ_E is the variable, and the two-cosine scan says of what: by Proposition 3.7 the residual is a *moment defect*, the gap between the measure ν_v that f integrates and the single atom $\hat{\nu} = (c^2/t)\delta_{1/t}$ that cf integrates. The correct reading is therefore not “the angles are spread” but “ ν_v carries more than one atom, and the leading term (17) beats the remainder” — in which a second cosine is neither necessary nor sufficient. I wrote that it was necessary (retraction (m) of Section 7): with the geometry held multi-atom, the same scan measures the flip in 100/100 draws at a second cosine of 0.087 and again in 100/100 at $\text{cf} = 0.133$, while the *same* spectra placed on a single-atom geometry leave every gap on the eigvalsh floor (12 cells, slack = 0 to round-off). So That design’s tally is a statement about size, not about which hypothesis licenses a flip: $\text{cf} - \lambda_{\min}$ is ≥ 0 on all 240 equal-angle draws and ≤ 0 on 100/100 ($5\text{--}40\text{--}75^\circ$) and 99/100 ($2\text{--}45\text{--}88^\circ$) of the spread draws, whereas $30\text{--}88\text{--}89^\circ$, spread but with secondary cosines ≈ 0.03 , does not flip (0/100) — there too slack > 0 , but $\text{cf} \cdot \text{slack}$ is smaller than the gap by a factor 126–394, which is the remainder winning, exactly as (17)’s $o(\lambda)$ allows. The anisotropy design b44 also closes the audit the prescribed-spectrum run left open: the smallest violating gap in the two flipping cells is $9.1 \cdot 10^5$ and $1.1 \cdot 10^4$ times the per-draw eigvalsh floor $100\epsilon\|\mathcal{G}\|_2$, with no draw below its own floor, so the flip is algebraic and not round-off — while at $\Lambda_E = I_p$, where the true gap is identically zero, the same tally comes out as a coin flip (35–50 of 100), which is what a null should produce. The upper-side reading of (10) is an equal-angle statement too — which is consistent with Remark 3.4: no estimate in this chapter uses cf , and a future one that called it a bound for a learned dictionary would be overstepping the hypotheses of Proposition 3.2. (iii) *The root find is accurate in absolute terms, and the conditioning scan measures where that stops being enough.* Over 60 draws per cell at $p = q = 3$ and $\alpha = 45^\circ$ down to 0.001° , the relative deviation of the bisection from $\text{eigvalsh}(\mathcal{G})$ obeys $\text{relerr} \times \lambda_{\min}(\mathcal{G}) \in [1.6, 1.8] \cdot 10^{-15}$ with no trend across six decades of $\lambda_{\min}(\mathcal{G})$: the method pins $\lambda_{\min}(\mathcal{G})$ to a few units of the pencil’s absolute round-off, so its *relative* accuracy inherits the very $1/\sin^2 \alpha$ amplification this chapter proves for the error ($\lambda_{\min}(\mathcal{G}) \approx \sin^2 \alpha / (1 + \cos^2 \alpha t)$). Two operational consequences. The bracket endpoint must sit a distance $\gg \epsilon\|\Lambda_E\|$ below the pole $\lambda_{\min}(\Lambda_E)$: a relative gap of 10^{-12} makes the resolvent return garbage of order 1, and the “no root, report the cap” branch then fires silently — that, not (14), was the source of the order-100 errors of the first version of that scan, and the guarded endpoint restores the absolute floor down to $\lambda_{\min}(\mathcal{G}) \approx 10^{-9}$. Below $\lambda_{\min}(\mathcal{G}) \approx \epsilon\|\Lambda_E\|$ no tuning of the bracket helps, which is exactly the regime the high-relative-accuracy arrowhead solvers of [4] are built for. None of this touches $(\star)$, which uses the closed-form lower side LB and not $\lambda^\star$ (Remark 3.11); what it constrains is any future use of $\lambda^\star$ as a *reported number* at small α .

Proposition 3.9 (the chord surrogate is the first member of a certified moment hierarchy). *Keep the notation of Proposition 3.8 and put, for a unit v and $j \geq 0$, $m_j := v^\top K^\top \Lambda_E^{-(j+1)} K v = \int y^j d\rho_v(y)$ (so $m_0 = B(v)$, $m_1 = C(v)$). Run the Stieltjes procedure on ρ_v : $h_0 := m_0$, $\pi_0 := 1$, $\pi_{j+1} := (y - a_j)\pi_j - b_j\pi_{j-1}$ with $a_j := \int y\pi_j^2 d\rho_v/h_j$ and $b_j := h_j/h_{j-1}$, $h_j := \int \pi_j^2 d\rho_v$, and let $J_n(v)$ be the resulting symmetric tridiagonal matrix with diagonal $a_0, \dots, a_{n-1}$ and off-diagonals $\sqrt{b_1}, \dots, \sqrt{b_{n-1}}$. Assume $h_0, \dots, h_{n-1} > 0$, i.e. ρ_v carries at least n atoms, and let $x_1, \dots, x_n$ be the eigenvalues of $J_n(v)$ with weights $w_j := m_0(q_j)_1^2$ ($(q_j)_1$ the first component of the j -th normalised eigenvector). Then $G_n^v(\lambda) := \sum_j w_j/(1 - \lambda x_j)$ is the n -point Gauss rule for $y \mapsto (1 - \lambda y)^{-1}$, it is finite and strictly*

increasing on $[0, \bar{\lambda})$, satisfies $G_n^v(0) = B(v)$ and $G_n^v(\lambda) \rightarrow \infty$ as $\lambda \uparrow \max_j 1/x_j (\geq \bar{\lambda})$, and its unique crossing $\lambda_n(v)$ with $1 - \lambda$ obeys

$$\lambda_{\min}(\mathcal{G}) \leq \lambda_n(v) \quad \text{for every unit } v \text{ and every such } n \geq 1, \quad (21)$$

and $\lambda_1(v)$ is the smaller root of (20), so that $\text{cf}_1 = \lambda_1(v_1)$ is the member $n = 1$. Explicitly, $\lambda_2(v)$ is the smallest non-negative root of

$$p_2 \lambda^3 - (p_2 + s_1) \lambda^2 + [(s_1 + 1) - (m_0 s_1 - m_1)] \lambda + (m_0 - 1) = 0, \quad (22)$$

where $a_0 = m_1/m_0$, $b_1 = m_2/m_0 - a_0^2$, $a_1 = (m_3 - 2a_0 m_2 + a_0^2 m_1)/(m_0 b_1)$, $s_1 = a_0 + a_1$, $p_2 = a_0 a_1 - b_1$, and $G_2^v(\lambda) = [m_0 - (m_0 s_1 - m_1) \lambda] / (1 - s_1 \lambda + p_2 \lambda^2)$.

Proof. For $\lambda \in [0, \bar{\lambda})$ every atom has $\lambda y < 1$ and $h_\lambda(y) := (1 - \lambda y)^{-1} \in C^\infty$ of the support of ρ_v , with all even derivatives $h_\lambda^{(2n)}(y) = (2n)! \lambda^{2n} (1 - \lambda y)^{-2n-1} \geq 0$. The classical Gauss error formula $\int h_\lambda d\rho_v - G_n^v(\lambda) = h_\lambda^{(2n)}(\xi)/(2n)! \cdot h_n \geq 0$ is the Gauss–Markov sign statement for rules whose integrand has $2n$ nonnegative even derivative; it is recorded in this form in the standard treatments of Gauss quadrature and orthogonal polynomials [6, 7, 8], and we cite it as a classical theorem rather than reproducing it with a section number, because we have verified those records bibliographically but not read the displayed formula in them. Nothing in what follows rests on that gap: the consequence actually used — $G_n^v \leq \phi_v$ — is checked instead by the moment identities $\sum_j w_j x_j^k = m_k$, $k < 2n$, on every draw (Remark A.2), so the inequality is verified inside the construction that uses it $G_n^v \leq \phi_v$ on $[0, \bar{\lambda})$. The nodes are in the convex hull of the support, $x_j \leq 1/\lambda_{\min}(\Lambda_E)$, hence $\max_j 1/x_j \geq \bar{\lambda}$ and G_n^v is finite and strictly increasing there; $G_n^v(0) = \sum_j w_j = m_0 = B(v) \leq 1$ and $G_n^v \rightarrow \infty$ at the pole give a unique crossing $\lambda_n(v)$ with $1 - \lambda$, and $\lambda_n(v) \geq \bar{\lambda}$ already implies $\lambda_n(v) \geq \bar{\lambda} \geq \lambda_{\min}(\mathcal{G})$ by the second inequality of Proposition 3.7. Otherwise $\lambda_n(v) \in [0, \bar{\lambda})$ and $f(\lambda_n(v)) \geq \phi_v(\lambda_n(v)) \geq G_n^v(\lambda_n(v)) = 1 - \lambda_n(v)$, which as in Proposition 3.8 places $\lambda_n(v) \geq \lambda^* \geq \lambda_{\min}(\mathcal{G})$ — the only step replaced is $s_v \leq \phi_v$ by $G_n^v \leq \phi_v$, so the case split is the same one. For $n = 1$: $J_1 = [m_1/m_0]$ and $w_1 = m_0$, so $G_1^v(\lambda) = m_0/(1 - \lambda m_1/m_0) = B(v)^2/(B(v) - \lambda C(v)) = s_v$, whose crossing is (20); taking $v = v_1 \in E_0$ with $B = \cos^2 \alpha$ gives cf_1 . For $n = 2$: $\sum_j w_j = m_0$ and $\sum_j w_j x_j = m_0 e_1^\top J_2 e_1 = m_0 a_0$ are exact (degree ≤ 3 is integrated exactly), $\sum_j w_j x_j^2 = m_0(a_0^2 + b_1) = m_2$ identifies b_1 , and $\text{tr } J_2 = a_0 + a_1 = s_1$, $\det J_2 = a_0 a_1 - b_1 = p_2$ give $\{x_j\}$; the numerator of G_2^v is $m_0 - \lambda(w_1 x_2 + w_2 x_1)$ with $w_1 x_2 + w_2 x_1 = m_0 s_1 - m_1$ from $(w_1 + w_2)(x_1 + x_2) = \sum w_j x_j + \sum w_j x_j'$. Clearing the denominator, which is $(1 - \lambda x_1)(1 - \lambda x_2) = 1 - s_1 \lambda + p_2 \lambda^2 > 0$ at the crossing, gives (22). $\square$

Proposition 3.10 (the fibre bound is exact at the minimal eigenvector, and that is all it is). *Write $\mathcal{G}[x; y] = \lambda[x; y]$ in the blocks $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$, and let $\lambda := \lambda_{\min}(\mathcal{G})$. If $\lambda < \lambda_{\min}(\Lambda_E)$ — which holds whenever $\lambda_{\min}(\Lambda_E) > 1$, since $\lambda_{\min}(\mathcal{G}) \leq \bar{\lambda} \leq 1$ by Proposition 3.7 — then $y \neq 0$ and*

$$K^\top (\Lambda_E - \lambda I)^{-1} K y = (1 - \lambda) y, \quad \text{so for } v_* := y/\|y\| : \quad \phi_{v_*}(\lambda) = 1 - \lambda. \quad (23)$$

The crossing $\lambda_{\text{ex}}(v_*)$ of ϕ_{v_*} with $1 - \lambda$ therefore obeys $\lambda_{\text{ex}}(v_*) = \lambda_{\min}(\mathcal{G})$: the family $\{\lambda(v)\}$ of (20), and the hierarchy of Proposition 3.9 once n reaches the atom count of ρ_{v_*} , attain the certificate. Nothing here makes them computable: v_* is the q -block of the eigenvector whose eigenvalue is being bounded, and the member $n = 1$ does not attain it — measured on 320 draws, $(\lambda_1(v_*) - \lambda_{\min}(\mathcal{G})) / (\text{cf}_1 - \lambda_{\min}(\mathcal{G}))$ has median 0.50–1.01 in all eight cells (Remark A.1).

Proof. The two block rows of $\mathcal{G}[x; y] = \lambda[x; y]$ read $(\Lambda_E - \lambda I)x = -Ky$ and $K^\top x = (\lambda - 1)y$. If $y = 0$ the first gives $\Lambda_E x = \lambda x$ with $x \neq 0$, contradicting $\lambda < \lambda_{\min}(\Lambda_E)$, so $y \neq 0$; and $\Lambda_E - \lambda I \succ 0$,

so $x = -(\Lambda_E - \lambda I)^{-1}Ky$ and the second row becomes (23). Divide by $\|y\|^2$ and use $a_i = (k_i \cdot v_*)^2$: the left side of (23) is $\sum_i a_i/(\ell_i - \lambda) = \phi_{v_*}(\lambda)$, the right side is $1 - \lambda$. On $[0, \lambda_{\min}(\Lambda_E))$ the map $\mu \mapsto \phi_{v_*}(\mu) - (1 - \mu)$ is strictly increasing, is ≤ 0 at $\mu = 0$ (where it equals $B(v_*) - 1$) and diverges at the pole $\mu = \ell_{\min}$, so it has exactly one zero there; λ is a zero, hence $\lambda = \lambda_{\text{ex}}(v_*)$. The displayed inequality is Proposition 3.9 at $n = \#\text{supp } \rho_{v_*}$, where $G_n^{v_*} = \phi_{v_*}$ exactly. $\square$

Lemma 3.11 (ordering two members of the family: the margin). *For a unit v with $B := B(v) \in (0, 1)$ put $t := C(v)/B(v)$, and let $r(B, t)$ be the smaller root of $(1 - x)(1 - tx) = B$, so that $\lambda_1(v) = r(B(v), t(v))$ is the $n = 1$ member of (20) and $\text{cf}_1 = r(c^2, t_1)$. Then on the smaller-root branch*

$$\frac{\partial r}{\partial t} = \frac{r(1 - r)}{2tr - 1 - t} < 0, \quad \frac{\partial r}{\partial B} = \frac{1}{2tr - 1 - t} < 0, \quad (24)$$

and for every $\mu \in [0, (1 + t)/(2t)]$ the comparison of a family member with a candidate value is an exact criterion,

$$r(B, t) \leq \mu \iff (1 - \mu)(1 - t\mu) \leq B \iff t \geq t_1 + \frac{c^2 - B}{c^2} \cdot \frac{1 - t_1\mu}{\mu}, \quad \mu = \text{cf}_1. \quad (25)$$

Moreover the two knobs collapse to one at first order: since at $\mu = \text{cf}_1$ one has $\partial r/\partial t = \mu(1 - \mu)\partial r/\partial B$ and $(1 - \mu)(1 - t_1\mu) = c^2$, the intercept term equals $\delta := c^2 - B$ times $\partial r/\partial B$ exactly, and

$$\lambda_1(v) - \text{cf}_1 = \frac{\mu(1 - \mu)}{2t_1\mu - 1 - t_1} \left(t - t_1 - \frac{\delta}{c^2} \frac{1 - t_1\mu}{\mu} \right) (1 + o(1)) : \quad (26)$$

the sign of the bracket — the margin — decides which of the two members is the better bound.

Proof. Both partials are implicit differentiation of $Q(x; t, B) := tx^2 - (1 + t)x + 1 - B$ along its smaller root: $Q_x = 2tx - 1 - t$, and the smaller root lies left of the vertex $(1 + t)/(2t)$, so $Q_x < 0$ there; $Q_t = x^2 - x = -x(1 - x)$ and $Q_B = -1$ give (24) with $0 < r < 1$. Since Q opens upward with $Q(0) = 1 - B > 0$, and is strictly decreasing on $[0, (1 + t)/(2t)]$, for μ in that interval $r \leq \mu$ reads $Q(\mu) \leq 0$, i.e. $(1 - \mu)(1 - t\mu) \leq B$. Taking $\mu = \text{cf}_1$, dividing by $1 - \mu > 0$ and substituting $(1 - \mu)(1 - t_1\mu) = c^2$ — the root equation at (c^2, t_1) — turns that into $1 - t\mu \leq (1 - \delta/c^2)(1 - t_1\mu)$, which is the right hand side of (25); every step is reversible, so the only side conditions are $\mu \leq (1 + t)/(2t)$, $\mu < 1$ and $1 - t_1\mu > 0$, and the ray scan b55 verifies each on every draw. Finally $\frac{\partial r}{\partial t} \cdot \frac{\delta}{c^2} \frac{1 - t_1\mu}{\mu} = \delta \frac{\partial r}{\partial B}$ by the two displayed identities, which is the collapse in (26). $\square$

Corollary 3.12 (the direction question is closed when the angles are equal). *If $M := \Lambda_E^{-1/2}K$ satisfies $M^\top M = c^2 I_q$ — an equal cosine spectrum does, since then $M = Q_1 \text{diag}(c, \dots, c) Q_2^\top$ — then $B(v) \equiv c^2$ on the unit sphere, $E_0 = \mathbb{R}^q$, and*

$$\min_{\|v\|=1} \lambda_1(v) = \text{cf}_1,$$

attained exactly at the unit maximisers of $C(v)$; every other direction, v_* included, gives a strictly larger member. So in those cells the certificate is not attained by any $n = 1$ member, however the direction is chosen — and $\delta = 0$ makes (25) read $\lambda_1(v) \leq \text{cf}_1 \iff t(v) \geq t_1$, which no direction satisfies.

Proof. $B(v) = v^\top M^\top M v = c^2 \|v\|^2$, so $\delta = 0$ and every unit vector saturates the intercept, which is $E_0 = \mathbb{R}^q$; then $t_1 = \max_{\|v\|=1} C(v)/c^2$ by (18), and $\lambda_1(v) = r(c^2, C(v)/c^2)$, which (24) minimises exactly where C is maximal, with value $r(c^2, t_1) = \text{cf}_1$. Strictness is the strict negativity of $\partial r/\partial t$. $\square$

Lemma 3.13 (the descent set on a ray is a cubic). *Fix a unit $v_1 \in E_0$ maximising C over E_0 , so $B(v_1) = c^2$, $t(v_1) = t_1$ and $\lambda_1(v_1) = \text{cf}_1 =: \mu$, and let u be any unit vector with $u \perp v_1$. Put*

$$a := B(u), \quad b := C(u), \quad g := v_1^\top K^\top \Lambda_E^{-2} K u, \quad D := c^2 - a, \quad P := \frac{Dc^2}{\mu(1-\mu)} - b + t_1 a,$$

in which the coefficient of the cubic below uses only μ : $\sigma := \frac{1-t_1\mu}{c^2\mu} = \frac{1}{\mu(1-\mu)}$, since $(1-\mu)(1-t_1\mu) = c^2$. For $\theta \in (0, \pi/2)$ and $v_\theta := \cos \theta v_1 + \sin \theta u$,

$$B(v_\theta) = c^2 - D \sin^2 \theta, \quad C(v_\theta) - t_1 B(v_\theta) = 2g \sin \theta \cos \theta + (b - t_1 a) \sin^2 \theta,$$

and with $\Phi(x) := (1+x^2)(Px - 2g) - \sigma D^2 x^3$,

$$\lambda_1(v_\theta) \leq \text{cf}_1 \quad \Longleftrightarrow \quad \Phi(\tan \theta) \leq 0. \quad (27)$$

So the boundary of the descent set on the ray is a positive root of a cubic in $\tan \theta$ with four coefficients built from five scalars; leaving v_1 descends at all iff $g > 0$, since $\Phi(0) = -2g$; dropping σD^2 makes the boundary exactly $\tan \theta = 2g/P$, which is t_{hi} of [b52](#) read analytically; and if $P > 0$ that two-jet value pertains to the true root as $x_{\text{hi}} = x(1 + \frac{\sigma D^2 x^2}{P(1+x^2)} + O((\sigma D^2)^2))$, $x = 2g/P$.

Proof. $K^\top \Lambda_E^{-1} K v_1 = c^2 v_1$ gives $v_1^\top K^\top \Lambda_E^{-1} K u = c^2(v_1 \cdot u) = 0$ for every $u \perp v_1$, so the cross term in $B(v_\theta)$ vanishes and $B = v_1$ -term $\cos^2 \theta + a \sin^2 \theta = c^2 - D \sin^2 \theta$; the same expansion with Λ_E^{-2} keeps the cross term, which is g , and subtracting $t_1 B$ cancels the $c^2 \cos^2 \theta$ against $t_1 c^2 \cos^2 \theta$ to leave the displayed $C - t_1 B$. Now (25) is $\lambda_1(v_\theta) \leq \mu \iff T \geq 0$ for $T := t - t_1 - \sigma(c^2 - B)$, and $B > 0$ lets us multiply it by B : $0 \leq C - t_1 B - \sigma B(c^2 - B) = 2g s c + b s^2 - t_1 a s^2 - \sigma s^2(c^2 - D s^2) = s^2[2g c/s - P + \sigma D^2 s^2]$ with $s := \sin \theta$, $c := \cos \theta$. Dividing by $s^2 > 0$ and multiplying by $x(1+x^2) > 0$ at $x = \tan \theta$ (so $c/s = 1/x$, $s^2 = x^2/(1+x^2)$) turns the bracket into $-\Phi(x)$, which is (27). Setting $\sigma D^2 = 0$ in Φ leaves $(1+x^2)(Px - 2g)$, whose only real root is $2g/P$; and at that root the σD^2 -free part of Φ has derivative $P(1+x^2)$, while Φ itself changes by $-\sigma D^2 x^3$, so the root shifts by $\sigma D^2 x^3/(P(1+x^2))$ to first order in σD^2 — the stated perturbation, and it is $O(x^3)$ in the layer width, which is why the two-jet value is the right one to quote when x is small. $\square$

Remark 3.10 (What the direction leg did not buy). The direction leg was run to look for a cheap way to close the residual gap $\text{cf}_1 - \lambda_{\min}(\mathcal{G})$ by optimising the direction v alone, and it bought nothing: the $n = 1$ chord loses 0.50–1.01 of that gap *even at v_** , the direction that attains the exact fibre crossing of Proposition 3.10, whereas one additional scalar moment removes 1.00 of it on the same tables (median *and* max, 20/20 cells). The residual of cf_1 is therefore truncational, not directional, and the angle $\theta = \angle(v_*, v_b)$ that a reader might otherwise take for a convergence error is a model-discrepancy angle: v_* is not a critical point of λ_1 at all, so no stationary-point story explains cf_1 . The measurement is recorded in full in Appendix A — direction nets and their resolution limits, the two independent routes to the descent-set boundary, the cubic-share attribution of the residual, and the moment hierarchy — because a negative result is only as credible as the resolutions at which it was tested.

Remark 3.11 (what the exact formula does for the certificate). $\lambda_{\min}(\mathcal{G})$ is now available as an *equality*, so the dictionary factor q_4 of §6.2 measures the price of the closed form that $(\star)$ uses, not an intrinsic limitation: replacing LB in $(\star)$ by λ^* of (12) removes the 5% at $\alpha = 2^\circ$ entirely and leaves only the sampling factors. We keep the closed form, because it needs no root find, no eigenvalue, and no extra calibration — and because q_4 is the quantity that says what that choice costs.

4 The sampling side: a remainder that does not know about α

The certificate needs $\sigma_{\min}(G)$, but only $\mathcal{G}$ is a property of the dictionary. The bridge is $\mathbb{E}[G^\top G] = n\mathcal{G}$ *exactly* (the expectation identity is analytic; the finite-sample version was verified numerically (script b11)), so all that can go wrong is the concentration of $G^\top G/n$ around $\mathcal{G}$ in the small directions.

Lemma 4.1 (per-direction quadratic form). *For a fixed unit coefficient vector x put $M_x = \sum_a x_a M_a$ and $r(M) := \|MM^\top\|_F/\|M\|_F^2 \leq 1$. Then*

$$\frac{\|M_x U\|_F^2}{n\|M_x\|_F^2} = 1 + O\left(r(M_x)\sqrt{\frac{1}{n}}\right) \quad \text{with} \quad \text{Var}(\|M_x u_i\|^2) = 2\|M_x M_x^\top\|_F^2 \leq 2\|M_x\|_F^4. \quad (28)$$

Proof. $\|M_x U\|_F^2 = \sum_{i=1}^n \|M_x u_i\|^2$, and for standard Gaussian u , $\mathbb{E}\|M_x u\|^2 = \|M_x\|_F^2$ while $\|M_x u\|^2$ is a Wishart quadratic form with variance $2\|M_x M_x^\top\|_F^2$. Averaging n independent copies and dividing by $n\|M_x\|_F^2$ gives the relative fluctuation $2^{1/2}r(M_x)/\sqrt{n}$. Covering the unit sphere of the $(p+q)$ -dimensional coefficient space by a net converts the fixed-direction statement into a uniform one at the price of a factor $\sqrt{p+q}$. $\square$

Theorem 4.2 (angle-free sampling remainder). *With the notation above, uniformly over coefficient directions,*

$$\left| \frac{\|M_x U\|_F^2}{n\|M_x\|_F^2} - 1 \right| \leq \kappa r(M_x) \sqrt{\frac{p+q}{n}}, \quad \text{hence} \quad \sigma_{\min}(G) = \sqrt{n} \sigma_{\min}(\mathcal{G}^{1/2}) \left(1 \pm \kappa \sqrt{\frac{p+q}{n}}\right), \quad (29)$$

*and the right-hand side contains **no** α and **no** d . The statement is valid when $\kappa\sqrt{(p+q)/n} \ll 1$, i.e. $n \gg p+q$, independently of $\alpha \rightarrow 0$.*

Remark 4.1 (what this buys, and what it costs). Theorem 4.2 is the hinge of the whole chapter: it says the $1/\sin \alpha$ law survives going from the exact dictionary to finite samples. An earlier attempt at this proof used an operator-norm perturbation bound and produced $O(d/(\sqrt{n}\sin^2 \alpha))$, which at $\alpha = 0.3^\circ$, $d = 24$, $n = 200$ predicts a relative deviation of 6.2×10^4 against a measured 0.0125 — wrong by seven orders of magnitude, and structurally so, because a smallest-eigenvalue perturbation is quadratic, not linear. Singular-value lower bounds must not be obtained from global operator-norm perturbations.

Remark 4.2 (the displayed rate is provable but loose). The net argument in Lemma 4.1 gives $\sqrt{p+q}$; the measured fluctuation does not scale that way. Theorem 4.3 below replaces it by the dimension of the near-null subspace, $D_{\text{eff}} = O(1)$. We state both because the first is what we can prove and the second is what the data say; publishing only the first would misreport the size of the remainder, publishing only the second would overclaim a theorem.

Numerical status of Theorem 4.2. The mechanism (r flat in α , $\sim 1.4/\sqrt{d}$):

	$\alpha = 1^\circ$	$\alpha = 30^\circ$	isotropic $1/\sqrt{d}$
$d = 10$	0.4515	0.4500	0.316
$d = 24$	0.2910	0.2921	0.204

The rate (α across 90° at fixed n shows no trend; $n : 100 \rightarrow 3200$ decays $\approx 6\times$ against the $\sqrt{n}$ prediction $5.7\times$), with $|\varepsilon|$ averaged over 12 trials at $d = 10$, $p+q = 6$:

n	$\alpha = 1^\circ$	5°	30°	90°
100	0.0263	0.0347	0.0405	0.0346
200	0.0137	0.0171	0.0133	0.0194
800	0.0113	0.0111	0.0097	0.0111
3200	0.0059	0.0040	0.0034	0.0042

$|\varepsilon|/\sqrt{(p+q)/n} \in [0.07, 0.17] \Rightarrow \kappa \approx 0.15$. Worst case over 40 draws: relative downward deviation of $\lambda_{\min}$ is 12.7%/6.3%/2.7% at $n = 200/600/1800$, always below $\sqrt{(p+q)/n} = 17.3\%/10.0\%/5.8\%$. (Scripts: `b11`, `b14_g_alpha_kappa.py`.)

4.1 The remainder refined: a computable one-sided bias, not a symmetric fluctuation

Theorem 4.3 (shape of the remainder). *Write $H := G^\top G/n = \mathcal{G} + D$ and let (λ_1, v_1) be the smallest eigenpair of $\mathcal{G}$. Define the second-order displacement*

$$\beta_2 := \sum_{k \geq 2} \frac{\mathbb{E} |\langle v_1, Dv_k \rangle|^2}{\lambda_k - \lambda_1} = \frac{1}{n^2} \sum_{k \geq 2} \frac{\mathbb{E} |\langle v_1, \Delta v_k \rangle|^2}{\lambda_k - \lambda_1}, \quad \Delta := G^\top G - n\mathcal{G}. \quad (30)$$

Then, with $\varepsilon_n := \sigma_{\min}(G)/\sqrt{n\lambda_{\min}(\mathcal{G})} - 1$,

$$\varepsilon_n = -\frac{\beta_2}{2\lambda_1} \pm c_{\text{fluc}} r \sqrt{\frac{D_{\text{eff}}}{n}}, \quad \mathbb{E}[\lambda_{\min}(G^\top G)] - n\lambda_1 = -n\beta_2 + O(n^{-1}). \quad (31)$$

The leading term is (i) one-sided and negative — $\mathbb{E}[\lambda_{\min}(G^\top G)/n] < \lambda_{\min}(\mathcal{G})$, by concavity of $\lambda_{\min}$ together with $\mathbb{E}H = \mathcal{G}$; (ii) of order $1/n$ in β_2 for fixed $\mathcal{G}$ (each $\langle v_1, Dv_k \rangle$ is $O_p(n^{-1/2})$ by Lemma 4.1), hence relative to λ_1 it is $\beta_2/(2\lambda_1) = O(1/n)$, not $O(n^{-1/2})$; and (iii) computable from the dictionary alone, since β_2 needs only the eigendecomposition of $\mathcal{G}$ and the second moments of U . Note the two readings of the same quantity: the absolute displacement $n\beta_2$ is $O(1)$ in n , while the relative one is $O(1/n)$. The fluctuation term carries the near-null dimension D_{eff} , not $p+q$.

Proof outline. Second-order Rayleigh–Schrödinger expansion of $\lambda_{\min}$ at $\mathcal{G}$ under the symmetric perturbation D : the first-order shift $\langle v_1, Dv_1 \rangle$ has mean 0 because $\mathbb{E}D = 0$, so the mean displacement is the sum of the second-order terms, each of which is negative for the smallest eigenvalue since $\lambda_k - \lambda_1 > 0$ for $k \geq 2$; the displayed value of $n\beta_2$ is that sum, and the remainder requires the non-degeneracy condition $|\langle v_1, Dv_k \rangle| \ll \lambda_k - \lambda_1$, which is the regime where the expansion is valid. Concavity of $\lambda_{\min}$ makes the average strictly below the deterministic value: this is Jensen’s gap in operator form, and it is the same mechanism (a concave spectral functional, a zero-mean perturbation) that produces the one-sided residual in the coverage certificate of the companion chapter. The symmetric spread left over is Lemma 4.1 restricted to the set of directions that nearly attain λ_1 , whose dimension is D_{eff} . Not proved here and not claimed: an upper bound on β_2/λ_1 that is uniform in α . Since $\lambda_1 \asymp \sin^2 \alpha$ (Theorem 2.4), uniformity requires $\beta_2 = O(\sin^2 \alpha/n)$, i.e. the numerator must vanish at exactly the same rate as the denominator; Theorem 4.4 records this as a measured fact over a scanned parameter range, with the parameter grid stated there. $\square$

n	100	200	600	1800	5400
$1 - \mathbb{E}[\lambda_{\min}(G^\top G)/n]/\lambda_{\min}(\mathcal{G})$	0.1342	0.0840	0.0344	0.0102	0.0052
$\times n$	13.4	16.8	20.6	18.4	27.8

$\times n$ is flat within a logarithmic drift, while $\times \sqrt{n}$ would have to decrease by a factor 5 over the table and does not (1.34, 1.19, 0.84, 0.44, 0.38): the rate is $1/n$. Quantiles of ε_n ($p + q = 24$, $d = 10$, $\alpha = 5^\circ$, 400 trials) confirm one-sidedness — at $n = 600$ the 95% quantile is $+0.0042$, so $\sigma_{\min}(G)$ falls short of the deterministic prediction with probability ≈ 0.95 :

n	$q_{0.0025}$	$q_{0.01}$	$q_{0.05}$	median	$q_{0.95}$	$q_{0.99}$	mean	sd
200	−0.0945	−0.0885	−0.0766	−0.0413	−0.0111	+0.0030	−0.0432	0.0201
600	−0.0461	−0.0436	−0.0355	−0.0149	+0.0042	+0.0135	−0.0155	0.0123
1800	−0.0245	−0.0208	−0.0165	−0.0058	+0.0056	+0.0105	−0.0055	0.0071

And the predicted-versus-measured displacement $-n\beta_2$, with *no* fitted parameter:

n	measured mean shift $\mathbb{E}[\lambda_{\min}(G^\top G)] - n\lambda_1$	predicted $-n\beta_2$	ratio
200	-1.06×10^{-1}	-1.28×10^{-1}	0.825
600	-1.20×10^{-1}	-1.38×10^{-1}	0.870
1800	-1.35×10^{-1}	-1.43×10^{-1}	0.947

The absolute shift is nearly n -independent, exactly as $n\beta_2 = O(1)$ predicts; agreement improves with n because the perturbation-series validity condition $|\langle v_1, \Delta v_k \rangle| \ll \lambda_k - \lambda_1$ degrades at small n , which is also the source of the residual 5–18% gap. (Scripts: `b15_bias_not_noise.py`, `b24_ggamma.py`, `b25_pqn_law.py`.)

4.2 Uniformity in α and the size of the constants

The question that decides whether the certificate needs an applicability domain is whether the relative bias decays more slowly than $\lambda_{\min}(\mathcal{G}) \asymp \sin^2 \alpha$ as $\alpha \rightarrow 0$. It does not.

Theorem 4.4 (α -uniform bias; measured). *For $d \in \{3, \dots, 24\}$, $p + q \in \{2, \dots, 14\}$, $n \in \{200, 600, 1200, 2400\}$ and $\alpha \in (0, 30^\circ]$,*

$$\left| \frac{\mathbb{E}[\lambda_{\min}(G^\top G/n)]}{\lambda_{\min}(\mathcal{G})} - 1 \right| \leq \frac{c}{n}, \quad c \leq 9, \quad (32)$$

while for α up to 89° , $c \leq 20$. The fluctuation satisfies $\text{sd}(\lambda_{\min}(G^\top G/n)/\lambda_{\min}(\mathcal{G})) = 0.68/\sqrt{n}$ in 21/21 cells, independent of $p + q$. Consequently $\sigma_{\min}(G) \geq \sqrt{n\lambda_{\min}(\mathcal{G})} (1 - c/2n - 0.88/\sqrt{n})$ with 0.5% failure probability.

n	mean relative bias, $\alpha \leq 20^\circ$	theory $-(p + q)/n = -7/n$	$\alpha = 85^\circ$
200	−0.0356	−0.0350	−0.0574
600	−0.0125	−0.0117	−0.0315
2400	−0.00313	−0.00292	−0.0150

Three parameter scans pin the constant c and kill two conjectures of ours:

- α , fixed $p + q = 7, d = 8, n = 600$: c by $|\text{bias}| \cdot n$ equals 7.79, 7.20, 8.58, 7.15, 7.80 at $\alpha = 0.1^\circ, 0.5^\circ, 2^\circ, 10^\circ, 30^\circ$ and then rises to 14.17 (60°) and 19.26 (89°). Flat exactly where the certificate needs to hold, worst where $\lambda_{\min}(\mathcal{G}) \approx 1$ anyway.

- $p + q$, at $n = 200/600/1200$: 0.38/0.94/ − 0.76 ($p + q = 2$); 5.70/8.53/6.25 ($p + q = 7$); 9.45/11.39/12.75 ($p + q = 14$). So $c \neq p + q$: it grows sublinearly and saturates. A law of the form $(p + q)/n$ is falsified.
- d , at $p + q = 7, n = 600$: 3.01, 4.05, 5.87, 8.68, 11.27, 9.06, 8.84 for $d = 3, 4, 6, 8, 12, 16, 24$. c increases with d and saturates around 9; our predicted $O(1/d)$ correction had the wrong sign.

5 The leakage theorem and the tight worst case

The first-order expansion of the correction coefficients is $\hat{\psi} - \psi^* = (G^\top G)^{-1} G^\top r$ with $r = \text{vec}(\Delta U + \Sigma)$, a residual independent of the fitted class. Projecting on the smallest singular triple $(u_{\min}, \sigma_{\min}, v_{\min})$ of G and using $\|\sum_j v_j F_j\|_F / \|v\| = 1$ from Definition 1.1 gives the exact statement we need.

Theorem 5.1 (identifiability of the correction, worst case). *For every dictionary, every α , every $n > d$, and every Δ, Σ ,*

$$\|\hat{C} - C^*\|_F \leq \frac{\|\Delta U + \Sigma\|_F}{\sigma_{\min}(G)} \leq \frac{\|\Delta\|_F + \sigma\sqrt{d}}{\sin \alpha_{\min}} \sqrt{1 + 2 \cos^2 \alpha_{\min} \|\Lambda_E^{-1} H\|_2^2} (1 + o(1)), \quad (33)$$

where $C^* = \sum_j \psi_j^* F_j$ and $\hat{C} = \sum_j \hat{\psi}_j F_j$. The leading constant is exactly 1 and cannot be improved: for $\Delta := W U^\top (U U^\top)^{-1}$, rescaled to any prescribed $\|\Delta\|_F$ (solvable whenever $n > d$), one has $\text{vec}(\Delta U) \parallel u_{\min}$ and (33) is attained.

Corollary 5.2 (average case is a dimensional artifact). *If the direction of Δ is isotropic, then a single direction contributes $|\langle \text{vec}(\Delta U), u_{\min} \rangle| \approx \|\Delta U\|_F / d$, but the correction block accumulates q such components: by rotational invariance $\|\mathcal{P}_{\mathcal{F}} \Delta\|_F^2 / \|\Delta\|_F^2 \sim \text{Beta}(q/2, (d^2 - q)/2)$, whence, to leading order in $1/d$,*

$$\mathbb{E} \|\hat{C} - C^*\| \approx \frac{\mathbb{E} \|\mathcal{P}_{\mathcal{F}} \Delta\|_F}{\sin \alpha_{\min}} = \kappa(q) \frac{\sqrt{q}}{d} \frac{\|\Delta\|_F}{\sin \alpha_{\min}}, \quad \kappa(q) := \sqrt{\frac{2}{q}} \frac{\Gamma(\frac{q+1}{2})}{\Gamma(\frac{q}{2})} \uparrow 1.$$

The average case thus vanishes as $d \rightarrow \infty$ at a $\sqrt{q}/d$ rate — the $\sqrt{q}$ is block aggregation, not physics ($\kappa(1) = 0.798$, $\kappa(3) = 0.921$, $\kappa(6) = 0.960$), while the worst-case bound (33) does not dilute at all. Noise, by contrast, does not dilute: $\text{vec}(\Sigma)$ is isotropic in dn dimensions, giving $\approx \sigma / \sin \alpha_{\min}$ with no d .

Remark 5.1 (consequence for reporting). In high- or infinite-dimensional SciML (PDEs, rich dictionaries) “random misspecification is harmless” is a statement about the dimension, not about the physics. Only the worst-case normalization of Theorem 5.1 is a certificate.

Remark 5.2 (fitted constants in prior versions of this work are not free — and one decomposition was wrong). The measured ratio $\text{err}/\text{bound} = 0.144$ at $d = 10$, $q = 3$ is a $\sqrt{q}$ -type dilution times the dictionary multiplier: $\kappa(3)\sqrt{3}/10 = 0.160$ against the $c_1 \approx 0.155$ –0.16 fitted in two independent designs (the adversarial table, and A’s two-arm least-squares experiment; scripts **b40**, **b41**). An earlier version of this Remark wrote that number as $(1/d) \times \sqrt{1 + 2 \cos^2 \alpha} \|\Lambda_E^{-1} H\|_2^2 = 0.10 \times 1.08$, which is 0.108, not the 0.155 it claimed to reproduce; the missing factor is $\sqrt{q}$, i.e. block aggregation, and the q -scan of the q -scan **b41** decides the question rather than the arithmetic: c_1 at $q = 1, 2, 3, 4, 6$ is 0.077, 0.124, 0.151, 0.183, 0.227, so it is not q -blind, and $c_1/(\kappa(q)\sqrt{q}/d) \in [0.94, 0.99]$ with no trend in q . Both factors are computable before seeing data ($\|\Lambda_E^{-1} H\|_2^2$ measured 0.123–0.134 $\approx 1/d$).

The companion constant $c_2 \approx 0.063$ is $\sqrt{q/n} = \sqrt{3/600} = 0.0707$ to within 13% — a Gaussian quantity over $\sqrt{n}$ with the same block factor, and independent of d — not a physical constant. Hence the two-scale law $c_1\|\Delta\| + c_2\sigma$ of an earlier draft should be read as “one tight bound plus one dilution factor, each carrying its own $\sqrt{q}$ ”, and c_1, c_2 must not be presented as universal.

6 The certificate and what it costs to use it

$$\boxed{\|\hat{C} - C^*\|_F \leq \frac{\|\Delta\|_F + \sigma\sqrt{d}}{\sin \alpha_{\min}} \cdot \sqrt{1 + 2 \cos^2 \alpha_{\min} \cdot \|\Lambda_E^{-1} H\|_2^2}} \quad (\star)$$

Every quantity on the right is computable from the dictionary, the declared model-form-error budget $\|\Delta\|_F$, and the noise level: there is no fitted constant, no resampling, and no unknown spectral quantity. The sample-size factor of Theorem 4.4 may be appended as $(1 - c/2n - 0.88/\sqrt{n})^{-1}$; Section 6.2 shows where it belongs and where it does not.

6.1 $(\star)$ holds with no sample-size margin, in the only test that can detect a failure

Random-direction Δ tests of a certificate are *vacuously true*: at $p + q = 7$, $n = 600$, $\sigma = 10^{-4}$, the ratio err/bound sits in 0.13–0.17 and is flat in n from 60 to 1800 — a $30\times$ range in which the remainder of Theorem 4.4 changes by more than 10%. Corollary 5.2 explains it: the leakage is diluted by $1/d$, so the test never gets close to the bound. Any claim of the form “the certificate passes at $n = \dots$ ” made on a random arm carries no information about the margin. The adversarial arm of Theorem 5.1 is required.

n	$\alpha = 2^\circ$	5°	20°	45°	85°	$\max(85^\circ)$	violations
30	0.880	0.879	0.890	0.927	0.983	0.992	0/120
60	0.883	0.884	0.895	0.930	0.988	0.992	0/120
100	0.884	0.885	0.896	0.931	0.990	0.993	0/120
200	0.887	0.887	0.896	0.933	0.991	0.993	0/120
600	0.887	0.888	0.897	0.935	0.992	0.993	0/120
1800	0.887	0.886	0.897	0.933	0.993	0.993	0/120

36 cells $\times$ 120 trials = 4320 adversarial runs ($\text{vec}(\Delta U) \parallel u_{\min}(G)$, $\|\Delta\|_F = 0.05$, $d = 10$, $p = 4$, $q = 3$): **zero violations with the margin set to 1**, maximum observed ratio 0.993. Note the counterintuitive ordering: the bound is *tightest* at large α ($0.983 \rightarrow 0.993$), not small, because at small α the bound itself shrinks. (Script: `b27_adversarial_margin.py`.)

6.2 Why the ratio is flat in n : an identity, measured

Under the adversarial construction the achieved ratio factorizes

$$\frac{\text{err}}{\text{bound}} = q_4 \cdot q_1 \cdot q_2 \cdot q_3, \quad q_4 = \sqrt{\frac{\text{LB}}{\lambda_{\min}(\mathcal{G})}}, \quad q_1 = \sqrt{\frac{\lambda_{\min}(\mathcal{G})}{\lambda_{\min}(G^\top G/n)}}, \quad q_2 = \frac{\|\Delta U\|_F}{\sqrt{n} \|\Delta\|_F}, \quad q_3 = \|v_{\mathcal{F}}\|, \quad (34)$$

where $\text{LB} = \sin^2 \alpha / (1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2)$ is the lower branch of Theorem 3.1 and $v_{\mathcal{F}}$ is the correction-block part of the minimum right-singular direction of the design matrix: the leakage is the *amplitude* divided by $\sigma_{\min}(G)$, but only the fraction $\|v_{\mathcal{F}}\|$ of that amplitude lands in the

correction block. Thus q_4 is *the slack of the dictionary theorem*, q_1 *the sampling shortfall* measured in Theorems 4.3 and 4.4, q_2 *the price of realizing ΔU with a minimum-norm operator*, and q_3 *the block mass*. The four factors multiply to the observed ratio per trial (script `b32`: $q_4 q_1 q_2 q_3 = 0.8905$ against a measured 0.8905 at $\alpha = 2^\circ$, $n = 200$); an earlier draft of this chapter omitted q_3 and left an unexplained “residual” column of $\approx 5\%$ in the table below, which is $1 - q_3$ read off column means ($q_3 = 0.943\text{--}0.950$ at $\alpha = 2^\circ$, $0.9985\text{--}0.9995$ at 85° ; the direct per-trial measurement over 60 trials gives 0.9466 (per-trial sd 0.0066) and 0.9993) — see Section 7, item (h).

n	α	err/bnd	q_4	q_1	q_2	$q_1 q_2$	$1 - q_3$
30	2°	0.8859	0.9482	1.0733	0.9190	0.9863	0.0464
60	2°	0.8908	0.9484	1.0453	0.9475	0.9904	0.0473
200	2°	0.8916	0.9478	1.0187	0.9754	0.9936	0.0498
600	2°	0.8920	0.9478	1.0050	0.9905	0.9955	0.0514
1800	2°	0.8920	0.9476	1.0018	0.9959	0.9977	0.0534
30	20°	0.8945	0.9531	1.0656	0.9269	0.9871	0.0433
600	20°	0.9029	0.9538	1.0061	0.9894	0.9954	0.0464
1800	20°	0.9010	0.9528	1.0035	0.9941	0.9976	0.0495
30	85°	0.9901	0.9996	1.0896	0.9129	0.9946	0.0015
1800	85°	0.9988	0.9996	1.0086	0.9913	1.0000	0.0005

The last column is $q_4 q_1 q_2 - \text{err}/\text{bound}$ under column means, which is why it is not exactly $1 - q_3$; the per-trial closure of (34) is the statement in the text above.

Theorem 6.1 (tightness is a property of $\mathcal{G}$; measured). *q_4 is independent of n , σ and the dictionary scaling: at $\alpha = 2^\circ$ it is 0.9476–0.9484 over $n = 30 \dots 1800$ (0.08% spread), and 0.9996 at $\alpha = 85^\circ$. Moreover $q_1 - 1 \approx 3/n$ and $q_2 = 1/q_1 \cdot (1 \pm 1\%)$ cell by cell, so $q_1 q_2 \in [0.986, 1.000]$ throughout the i.i.d. grid. Section 7, item 3, measures the correlated (trajectory) design arm, where the product drifts down to 0.975: the cancellation degrades but always in the safe direction.*

Mechanism. Both q_1 and q_2 are lower-edge Marchenko–Pastur effects of the *same* data matrix U : $\lambda_{\min}$ of the design Gram is short by the same amount that the minimal stretch of the map $\Delta \mapsto \Delta U$ is short, and a minimal-norm realization of a fixed ΔU becomes exactly that much *cheaper* in $\|\Delta\|_F$. The two factors are one phenomenon read from two sides, so their product is ≈ 1 . $\square$

Remark 6.1 (q_4 is not measured — it is a formula). Because the upper side of Theorem 3.1 is attained per sample (Section 3, script `b29`), the dictionary factor is available in closed form:

$$q_4 = \sqrt{\frac{\text{LB}}{\lambda_{\min}(\mathcal{G})}} \approx \left(\frac{1 + \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2}{1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2} \right)^{1/2}, \quad (35)$$

and the right-hand side uses only the dictionary. With $\|\Lambda_E^{-1} H\|_2^2 = 0.12809$ (median over 300 draws at $d = 10$, $p + q = 7$) it predicts $q_4 = 0.94770$ against the measured 0.9478 at $\alpha = 2^\circ$, and 0.99951 against 0.9996 at $\alpha = 85^\circ$ — both to 10^{-4} (script `b30`). The 5.2% that the dictionary factor removes from $(\star)$ is thus *exactly* the $\varepsilon = 1$ Young step of the lower branch: no sampling, no noise, and no dictionary scale enters it. The claim was then checked over the whole angle range, not just at two points: 4400 dictionaries at $d = 10$ over eleven α ’s give per-sample deviation of the closed form (35) from the exact factor of 0.0000–0.0020 in the mean (worst single draw 0.0040, again at $\alpha = 45^\circ$) with zero upper-side violations, and the fitted q_4 at $\alpha = 20^\circ$ in the table above — obtained from a least-squares solve, hence not equivalent to the closed form by construction — is 0.9528–0.9538

against a predicted 0.9525. (Script b31.) The deviation is 1.5% at $d = 8$ (script b29) and 0.4% at $d = 10$: smaller in better-conditioned dictionaries, although we have not established the scaling law. The fit-based check exists at three angles only; the other eight are checked against the exact dictionary factor, which is the same statement one step earlier in the chain. Proposition 3.2 sharpens the display in two ways at once. Since $\lambda_{\min}(\mathcal{G}) \leq \text{cf}(t, \alpha)$, the dictionary-only expression $\sqrt{\text{LB}/\text{cf}}$ is a *proven lower bound* for q_4 rather than an approximation — so no sampling, noise or dictionary scale accounts for the looseness q_4 measures, which is carried entirely by the dictionary side — and because cf sits closer to $\lambda_{\min}(\mathcal{G})$ it approximates q_4 to half that slack under the square root: $\leq 1.1 \times 10^{-3}$ relative at $d = 8$, against the 1.5% of (35)’s right-hand side at the same d . Theorem 3.3 then removes the gap altogether: substituting its root λ^* for $\lambda_{\min}(\mathcal{G})$ makes q_4 a computed quantity rather than an estimated one, and Corollary 3.4 shows the closed form *is* exact whenever $p = q$. Remark 3.11 states what that choice costs and why we did not take it.

Corollary 6.2 (the observed looseness of $(\star)$ is one dictionary scalar). *Combining (35) with $q_3 \approx \sqrt{\lambda_{\min}(\mathcal{G})}/\sin^2 \alpha$ (the dictionary-side surrogate for the block mass; it agrees with $\|v_{\mathcal{F}}\|$ to 0.6% in the mean and 2.1% on the worst trial, script b32) and $q_1 q_2 \in [0.986, 1.000]$ gives*

$$\frac{\text{err}}{\text{bound}} \approx \left(\frac{\text{LB}}{\sin^2 \alpha} \right)^{1/2} = (1 + 2 \cos^2 \alpha \|\Lambda_E^{-1} H\|_2^2)^{-1/2}, \quad (36)$$

which contains neither $\lambda_{\min}(\mathcal{G})$ nor any sampled quantity. Predictions against the fitted values of the table above: 0.8923 vs 0.8859–0.8920 ($\alpha = 2^\circ$), 0.9031 vs 0.8945–0.9029 (20°), 0.9990 vs 0.9901–0.9988 (85°); the largest discrepancies are the $n = 30$ cells, where $q_1 q_2$ is at its minimum (0.986–0.987). In words: how loose the certificate is in practice is decided by the same $\varepsilon = 1$ Young step, and it is readable off the dictionary before the fit.

Remark 6.2 (where the sample-size remainder belongs — and where it must not appear). The proof of Theorem 6.1 is unavailable to the certificate’s derivation: the proof chain goes through the one-sided inequality $\sigma_{\min}(G) \geq \sqrt{n \lambda_{\min}(\mathcal{G})} (1 - c/2n - 0.88/\sqrt{n})$, which cannot see the q_2 compensation. Hence the factor $(1 - c/2n - 0.88/\sqrt{n})^{-1}$ **must appear in the statement of the theorem and must not appear in any numerical table**. Extrapolating it to $n = 30$ predicts $\text{err}/\text{bound} \approx 1.02$ (a violation) against a measured 0.886; including it in a results table would tell a reader that the certificate degrades with sample size, which is the opposite of the measurement. We flag this explicitly because our own draft made the prediction before the measurement and was wrong in direction twice: once on the sign of the effect (a larger margin enlarges the bound and *lowers* the ratio; all 25 cells satisfied $\text{new} < \text{old} < \text{raw}$) and once on its size (the $c/n + 1.75/\sqrt{n}$ extrapolation overestimates $q_1 - 1$ by a factor 1.6).

7 Limitations, retracted claims, and open items

1. **Constants are calibrated, not derived.** $\kappa \approx 0.15$ in Theorem 4.2 and $c \leq 9/c \leq 20$ in Theorem 4.4 are empirical envelopes over scanned ranges ($d \in [3, 24]$, $p + q \in [2, 14]$, $\alpha \in (0, 89^\circ]$, $n \in [200, 2400]$). A proof of $c(d, p + q)$ exists only as a claim that a net argument gives *some* constant; we judged the analytic derivation not worth the effort, since it would replace a 21-cell-supported number with an uglier, looser bound. The honest label is “explicit constant, empirically validated”.
2. **A regime we have not tested: n comparable to $p + q$.** Our tables keep $n > p + q$ by a factor of at least 14. With an effectively infinite dictionary ($p + q \sim 100$ at $n = 600$) the remainder of Theorem 4.2 is 0.19 and the theorem is heuristic.

3. **Gaussian U is load-bearing for the tightness, not for the safety — now measured.**

The identity $q_1 q_2 \approx 1$ uses the lower edge of a Gaussian design, and the hand-off prediction was that time-series designs break it. We broke it ourselves first: columns of U taken as consecutive states of d independent AR(1) series (unit stationary variance, so $\rho = 0$ is exactly the arm of Section 6.2) give, at $\alpha = 2^\circ$,

ρ	n	q_1	q_2	$q_1 q_2$	err/bound
0.00	200	1.017	0.977	0.9935	0.8923
0.50	200	1.026	0.967	0.9916	0.8886
0.80	200	1.059	0.934	0.9877	0.8879
0.95	200	1.127	0.871	0.9747	0.8776
0.95	600	1.062	0.930	0.9855	0.8852

Both factors move as predicted (q_1 up, q_2 down) and the product *leaves 1 downward*: the cancellation degrades from 0.9935 to 0.9747 at $\rho = 0.95$, $n = 200$, and $q_1 q_2$ stays monotone in ρ and closer to 1 at $n = 600$. Every cell satisfies $\text{err/bound} < 1$, and the four-factor identity (34) still closes per trial (0.8776 against 0.8776). So correlation makes the certificate *looser* than the dictionary scalar (36) predicts (1.6% at $\rho = 0.95$, $n = 200$), which is the safe direction, and it does not resurrect any need for a calibrated term. We do not claim a rate: the departure is not consistent with $(p + q)/n_{\text{eff}}$ for $n_{\text{eff}} = n(1 - \rho)/(1 + \rho)$ (that would be 1.36 where we measure 0.025), and the same ordering holds at $\alpha = 20^\circ$. One diagnostic worth recording: at $\rho = 0.95$, $n = 200$ the mean of the per-trial product $q_1 q_2$ is 0.9747 while the product of the two means is 0.9814 — that 0.7% gap *is* the cancellation, since q_1 and q_2 are negatively correlated trial by trial, and it is why every closure claim in this chapter is stated per trial rather than from column means. What remains genuinely untested is a *learned* dictionary or an MLP-regression design, where q_2 can lose its structure in a direction we cannot parameterize by ρ ; we still consider small-sample overconfidence there to be the disappearance of q_2 , not a generic small- n effect (script `b34`).

4. **The direction dichotomy is exact only for the constructed angle model.** Definition 1.1 prescribes $F_j = \cos \alpha a_j + \sin \alpha b_j$ with b_j exactly orthogonal to $\text{span}\{E_i\}$. Real adaptive dictionaries (learned corrections, kernel EDMD dictionaries) do not satisfy this; α then becomes a distribution of principal angles and $\sin \alpha_{\min}$ must be estimated, which is a different (statistical) problem, addressed in the companion chapter on coverage rather than here.

5. **What we retracted.** (a) $\sigma_{\min}(\mathcal{G}^{1/2}) = \sin \alpha / \sqrt{1 + \cos^2 \alpha} \|\Lambda_E^{-1} H\|_2^2$ as an equality — it is an upper bound, unsafe direction (Remark 3.1); (b) the operator-norm remainder $O(d/(\sqrt{n} \sin^2 \alpha))$ — wrong by 7 orders of magnitude; (c) the remainder shape $\kappa \sqrt{(p + q)/n}$ as a *prediction of the size* — provable but loose, real fluctuation is $0.68/\sqrt{n}$; (d) the conjectured law $|\text{bias}| = (p + q)/n$ and an $O(1/d)$ correction — both falsified by the scans of Section 4.2, the second with the wrong sign; (e) the extrapolated sample-size margin as a numerical-table entry — see Remark 6.2; (f) the relative-bias shape $|\text{bias}| \propto B_n/(n \lambda_{\min}(\mathcal{G}))$ written with an extra factor $1/n$ — it conflated the absolute displacement $n\beta_2 = O(1)$ with the relative one $\varepsilon_n = -\beta_2/(2\lambda_1) = O(1/n)$, and the wrong form predicts a $\alpha \rightarrow 0$ divergence that Section 4.2 measures as flat; (g) a reported value $\|\Lambda_E^{-1} H\|_2^2 = 1.06$ produced by a Cholesky half-inverse rather than a solve (see the pitfall note in Section 1); the correct figure at $d = 8$ is 0.17; (h) (34) as written in an earlier draft, with three factors and the claim that it “factorizes exactly” — the identity closes only after the block-mass factor $q_3 = \|v_{\mathcal{F}}\|$ is included, and the $\approx 5\%$ column we then labelled “residual” is precisely the missing factor. Discovered by checking the sentence “the observed looseness is

q_4 alone” against the table underneath it, not by an experiment; (i) the conjecture that the sphere minimisation behind cf is attained at the top right-singular direction w_0 — falsified by an exhaustive circle search at $q = 2$, after 3200 random-direction draws had missed it because random points do not resolve S^2 (Remark 3.5); (j) the decomposition of the fitted $c_1 \approx 0.16$ as $(1/d) \times \sqrt{1 + 2 \cos^2 \alpha} \|\Lambda_E^{-1} H\|_2^2$ — that product is 0.108, and the missing $\sqrt{q}$ is block aggregation, settled by the q -scan **b41** rather than by arithmetic; Corollary 5.2 carried the same omission; (k) the sentence that $\lambda_{\min}(\mathcal{G})$ is “obtainable to round-off” by the root find of Theorem 3.3 — it is obtainable to *absolute* round-off, and the conditioning scan **b43** measures the law $\text{relerr} \times \lambda_{\min}(\mathcal{G}) \approx 1.7 \cdot 10^{-15}$ over six decades, so at $\alpha = 0.001^\circ$ the same code is good only to 10^{-5} relative. The overclaim survived three rounds of QA because every script that checked it ($\alpha \geq 1^\circ$) sat in the regime where absolute and relative agree; (l) the *attribution* of that $\leq 3.4\%$ residual and of the cf $-\lambda_{\min}$ sign flip to “the angle distribution”. The anisotropy design **b44** holds the cosine spectrum fixed and switches Λ_E from a generic Gram to γI_p , which removes eight to twelve orders of the gap, and holds Λ_E fixed and switches the spectrum, which moves it by a factor 13–300: the variable is the anisotropy of Λ_E , and the flip additionally requires a second substantial cosine. The numbers were not in question; the label was, and a wrong label sends a future reader to the wrong relaxation. The same script’s attempt to separate the *alignment* of the weak dictionary direction from its size failed — its “aligned” and “transverse” cells are one and the same ensemble, because the coupling’s singular vectors were drawn independently of Λ_E — so alignment is untested here, not ruled out; (m) the surviving clause of (l) itself, that the flip “*additionally* requires a second substantial cosine”. The two-cosine scan **b45** fixes the geometry and runs the two cosines as independent axes — a clean 2×2 plus two seven-point ladders, 1800 draws — and cf lands below $\lambda_{\min}(\mathcal{G})$ in 100/100 draws at a second cosine of 0.087, and again in 100/100 at cf = 0.133. Neither “a second substantial cosine” nor the substitute I had ready (“cf small enough”) is the criterion. What decides is the atom structure of the measure ν_v of Proposition 3.7: with Λ_E ’s weak eigenvector along any single left singular direction of K the measure has one atom, slack = 0 to round-off in twelve cells, and the gap stays on the eigvalsh floor; at 45° to that basis the measure is genuinely multi-atom and every cell flips, with $|\text{gap}/(\text{cf} \cdot \text{slack})| = 1.00$. The spectrum scales slack; it does not select its sign. That run also repaired (l)’s broken control — drawing Λ_E *after* K and pinning the weak eigenvector to K ’s own left singular structure — so the two families agree: alignment is not a competing explanation, it is the atom count seen from the other side; (n) the reading of the family scan **b47** as evidence that no direction beats cf₁. The polar-net search **b48** replaced sampling — a net of 96×240 directions per draw, 7,380,480 family values — and directions *do* beat it, at *first* order: along $v(t) = \cos t v_1 + \sin t u$ with $u \perp v_1$ one has $B'(0) = 0$ but $C'(0) = 2v_1^\top K^\top \Lambda_E^{-2} K u$, so $\lambda'(0^+) = -2v_1^\top K^\top \Lambda_E^{-2} K u \cdot \lambda(1 - \lambda)/|\partial_\lambda F|$ is generally nonzero, and cf₁ is a saddle of the family rather than its minimum (84/84 generic draws, $\|P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1\| = 1.5 \cdot 10^{-3} \dots 2.1 \cdot 10^{-1}$, measured slope matching the formula to $1.7 \cdot 10^{-2}$ relative). The correct statement is a criterion: v_1 is stationary iff $K^\top \Lambda_E^{-2} K v_1 \parallel v_1$, which holds on the slack = 0 loci ($\dim E_0 = q$: equal angles, isotropy; there the search finds 0 competitors and $\|P_{v_1^\perp} \cdot\| \leq 6.2 \cdot 10^{-16}$) and fails generically. Two artifacts of my own are retracted with it: the first version of that search took v_1 to be an arbitrary eigenvector of E_0 , which at $\dim E_0 = 3$ is not the C -maximiser — its self-check caught a $1.4 \cdot 10^{-2}$ discrepancy; and a degenerate-cell “violation” of Proposition 3.8 (15,793 of them) came from normalising a round-off-sized vector, which amplifies its leftover v_1 -component so that $v(t)$ leaves the sphere, $B > 1$, and the smaller root of (20) turns negative. Proposition 3.8 itself is *not* in trouble: on the corrected net and on 144,000 ray points no family member falls below $\lambda_{\min}(\mathcal{G})$, and every

competitor pays intercept ($B/c^2 \leq 1 - 10^{-6}$) for slope ($C/(c^2 t_1) = 1.000005 \dots 1.000499$), as the implicit-function signs predict. (o) the attribution of cf_1 's residual gap to the *direction* v_1 , and with it the conjecture that $\min_v \lambda_1(v) = \lambda_1(v_*) = \lambda_{\min}(\mathcal{G})$. The identity (23) is true — verified as an identity on 320/320 draws to $2.5 \cdot 10^{-15}$ — but its content is the *exact* fibre crossing $\lambda_{\text{ex}}(v_*)$, not the $n = 1$ member: $\lambda_1(v_*)$ stays 0.50–1.01 of the gap above the certificate, and in the one $\dim E_0 = 3$ cell even above $\text{cf}_1 = \lambda_1(v_1)$ on all 40 of its draws ($\max(\text{cf}_1 - \lambda_1(v_*))/\text{gap} = -4.9 \cdot 10^{-5}$), which violates nothing (Proposition 3.8 never claims $\lambda_1(v)$ is smaller at good v) but removes the motivation for the search. The residual is truncational (Remark A.1). (p) the use of uniform or semi-uniform direction nets as evidence about $\min_v \lambda_n(v)$, and with it the acceptance thresholds tested by the direction-net run **b51**. Its $\min_{\text{net}} \lambda_1 = 0.0000$ in 7 of 8 cells is a property of the angle list, not of the functional: the left endpoint of a $\text{linspace}(10^{-3}, \pi/2, n_t)$ list is that statistic's ceiling once the descent set has width $t_{\text{hi}} \leq 1.6 \cdot 10^{-3}$, so flatness in n_t was construction, and the $> 0.99 / \leq 0.90$ thresholds of the previous draft measured the grid. Uniform nets are retired as a measurement device for this question; grid-free routes (the analytic ray, the eigenvector block) replace them. (q) the strength of the ray leg's gap fraction, and with it the reading of (iv) as a per-draw guarantee. The ray scan **b55** falsified the claim exactly as it had been falsified-in-advance: on new tables (doubled p , doubled q , new seed) the analytic two-jet ray's λ_2 value drops to 0.7387 (median) in one $q = 6$ cell, and along a q -ladder the per-cell medians fall from 1.0000 at $q = 2$ to 0.6547 at $q = 8$ with 20/20 draws below 0.90 at the top of the ladder. Since the ranges I quoted in (iv) were *per-cell medians* and are now labelled as such, no number in this chapter changes sign, but the sentence “the grid-free route lifts λ_2 to ≥ 0.9965 ” must be read as a statement about $q \leq 3$ tables, and the ray leg is closed as a research direction rather than extended. What survives on the same tables is the eigenvector block: $(\text{cf}_1 - \lambda_2(v_*))/\text{gap}$ prints 1.00 at both its median and its maximum in all 20 cells, i.e. the deficit is below $5 \cdot 10^{-3}$ of the gap throughout (b53's tighter readout of the same quantity gave $3.3 \cdot 10^{-7}$ to $5.0 \cdot 10^{-4}$).

6. **What this chapter does not do.** It gives no real-data validation. An attempted cyclostationary ENSO leg was closed after three independent failure modes were identified (a small-denominator statistic, non-replication across window choices, and a delay-embedding basis artifact that a zero-physics control reproduced in full); the surviving negative content — a seasonal-normal null genuinely cannot produce the observed numbers, and $\|A^n\|_2/\rho^n$ being two orders of magnitude more stable than trace-normalized alternatives under basis changes — is reported as a separate methodological note, not as evidence for the certificate.
7. **Closed, then retracted, then closed again: the direction w_0 .** The quantity $\text{cf}(t, \alpha)$, $t = \|\Lambda_E^{-1} H\|_2^2$, first appeared in a draft as a numerically suggested candidate; Proposition 3.2 derives it as the Rayleigh minimum on an explicit plane. The conjecture that followed — that the minimiser over the correction sphere is the top right-singular direction w_0 of C — is *false*, and we found that by searching instead of by sampling: a dense circle search beats $\mu(w_0)$ in 143 of the 180 exhaustively-searched $q = 2$ dictionaries at $\alpha \geq 20^\circ$ — 79% of them, so the failure is typical rather than exceptional (Remark 3.5, script **b37**). What replaced it is stronger than what it replaced. Theorem 3.3 gives $\lambda_{\min}(\mathcal{G})$ as the unique root of one monotone scalar equation in $\lambda_{\max}$ of an explicit $q \times q$ matrix — exact to $1.8 \cdot 10^{-11}$ over 6480 dictionaries, nine (d, p, q) configurations, $\alpha = 1^\circ \dots 89^\circ$ — and Corollary 3.4 shows that in the square case $p = q$ that root is *exactly* $\text{cf}(t, \alpha)$ (verified to $1.4 \cdot 10^{-11}$ on seven square configurations $\times$ five angles, script **b39**). *What remains open:* no closed form is known for $\lambda_{\max}(B(\lambda))$ when $q < p$; the measured defect of the cf surrogate there is $\leq 1.9 \cdot 10^{-3}$ at $(d, p, q) = (8, 4, 3)$ and $\leq 6.9 \cdot 10^{-3}$ at $(6, 4, 3)$, both at $\alpha = 45^\circ$. So the practical question (is

the closed form adequate without a root find?) is answered yes in the scanned regime, and the theoretical one (a $q < p$ closed form) is not answered. Proposition 3.5 sharpens which of the two is left: the *root find* needs no equal-angle hypothesis at all, so the open question concerns the closed form only. And the prescribed-spectrum run **b42** put a price on the other relaxation one might reach for — a distribution of principal angles in place of the constructed single one: $\lambda_{\min}(\mathcal{G})$ moves by $\leq 3.4\%$, while the *sign* of $\text{cf} - \lambda_{\min}$ moves by a full 180° . That price is right and its label was wrong (retractions (l) and (m) below): the 2×5 design **b44** attributes both effects to the *anisotropy* of Λ_E — at $\Lambda_E = \gamma I_p$ the gap is identically zero for every spectrum (Corollary 3.6) — and the two-cosine scan **b45** names the anisotropy: the sign of the gap is the sign of the moment defect $D(\text{cf})$ of Proposition 3.7, whose leading term is $\text{cf} \cdot \text{slack}$ with $\text{slack} = c^2 t - \lambda_{\max}(E_0^\top K^\top \Lambda_E^{-2} K E_0)$, one joint functional of Λ_E 's spectrum and K 's weights, and *not* a condition on the cosine spectrum — the flip survives a second cosine of 0.087 and a cf of 0.133 alike. An angle-distribution theory of the smallest eigenvalue is therefore a low-yield target for two independent reasons: at isotropy the distribution is invisible to $\lambda_{\min}(\mathcal{G})$, and where it is visible it enters only through slack, a quantity that is one eigenvalue of an explicit matrix. The functional the previous draft listed as open is thus identified. Pairing the intercept c^2 with the slope that actually attains it — t_1 of (18) — turns the diagnosis into a replacement: cf_1 bounds $\lambda_{\min}(\mathcal{G})$ from above with *no* angle, isotropy or simplicity hypothesis (Proposition 3.8), and its residual is second order (measured log-log slope 2.081, pre-registered as 2). Two items stay open and the first has been settled in the direction that matters for practice while staying open as a closed form: the polar-net search **b48** settled the question “is cf_1 the smallest member of (20)?” in the *negative* (retraction (n)), and the ray computation **b49** sized what the negative costs and buys. Along the eigenvalue-free ray $u \parallel P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1$ the depth of the descent is $\text{dip} = -L_1^2/(4L_2)$ of Remark 3.8, verified to 10^{-3} relative on 280 draws, and one step on that ray removes 0.0004–0.50 of cf_1 's gap at the median (0.76 worst draw, absolute $|\text{dip}| \leq 1.2 \cdot 10^{-7}$) while a 64×64 net on the same draws gets at most $\sim 10\%$ deeper still. Evaluating (20) at the resulting direction gives $\lambda_{\min}(\mathcal{G}) \leq \text{cf}_2$ by Proposition 3.8 — *no new proof*, one 2×2 root (tightening it below cf_1 additionally needs $L_2 > 0$, observed 280/280 and not proved). What is deliberately *not* claimed is that $\text{cf}_1 + \text{dip}$ bounds anything on either side — the third jet of $\lambda(t)$ is unbounded and the measured drift in $|c_1|$ has no consistent sign — and the family minimum over the whole sphere stays without a closed form: the one-dimensional fractional programme is open, but its remaining marginal is now measured, and we stopped on it for that reason rather than for lack of a technique. Proposition 3.9 changes the shape of that open item, not its status: since $\lambda_2(v)$ is a certified bound for *every* unit v and an explicit cubic root in four scalars, $\min_v \lambda_2(v)$ over any finite set of directions is again certified (and at every direction tested by **b50**, $\lambda_2(v) \leq \lambda_1(v)$; the pointwise ordering of successive Gauss rules is measured, not proved). What **b50** measures is that at fixed $v = v_1$ the moment side is already exhausted ($|\lambda_2 - \lambda_{\text{ex}}| \leq 2 \cdot 10^{-3}$ of the gap). The three direction runs **b51**–**b53** then settle the direction half of that alternative, and settle it against the programme: the descent set of $v \mapsto \lambda_n(v)$ from v_1 has angular width $t_{\text{hi}} \approx 2.5 \cdot 10^{-5} \dots 1.6 \cdot 10^{-3}$ (so the 64×64 net above, and every net starting at 10^{-3} , is blind to it by construction — retraction (p)), and at the fibre direction v_* — the minimiser of the *exact* crossing λ_{ex} , not of λ_1 — the $n = 1$ member is still 0.50–1.01 of the gap short while $\lambda_2(v_*)$ is within $5 \cdot 10^{-4}$ (Remark A.1). The ray scan **b55** then replaces the trade-off that these runs described with the criterion of Lemma 3.11, closes the direction question in the equal-angle cells (Corollary 3.12: $\min_v \lambda_1(v) = \text{cf}_1$, attained at the C -maximisers and *not* at v_*), and falsifies the small- q universality of the ray (retraction (q)). What remains open

is therefore not “how fast does $\min_v \lambda_n(v)$ descend on a net” — that question was a grid artifact — but two sharper ones: whether $\inf_{\|v\|=1} \lambda_1(v)$ ever reaches $\lambda_{\min}(\mathcal{G})$ in a generic table (the inequality $\lambda_1 \geq \lambda_{\min}(\mathcal{G})$ is (19), and no draw has attained it; The leverage gate **b59** narrows it without closing it — on 264 rays through v_1 , minimised exactly and grid-free by the stationary numerator G of Remark A.1(vii), every ray infimum stays at least 0.2357 of the gap *above* $\lambda_{\min}(\mathcal{G})$, and 120 random rays on a $q = 4$ draw beat neither v_* nor that floor. The cross-lane run **b60** then replaced that sampling by the exact sphere gradient of Remark A.1(viii) — no grid, no finite differences, 60 starts per draw over five cells — and the sphere infimum stays at least 0.3378 of the gap above $\lambda_{\min}(\mathcal{G})$ on all 50 draws, while on 38 of them a direction strictly below $\lambda_1(v_*)$ clears that draw’s own evaluator floor. The infimum is therefore now *searched* rather than sampled, and still open as a proof; and Remark A.1(ix) prices the beat it finds — $\frac{1}{2}\kappa\theta^2$ with κ a measured sphere curvature and θ the angle by which v_* misses the minimiser — so the residual question is to control θ , not to keep hunting for a better direction), and whether the difference $\lambda_1(v) - \lambda_2(v)$ has a closed-form leading term along $v(t)$, which is the item **b53** did not reach. The moment question (“is $n = 2$ enough?”) is answered: it is enough to within $2 \cdot 10^{-3}$ of the gap at n equal to two, and it cannot be pushed past the atom count of ρ_v . The second item is unchanged: whether the $o(\lambda)$ remainder of (17) admits a closed-form bound, so that the residual of cf_1 could be sized without evaluating the defect. As before, no statement in this chapter uses cf or λ^* , and the competing-branch explanation for the old residual gap — the lower branch switching to $\lambda_{\min}(\Lambda_E)/2$ — stays ruled out: $\min \lambda_{\min}(\Lambda_E)/2 \geq 1.38$ against $\max \text{UB} \leq 0.99$.

8 Positioning relative to the closest work

- *Eigenvalue bounds for block and saddle-point matrices* [22, 24, 23, 25]. Bounding the spectrum of a block matrix by extremal eigenvalues and singular values of its blocks, with inertia and multiplicity statements attached, is *standard* in the preconditioning literature, and that literature already expresses the real eigenvalues as bounds in terms of the extremal roots of a family of parametric polynomials. Proposition 3.10 is therefore not offered as a technique. What it offers is that the bound is *attained* along the dictionary’s own fibre, and that it inverts in closed form — neither of which a preconditioning paper needs. This item was added in the round that ran the search, because the availability of the technique elsewhere lowers what may be claimed for it; the Gauss-rule side (Proposition 3.9) is searched as well, and its downgrade is the same one: the nodes of the n -point fibre rule are the eigenvalues of the Jacobi matrix of the fibre measure — the Golub–Welsch construction [5, 26, 27, 28, 29], including the Lanczos–Golub–Welsch route [28] and the question of when Lanczos iterations generate symmetric quadrature nodes [29] — and the inequality side of (19) is Rayleigh–Ritz at $n = 1$ and Cauchy interlacing of the Krylov compression at $n \geq 2$, so neither the rule nor the bound is offered as a technique; the interlacing side is used as a standing property in current numerical work too, where exact Ritz extraction is credited with “the interlacing property that makes each band energy an upper bound on the true one” [34] and a Ritz-value interlacing theorem is proved and extended to generalised singular values as a routine tool [35]. What survives on both sides is the same pair — attainment along the fibre and closed-form inversion — together with the measured losses of Remark A.1, which an interlacing statement does not supply. Small print, because it is the kind of inference I have had to take back before: the single hit for the phrase *Gauss–Stieltjes* on a preprint server [42], where the rule serves passivity-preserving reduction rather than certification, is *not* evidence that the statement is

unoccupied — the primary literature for this technique is in journals and monographs, and the judgement above rests on the seven Golub–Welsch hits and on the interlacing fact, not on a low count.

- *Error bounds for Krylov approximations of matrix functions* [30, 31, 32, 33]. This is the closest live machinery to what Remark A.1 does, and it cuts *against* this chapter: that work already derives both a priori and a posteriori bounds for Lanczos approximations of $f(\mathbf{A})\mathbf{b}$ and of the quadratic forms $\mathbf{b}^H f(\mathbf{A})\mathbf{b}$, with bounds that resolve clustered and isolated eigenvalues of $\mathbf{A}$ and are extended to finite precision [30] (abstract read), and is proposed as a practical stopping criterion [31]. Since cf_1 here is a two-moment (A_0, B_2) truncation of exactly that kind of object, “a certificate computable from low-order spectral data” is not offered as a contribution. What survives is the same pair as the item above — attainment along the fibre and closed-form inversion — plus the negative measurement of the *direction* side: that error has a spectrum-dependent part ($23.5\times$ shift of the median angle across 40 subspace draws at fixed cosines, distributions overlapping) and an orientation part (1.87–3.40 decades at fixed spectrum), no function of $\text{spec}(A_0) = \{c_i^2\}$ explains the second, and the angle is not an iteration error at all but the distance between the exact eigenvector’s fibre block and this functional’s own minimiser. The four entries above were read entry by entry (title, authors, abstract), not inferred from a search count.
- *Principal vectors for subspace pruning in Koopman approximation* [36]. They use principal angles to measure *invariance defect*, $\mathcal{K}(S)$ versus S . Our angles are between the prior and correction *blocks* and measure *attributability*. Contribution framing matters here: the claim of this note is not “principal angles in Koopman”, which is no longer novel, but “model-form error leaks into the correction block with worst-case amplification exactly $1/\sin \alpha$, constant 1, dimension-free, and attained”. The angle machinery itself is classical and we use it as classical: computation of the angles between subspaces [12], the perturbation and majorisation theory of those angles and of Ritz values [13, 14, 15], and the coherence obstacle to making $\sin \alpha_{\min}$ large by dictionary design, namely the Welch bound [16]. None of these is offered as a contribution here; what is offered is that the amplification factor of the attribution error is $1/\sin \alpha$, with constant one.
- *Eigenvector perturbation theory* [9, 10, 11]. A reader trained on $\sin\Theta$ theorems will look for one here, and it is not what this note contains: the angle we measure between the exact fibre block and the minimiser of the functional is a *model-discrepancy* angle, not an eigenvector perturbation, and it does not vanish under refinement of the estimator (Appendix A). The classical bounds are the right tools for a different question, and we say so rather than letting the notation suggest a theorem we did not prove.
- *Identifiability bounds for linear and nonlinear ODEs from solution data* [37]. Equation-versus-equation distinguishability via Hausdorff distance on solution sets and metric entropy, minimax over classes. We are complementary: attribution of one operator inside one decomposition $A^* = P + C + \Delta$, i.e. how the same data must be split between two blocks. The generic phrase “identifiability theory for SciML” is no longer available to us; the claim must read “decomposition-level identifiability”.
- *Learning physics from an imperfect ancestor* [38]. Structure prior from an imperfect neural operator plus residual refinement. It documents that the “imperfect prior + learned correction” architecture is active, and it contains no fidelity or identifiability theory for the correction term. This note supplies exactly that missing piece for the linear-operator case.

- *Residual pseudospectra for a physics-informed Koopman backbone* [39]. Kernel EDMD with residual minimization and pseudospectral analysis on ERA5/HadISST SST. Their certificate is spectral consistency; ours is a leakage bound in the dictionary decomposition, with no information quantity and no coverage interpretation. Together with the operator-identification monograph of Kutz *et al.* [21] (the standard reference for identification on standardised data, which is precisely the practice our methodological note criticizes), these fix the boundary of what is new here.
- *The least-squares projection structure and the Frisch–Waugh–Lovell theorem* [17, 18, 19]. Fitting the correction block after the prior block is, at the level of the normal equations, residualisation: the estimator of C is formed from the part of the data that P cannot explain, which is the content of that theorem, and [19] is the standard care taken with the resulting standard errors. Our object is different in kind: FWL says what the point estimate is, and this note bounds what an unmodelled operator Δ is *forced* to contribute to it, in the worst case, as a function of the dictionary’s principal angles. The distinction is stated because the two results are easy to conflate, and a reader who conflates them will conclude the leakage theorem is already in the regression literature.
- *Grey-box modelling and system identification* [20]. The “known structure plus parameterised residual” model class is the grey-box class of system identification, where the questions of identifiability and of variance reduction are well developed. This note’s contribution is confined to one quantity in that class that the classical theory does not price: the worst-case transfer of an *unmodelled* operator into the fitted correction block, which for the linear operator case here is exactly $1/\sin \alpha_{\min}$.
- *The chord bound of Proposition 3.8 against the truncated moment problem* ([40, 41]; abstracts read 2026-10-01, not the papers). Both describe the truncated (multi- and matricial) moment problem *through the Stieltjes transform* — as an interpolation problem solved step by step with continued fractions, and via a Schur-type algorithm respectively. Read against them, the function $s_v(\lambda) = B(v)^2/(B(v) - \lambda C(v))$ introduced in the proof of Proposition 3.8 is exactly the two-moment (one-atom) approximant of a Stieltjes transform, i.e. the kind of object produced by the *first* step of such an algorithm, and the inequality $\phi_v \geq s_v$ is Cauchy–Schwarz (equivalently Jensen) for a positive measure with prescribed mass and first moment: an argument I do not claim as new. What those papers do not do, and what this chapter does, is the *direction of use*: s_v lower-bounds $f(\lambda) = \lambda_{\max}(K^\top(\Lambda_E - \lambda I)^{-1}K)$, whose first crossing with $1 - \lambda$ is $\lambda_{\min}(\mathcal{G})$, so choosing v to maximise the slope on the top eigenspace yields a closed-form surrogate on a certified side of that eigenvalue — a statement about a block Gram matrix of a pair of subspaces, not about a moment solution set. Two things stay recorded as open rather than cited: whether the specific two-moment bound is stated classically under the name Chebyshev–Markov–Stieltjes (or as Markov’s theorem for Stieltjes transforms, or in Wall/Gautschi’s Gauss-rule error-sign treatment) is *not* verified entry by entry, so no reference is attached to Proposition 3.8; and if verification succeeds, the correct edit is to demote the inequality step of its proof to that citation while keeping the eigensystem construction, which is the part that would then remain. The two-atom check b50 sharpens that caveat in the unfavourable direction, and it is recorded here rather than only in the log: Proposition 3.9 replaces the one-atom approximant by the n -point Gauss rule on the *same* measure, whose lower-bound property is exactly the classical sign statement $\int h d\rho \geq G_n(h)$ for $h^{(2n)} \geq 0$. If the name-check comes back, what is being cited then covers the entire inequality machinery of (21), not just its $n = 1$ case; the part that could still remain mine is the identification of the crossing of the resulting rational function with $1 - \lambda$ as a certified surrogate for $\lambda_{\min}(\mathcal{G})$, the

direction family (20), and the $\Lambda_E^{-j}K$ moments as the computable ingredients — the quadrature itself would be classical.

9 Reproduction

The manuscript, the scripts, their saved output, the climate index tables, and the full collaboration log of this project are archived at <https://github.com/m-rui001/identifiability-leakage-principal-angles> (MIT licence). The layout is `manuscript/`, `scripts/agent_a/` and `scripts/agent_b/` (the numbered scripts below), `evidence/` (the stdout each script produced, named after it), `logs/` and `tools/`. Each number in the tables below is printed by the script named next to it, and the file under `evidence/` is that script’s saved output; the measurement scripts need numpy and scipy. The climate index tables are also archived, but they belong to the cyclostationary ENSO attempt that §7 records as *closed*: they are shipped for completeness and as evidence of what was tried, not as validation. Where a script ran and the claim was afterwards retracted, the retraction is in §7 and in the log, not quietly deleted.

Geometry, sampling, the certificate.

Script	Objects	Verifies
<code>b11.theorem_remainder.py</code>	Props. 2.1–2.3, Thm. 4.2	block Gram to 10^{-16} ; $r(M)$ flatness; $n^{-1/2}$ rate; $\kappa \approx 0.15$
<code>b13.two_sided_bound.py</code>	Thm. 3.1	900 draws, 0 violations on either side; tightness tables
<code>b12.certificate_no_constants.py</code>	($\star$), Cor. 5.2	24/24 cells pass, ratio 0.117–0.175; adversarial 0.876–0.919
<code>b14.g_alpha_kappa.py</code>	Thm. 4.2	κ calibration, dictionary-size scan
<code>b15.bias_not_noise.py</code>	Thm. 4.3	sign tails, Jensen gap, $n\beta_2$ prediction (0.825/0.870/0.947)
<code>b24.ggamma.py</code>	Thm. 4.4	$\alpha : 0.25^\circ \rightarrow 85^\circ$, $\lambda_{\min}(\mathcal{G})$ over 4 decades, bias flat in α
<code>b25.pqn_law.py</code>	Thm. 4.4	21 cells: $\text{sd}\sqrt{n} = 0.68 \pm 0.02$; falsifies $c = p + q$ and $O(1/d)$
<code>b26.margin_upgrade.py</code>	Remark 6.2	random arm is uninformative (0.113–0.174, flat in n)
<code>b27.adversarial_margin.py</code>	($\star$)	4320 adversarial trials, zero violations, no margin

The slack factorization.

Script	Objects	Verifies
b28_cancellation.py	Thm. 6.1	q_4 n -free to 0.08%, $q_1 q_2 \in [0.986, 1.000]$; leaves the $1 - q_3$ column of the table in §6.2
b29_upper_bound_check.py	Thm. 3.1	2400 per-sample draws, 0 violations of either side; $\lambda_{\min}/\text{UB} \geq 0.985$
b30_q4_analytic.py	Rem. 6.1	closed-form q_4 to 10^{-4} of b28's measured value; $\ \Lambda_E^{-1} H\ _2^2 = 0.128$ at $d = 10$
b31_q4_curve.py	(35)	eleven α 's, 4400 dictionaries: per-sample deviation ≤ 0.004 , 0 upper-side violations
b32_residual_diagnosis.py	(34), Cor. 6.2	per-trial closure $q_4 q_1 q_2 q_3 = 0.8905$ vs. measured 0.8905; $\ v_{\mathcal{F}}\ = 0.9466$ (sd 0.0066) at 2° , 0.9993 at 85°
b33_lambda_min_sharp.py	Prop. 3.2, Rem. 3.4	2700 draws: $\lambda_{\min}(\mathcal{G}) < \text{cf}$ on every one, relative gap $\leq 2.2 \cdot 10^{-3}$; $\min \lambda_{\min}(\Lambda_E)/2 = 1.38$ against $\max \text{UB} = 0.99$
b34_correlated_u.py	§7 item 3	AR(1) trajectory design: $q_1 \uparrow$, $q_2 \downarrow$, $q_1 q_2$ down to 0.9747 ($\rho = 0.95$, $n = 200$), all $\text{err}/\text{bound} < 1$, (34) closes per trial (0.8776/0.8776)

The exact $\lambda_{\min}(\mathcal{G})$ leg.

Script	Objects	Verifies
b35_mu_reduction.py	Thm. 3.3, Rem. 3.5	3200 draws, four (d, p, q) : 0 violations of $\lambda_{\min}(\mathcal{G}) \leq \mu(w_0)$, $ \lambda_{\min}(\mathcal{G})/\mu(w_0) - 1 \leq 1.0 \cdot 10^{-5}$; the row's original "0/3200 random w beating w_0 " was a sampling artifact, see b37
b36_cf_chain.py	Prop. 3.2	6300 draws, 42 cells: the chain $\lambda_{\min}(\mathcal{G}) \leq \mu(w_0) \leq \text{cf} < \text{UB}$ holds on every draw, and cf equals a brute-force plane minimisation to $4.5 \cdot 10^{-12}$
b37_w0_optimality.py	Rem. 3.5	exhaustive S^1 search at $q = 2$ vs. random directions at $q = 3$: 143/180 non-square $q = 2$ draws at $\alpha \geq 20^\circ$ admit $\mu(w) < \mu(w_0)$ (157/840 overall), separation up to 0.30 rad, gap $\leq 9.6 \cdot 10^{-7}$
b38_exact_scalar_formula.py	Thm. 3.3	6480 dictionaries, nine (d, p, q) , $\alpha = 1^\circ \dots 89^\circ$: worst $ \lambda^*/\lambda_{\min}(\mathcal{G}) - 1 = 1.8 \cdot 10^{-11}$, 0 violations; the $\lambda_{\min}(\Lambda_E)$ face never active
b39_square_case_cf_exact.py	Cor. 3.4	7 square configs $\times$ five angles, 200 draws: $ \lambda_{\min}(\mathcal{G})/\text{cf} - 1 \leq 1.4 \cdot 10^{-11}$, $\ \Lambda_E - C^{-\top} C^{-1}\ /\ \Lambda_E\ \leq 1.5 \cdot 10^{-15}$; rectangular defects $1.9 \cdot 10^{-3}$ (8, 4, 3), $6.9 \cdot 10^{-3}$ (6, 4, 3)

The two-constant audit, and the scope checks on cf.

Script	Objects	Verifies
b40_a_d10_dilution_audit.py (+b40b)	Intro, Cor. 5.2	audit of A's two-arm design, reproduced to the last digit on his seed: the model-form constant is the projection fraction (8% over $d = 6 \dots 20$, Beta mean verified to 0.5% by 10^4 draws), flat in n ; the noise constant is $\sqrt{q/n}$ (13%); $c_2/c_1 = d/\sqrt{n}$; A's "predicted c " column shown to be the same row's own fit
b41_q_scan_dilution.py	Cor. 5.2	$q = 1, 2, 3, 4, 6$ at fixed $d = 10$: $c_1 = 0.077\text{--}0.227$, so the dilution is $\kappa(q)\sqrt{q}/d$ (c_1 within $[0.94, 0.99]$ of it, no trend), not q -blind $1/d$; the α -tilt is shown <i>not</i> to discriminate
b42_multi_angle_exact.py	Prop. 3.5, Rem. 3.9	prescribed principal-cosine spectra in place of the constructed single angle: the root find holds to $\leq 1.6 \cdot 10^{-12}$ (and to $\leq 1.3 \cdot 10^{-15}$ against $\lambda_{\min} = 1 - c_1$ at $\Lambda_E = I_p$), (12) and (14) agree to $\leq 4.8 \cdot 10^{-13}$ on equal-angle cells, while cf stays within 3.4% but flips sign — ≥ 0 on 240 equal-angle draws, ≤ 0 on 119–120 of 120 per wide-spread cell
b43_rootfind_conditioning.py	Thm. 3.3, Rem. 3.9(iii)	accuracy of the reduction itself, 60 draws/cell, $\alpha = 45^\circ \dots 0.001^\circ$ and $\kappa(\Lambda_E) = 1 \dots 10^{12}$: $\text{relerr} \times \lambda_{\min}(\mathcal{G}) \in [1.6, 1.8] \cdot 10^{-15}$ flat over six decades (absolute floor, hence relative accuracy $\sim \sin^{-2} \alpha$); unguarded bracket endpoint at 10^{-12} below the pole gives order-100 errors, guarded restores the floor to $\lambda_{\min} \approx 10^{-9}$; at $p = q$ the cf deviation equals that floor, so it is implementation, not the closed form
b44_cf_exactness_attribution.py	Cor. 3.6, Rem. 3.9(ii), §7(l)	2×5 design (generic vs. isotropic Λ_E , five cosine spectra, 100 draws/cell) + 18 arbitrary- K cells: at $\Lambda_E = \gamma I_p$ the gap is at the floor ($\leq 2.1 \cdot 10^{-15}$) for <i>every</i> spectrum; the flip is resolution-checked ($9.1 \cdot 10^5$, $1.1 \cdot 10^4$ above $100\epsilon\ \mathcal{G}\ _2$, 0 unresolved); the sign condition this row first attached to that flip is retracted by b45

The one-pole reading and its replacement leg.

Script	Objects	Verifies
<code>b45_resolvervent_one_pole_diagnostic.py</code>	Prop. 3.7, Rem. 3.7, Cor. 3.6, §7(m)	~6650 draws: the sign of $\text{cf} - \lambda_{\min}(\mathcal{G})$ is the sign of $D(\text{cf})$, 0 exceptions in 500 resolved draws, and the gap equals $D/(1+f')$ to $\leq 1.1 \cdot 10^{-4}$ relative; $D \equiv 0$ at $\Lambda_E = \gamma I_p$ (15 cells, $\leq 1.3 \cdot 10^{-15}$) and for $ZK = 0$, $Z \neq 0$ ($ t-1 \leq 1.3 \cdot 10^{-15}$); flip in 100/100 draws at a second cosine of 0.087 and at $\text{cf} = 0.133$, absent in all 12 single-atom cells
<code>b46_true_slope_surrogate.py</code>	Prop. 3.8, Rem. 3.8, §7 open items	2100 draws on <code>b45</code> 's cells with the pole moved to the true slope: flip 100/100 $\rightarrow$ 0/100, 98/100 $\rightarrow$ 0/100, and 0/100 in all sixteen cells; $\text{cf} \leq \text{cf}_1$ in 100/100 wherever slack clears the floor, coin-flip where $E_0 = \mathbb{R}^q$; $\text{med} \text{gap}_1 / \text{med} \text{gap} = 0.001\text{--}0.31$ where the linear term dominates, 1.00–1.10 where it vanishes; log-log slope of $ \text{gap}_1 $ vs $\text{cf}_1 = 2.081$ (pre-registered 2, falsifier 1); nulls $ t_1 - 1/\gamma \leq 2.0 \cdot 10^{-15}$, $\max D_1 \leq 1.3 \cdot 10^{-15}$
<code>b47_theorem_probe.py</code>	Prop. 3.8	1600 draws: the Jensen step tested directly — $\phi_v - s_v \geq 0$ at $\lambda = 0.3, 0.7, 0.95$ of cf_1 for the chosen v_1 and for 200 random directions per draw, worst $-5.6 \cdot 10^{-16}$ (isotropy cells, $\text{gap} \equiv 0$); the family $\lambda(v)$ of (20) above $\lambda_{\min}(\mathcal{G})$ in 320,000/320,000 tests including the capped branch, 0 resolved sign flips of gap_1 ; records the degeneracy $\dim E_0 = 3$ at equal angles, where a single returned eigenvector misses $\max_{E_0} C$ by up to $6.6 \cdot 10^{-1}$

Searching the family (20) instead of sampling it.

Script	Objects	Verifies
<code>b48_family_min_search.py</code>	Prop. 3.8, Rem. 3.8, §7(n)	search instead of sample: polar net 96×240 per draw, 7,380,480 family values + 144,000 ray points, $\max \ v\ - 1 \leq 7.8 \cdot 10^{-16}$. 0 members below $\lambda_{\min}(\mathcal{G})$, but cf_1 is <i>not</i> the family minimum: the first variation along $v(t) = \cos t v_1 + \sin t u$ is $-2 v_1^\top K^\top \Lambda_E^{-2} K u \cdot \lambda(1 - \lambda)/ \partial_\lambda F $, negative in 84/84 generic draws (measured/predicted relative error $\leq 1.7 \cdot 10^{-2}$), while the $\dim E_0 = q$ cells give 0 competitors and $\ P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1\ \leq 6.2 \cdot 10^{-16}$; the best ray removes 0.07–0.50 of cf_1 's gap (worst draw 0.58), each competitor buying slope $C/(c^2 t_1) \in [1.000005, 1.000499]$ with intercept $B/c^2 \leq 1 - 10^{-6}$
<code>b49_two_jet_dip_formula.py</code>	Rem. 3.8, §7 open item 1	sizes that descent in closed form before searching for it: 280 saddle draws, $\sim 1.6 \cdot 10^6$ family values (504,000 on the steepest rays, the rest on a 64×64 net over the same draws), $\max \ v\ - 1 \leq 4.4 \cdot 10^{-16}$, $ \lambda(v_1) - \text{cf}_1 \leq 3.7 \cdot 10^{-16}$. The two-jet dip $= -L_1^2/(4L_2)$ matches the measured ray depth to $5.9 \cdot 10^{-6} \dots 1.0 \cdot 10^{-3}$ relative per cell, $t_{\text{meas}}/t^* \in [0.993, 1.008]$, $L_2 > 0$ in 280/280, 0 members below $\lambda_{\min}(\mathcal{G})$ and 0 draws where the predicted gain exceeds the gap; one ray step removes 0.0004–0.50 of the gap at the median (worst 0.7642, $ \text{dip} \leq 1.2 \cdot 10^{-7}$ absolute), reproducing b48 's 0.07–0.50 on a different seed by a different method. The net goes $\lesssim 10\%$ deeper (wins in 32/40, 31/40, 7/40, 2/40 and 0/40 cells, median ray / net = 0.906–1.009, and it does not contain the steepest direction at 5.6° — sampled, not exhaustive). The equal-angle cell is 40/40 stationary, where the formula's hypothesis $\ P_{v_1^\perp} K^\top \Lambda_E^{-2} K v_1\ > 0$ simply fails

Replacing the one-pole chord by a Gauss rule on the same measure.

Script	Objects	Verifies
<code>b50_gauss_moment_hierarchy.py</code>	Prop. 3.9, Rem. A.2, (22)	280 draws, seven cells: the rules integrate $1, y, \dots, y^{2n-1}$ exactly to $\leq 7 \cdot 10^{-15}$ relative (<i>checked before</i> any side claim — two earlier runs of this script mis-indexed the Stieltjes recurrence and are void), $\lambda_1 = \text{cf}_1$ to $\leq 7.2 \cdot 10^{-16}$, and $\lambda_2, \lambda_3, \lambda_{\text{ex}}, \lambda^* \geq \lambda_{\min}(\mathcal{G})$ in 280/280. Ordering $\lambda_1 \geq \lambda_2$ in 280/280, $\lambda_2 \geq \lambda_3$ in 239/239 finite pairs, pointwise $G_1 \leq G_2 \leq G_3 \leq \phi_v$ in 240/240. $\lambda_2(v_1)$ removes 0.26–0.9998 of cf_1 ’s residual gap (cell medians 0.50...0.999) against 0.0004–0.72 for b49 ’s ray; the explicit cubic (22) equals the eigenvalue-plus-bisection root to $2.5 \cdot 10^{-16}$ on 280/280 and reproduces a hand two-atom measure. Hierarchy terminates at the atom count: 39 of 80 $n = 3$ rules break down on the two near-isotropic cells (Λ_E has two eigenvalues), and there $\lambda_2 = \lambda_{\text{ex}}$ exactly

Direction versus moment order: what a net can and cannot see.

Script	Objects	Verifies
<code>b51_net_lambda2_direction_moment.py</code>	Prop. 3.9 on a net	320 draws, eight cells, same seed and call order as b52/b53 : gates pass (0 side violations, $\max v - 1 = 2.2 \cdot 10^{-16}$, moment exactness $\leq 6.6 \cdot 10^{-15}$ by two independent routes, $ \lambda_1(v_1) - \text{cf}_1 \leq 1.7 \cdot 10^{-16}$, 0 root-filter rejections), but $\min_{\text{net}} \lambda_1$ removes 0.0000 of the gap in 7 of 8 cells — and that column is <i>retracted as evidence</i> by b52 , retraction (p). Five defects were caught in a pre-run audit (three moment matrices where four are indexed, a non-unit $q = 2$ perpendicular frame, an RNG-consuming print, a dead nanmin line, and a false nesting premise); none of them survived into the run
<code>b52_boundary_layer_reconciliation.py</code>	Retraction (p), t_{hi}	Same 320 draws: the b50 ray reproduces, and b51 ’s own angle list on that same ray returns 0.0000 — the net is blind, not wrong. The descent set of $v \mapsto \lambda_n(v)$ from v_1 sits inside $t < t_{\text{hi}} = 2.5 \cdot 10^{-5} \dots 1.6 \cdot 10^{-3}$ with $t_{\text{hi}}(\lambda_2) \equiv t_{\text{hi}}(\lambda_1)$; a $\text{geospace}(10^{-8}, \pi/2, 300)$ list lifts λ_2 to ≥ 0.9994 in six of the seven ray cells, 0.9164 in the $q = 4$ one (residual to the analytic ray $\leq 8.4 \cdot 10^{-2}$, typically $3 \cdot 10^{-6}$), and rotating the azimuth by 0.2 rad retains 0.916–0.960 of the descent

The exact fibre direction, and the margin that orders the family at it.

Script	Objects	Verifies
b53_exact_direction_tightness.py	Prop. 3.10, Rem. A.1, (23)	Same 320 draws: (23) as an identity to $\leq 2.5 \cdot 10^{-15}$ on 320/320 plus the p -block row to $\leq 2.2 \cdot 10^{-15}$; the intercept defect $c^2 - B(v_*) \leq 4.6 \cdot 10^{-6}$ in the seven $\dim E_0 = 1$ cells and $\leq 3 \cdot 10^{-15}$ in the degenerate one; $\lambda_2(v_*) - \lambda_{\min}(\mathcal{G}) \leq 5 \cdot 10^{-4}$ of the gap (median per cell) with 0 side violations; $\lambda_1(v_*) - \lambda_{\min}(\mathcal{G})$ median 0.50–1.01 of the gap — the refutation of the $n = 1$ conjecture; $\text{angle}(v_*, v_1)$ median $1.3 \cdot 10^{-5} \dots 1.3 \cdot 10^{-3}$ (a second computation of t_{hi}); $\text{angle}(v_*, \text{ray}) \leq 5 \cdot 10^{-4}$ while $\lambda_1(\text{ray})$ removes only 0.0003–0.50 and $\lambda_2(\text{ray}) \geq 0.9965$ (six ray cells; the $q = 4$ one 0.9222), all of them per-cell <i>medians</i> ; their dispersion is added by b55, together with the q -trend that retracts them as a guarantee
b55_fibre_vs_chord_margin.py	Lem. 3.11, Cor. 3.12, §7(q)	550 draws in 20 cells (b53's 320, plus doubled- p and doubled- q tables on a new seed): the criterion (25) has 0 sign disagreements and all three side conditions hold on all 550; (26) reproduces $\lambda_1(v_*) - \text{cf}_1$ at ratio 0.9996–1.0003 per cell, and along the great circle $v(\vartheta)$ (24,000 points, $ T $ up to $1.5 \cdot 10^3$, $\max v - 1 \leq 3.3 \cdot 10^{-16}$) the sign never flips, the 1% band reaching $ T \approx 3.6 \cdot 10^2$ at $q = 3$ and $2.0 \cdot 10^1$ at $q = 4$ with departure $\leq 5.5 \cdot 10^{-3}$ for $ T \leq 1$. Equal-angle cells: $\max \ A_0 - c^2 I_q\ \leq 5.8 \cdot 10^{-15}$, $B(v) = c^2$ with 0 exceptions over 960,000 grid points, 0 points below cf_1 –floor, and $\lambda_1(v_*) > \text{cf}_1$ in 24/24. The ray leg fails its own registered test: $\lambda_2(\text{ray})$ medians 1.0000, 0.9995, 0.9715, 0.8932, 0.9034, 0.6547 at $q = 2, 3, 4, 5, 6, 8$ (20/20 draws below 0.90 at $q = 8$), and b53's own $q = 4$ cell has per-draw minimum 0.7804 with 13/40 below 0.90. Two defects of my own checks are recorded in the paired .txt rather than removed: a discriminant compared without its B^2 factor, and a registered count relation on the wrong side of the iff

The ray boundary as an algebraic set.

Script	Objects	Verifies
b57_descent_boundary_cubic.py, b57b_implied_outlier.py	Lem. 3.13, Rem. A.1(vi)	470 draws in thirteen cells (b53's 320, two deterministic mix tables, a five-point near-equal-angle ladder): the cross term $v_1^\top K^\top \Lambda_E^{-1} K u$ vanishes to $\leq 2.1 \cdot 10^{-15}$, B and C along the ray match their closed forms to $\leq 7.1 \cdot 10^{-15}$, $\Phi(\tan \theta)$ matches the margin to $\leq 4.7 \cdot 10^{-14}$ and the cubic vanishes at its own root to $\leq 3.1 \cdot 10^{-12}$; the two routes to the boundary agree in λ to $\leq 5.1 \cdot 10^{-16}$ apart from two draws pinned at the grid's left endpoint, which b57b shows is a bracket artefact (0 of 40 draws have more than one descent block). $L_1 = \frac{\partial r}{\partial t} A_1$ to $\leq 4.6 \cdot 10^{-14}$ relative, the two-jet shift holds to $2 \cdot 10^{-5}$ where measurable, $\text{angle}(v_*, v_1)/\theta_{\text{hi}}$ equal to 0.5000 in both deterministic tables and 0.493–0.551 on the sampled cells. Ladder: slopes falsified for x_{hi} and P , $P \leq 0$ in 13–18 of 30 near-equal-angle draws. Three defects of my own checks are recorded in the .txt : a dip ratio registered against a prediction of the wrong sign, a descent predicate at $10^{-12} \times \text{gap}$ that sits below the measured λ -noise floor, and a 10^{-10} left endpoint doing to a geometric grid what retraction (p) found in a linear one

The same family, ray-level and sphere-level (continued).

Script	Objects	Verifies
<code>b59_setlevel_gridfree.py</code>	Lem. 3.13 as a <i>set</i> statement, Rem. A.1(vii), §7 open item 1	791,736 ray points (264 rays $\times$ 2999 angles, eight cells): (27) has 13,757 sign exceptions, all with $ \lambda_1 - cf_1 \leq 4.6 \cdot 10^{-16}$, none above 10^{-15} , all at $\tan \theta \leq 2.2 \cdot 10^{-7}$, worst one 1.38 times the measured disagreement of the two independent root routes ($3.33 \cdot 10^{-16}$) — so no set-level exception at my evaluator's precision. $B(x), C(x)$ match direct evaluation to $3.9 \cdot 10^{-12}$; the stationary numerator G is verified exact-against-exact, $9.6 \cdot 10^{-16}$ relative to $-F_x/F_\lambda$, sign agreement 5221/5221, finite differences demoted to context (floor $6 \cdot 10^{-11}$ to $5 \cdot 10^{-10}$ near a stationary point). No ray's grid undercuts the exact ray min by more than $10\times$ that ray's own measured floor (0/264; median excess $8.6 \cdot 10^{-9}$ of gap), every ray minimum sits ≥ 0.2357 of the gap above $\lambda_{\min}(\mathcal{G})$ (0/264 below), and 0/120 random $q = 4$ rays beat v_* . Three measuring-stick defects in the <code>.txt</code> and the script's REGISTRATION HISTORY: a band six orders under the evaluator's error, a finite difference used as the reference for an identity, and a candidate list missing $x = 0$
<code>b60_sphere_gradient.py</code>	Rem. A.1(viii), §7 open item 1	Exact sphere gradient $\text{proj}_{T_v, S^{p-1}}(2\lambda_B A_0 v + 2\lambda_C B_2 v)$: verified against the closed-form ray derivative of (vii) ($6.8 \cdot 10^{-15}$ relative to $ \lambda /x$) and the root against <code>np.roots</code> ($8.9 \cdot 10^{-15}$). 50 draws $\times$ 60 starts: the sphere minimum is below $\lambda_1(v_*)$ on 50/50, claimed for the 38 whose beat exceeds $10\times$ that draw's floor (max $2.1 \cdot 10^{-6}$ of gap, = 2955 floors, angle $\leq 4.4 \cdot 10^{-5}$ rad); 0/50 below $\lambda_{\min}(\mathcal{G})$ beyond floor, deficit ≥ 0.3378 of gap; converged-start spread $\leq 7.1 \cdot 10^{-8}$ of gap (0/50 over 10^{-6} , single basin); 0/50 grid undercuts, nested ≤ 0.0625 floors; 19 stops above $10^{-6} \lambda $ in gradient, so only function values are claimed

The sphere minimum read as a law (continued).

Script	Objects	Verifies
<code>b61.geodesic.law.py</code>	Rem. A.1(ix), §7 open item 1	$\lambda_1(v_*) - \lambda_1(v_b) = d_1\theta + \frac{1}{2}\kappa\theta^2$ with each coefficient from two routes (κ): Richardson second difference of values vs centred difference of (viii)'s gradient; ratio med 1.0001 in every cell). Gates $1.7 \cdot 10^{-14}$, $2.8 \cdot 10^{-17}$; 0/50 negative curvature at a reported minimum; 0/50 with linear term $\geq 10\%$ of the beat. Law holds with leverage on 10/50 — ratio 1.0000 (0.9914–1.0008), residual ≤ 0.31 floor — and the other 40 are reported as <i>no leverage</i> , not as passes; the pooled slope there is 1.631 not 2 because κ and θ anticorrelate within a cell. Retracts [B-CLAIM-62]; the <code>.txt</code> records the v1 floor being inflated by a truncation term rather than noise, and the fix can only remove passes
<code>b62.newton.step.py</code>	(ix)'s Newton reading: tested and <i>stopped</i> , no claim added; §7 open item 1	$\theta = \ \nabla_S \lambda_1(v_*)\ /\kappa$ and $\text{beat} = \ \nabla_S \lambda_1(v_*)\ ^2/2\kappa$, coefficients at v_* only. Where the step clears its own floor the ratios are 1.0000 (0.9992–1.0039) and $\text{beat}/(g^{*2}/2\kappa^*) \leq 1.0037$; 0/50 violations of pure Newton, of the cubic-corrected step or of the value form; $\kappa^* > 0$ on 50/50 and no draw has $\ \nabla_S \lambda_1(v_*)\ $ exactly zero (min $4.5 \cdot 10^{-10}$). It is recorded as stopped because the recast raised the number of measurable draws from 10/50 to 11/50 — ten of them the same cell as the row above — which is this row's own pre-registered condition for dropping it, and because the two-route floor on g^* (max $1.4 \cdot 10^{-6}$) exceeds $ g^* $ itself (med $2.5 \cdot 10^{-7}$): the bottleneck is measuring g^* , not the law. Fired as registered: on 24/50 draws the descent started at v_* stops short of the six-start best by >10 evaluator floors, i.e. by $4.3 \cdot 10^{-11}$ of the gap at the median, $7.5 \cdot 10^{-9}$ at worst — b60's single-basin statement therefore holds at the gap scale, not at the evaluator's

What controls the angle, swept by design (continued).

Script	Objects	Verifies
<code>b63_theta_small_cosine_sweep.py</code>	§7 open item 1: designed sweep, both registered directions falsified	Two families, eight swept values, five draws each; leverage precondition met at all sixteen points (5/5 testable at 15 of 16, 3/5 at the sixteenth, whose largest stopping gradient is instead $6.10 \cdot 10^{-7}$ — which is why it is 3/5; the fifteen full ones top out at $1.73 \cdot 10^{-8}$). M1 fires: θ is not non-decreasing in the two-smallest-cosine ratio (Spearman +0.548, three sign flips, med θ from $8.96 \cdot 10^{-5}$ at $r = 10^{-4}$ to $2.91 \cdot 10^{-4}$ at $r = 0.999$ with a return to $6.39 \cdot 10^{-5}$ midway). M2 fires: θ is largest where the two <i>largest</i> cosines coincide ($4.56 \cdot 10^{-3}$ at $d = 10^{-4}$ against $3.12 \cdot 10^{-4}$ at $d = 0.40$, single-point max $7.58 \cdot 10^{-2}$). The <code>.txt</code> records my a-priori reason as wrong — it assumed Λ_E^{-1} weights the small cosines, but A_0 contains no Λ_E — and one header label as wrong (<code>lambda.1(v*)</code> printed where the code evaluates λ_-)
<code>b63b_spectrum_vs_subspace.py</code>	§7 open item 1: spectrum vs subspace at fixed spectrum; reframes θ as model discrepancy	Two cosine spectra, 40 subspace draws each, all 80/80 leverage-bearing. Identities checked before use: $\max \ A_0 - M^\top M\ = 3.33 \cdot 10^{-15}$, $\max \ B_2 - M^\top \Lambda_E^{-1} M\ = 3.89 \cdot 10^{-15}$, $\text{spec}(A_0) = \{c_i^2\}$ to $1.95 \cdot 10^{-14}$ (so $\rho_A = (c_1/c_2)^2$, hence fixed at fixed cosines). S1 holds as a designed contrast: median θ $4.912 \cdot 10^{-3}$ (coincident top pair) against $2.090 \cdot 10^{-4}$ (spaced), $23.5\times$, with overlapping ranges ($2.685 \cdot 10^{-5}$ vs $9.324 \cdot 10^{-4}$). S2 fires: 1.87–3.40 decades of θ at fixed spectrum. S3 is <i>withdrawn as void</i> , not fired — a Spearman against a regressor whose within-group range is $6.9 \cdot 10^{-15}$ and $1.8 \cdot 10^{-14}$ (relative $5.6 \cdot 10^{-15}$); the surviving statement is the invariance one. D2: $\ \nabla_S \lambda_1(v_*)\ = 8.0 \cdot 10^{-6}$ – $8.5 \cdot 10^{-4}$, $\geq 3.9 \cdot 10^8 \times$ its rounding scale, 0/80 draws stationary — so θ is a model-discrepancy angle, not a convergence error. The history block also withdraws a per-draw overclaim of my own and records that reading a within-group slope as a refutation of b61 broke b61 's own recorded caution

Run with `PYTHONIOENCODING=utf-8 python B/<script>`; each writes its table to the paired `.txt`. Expected numbers are the ones printed in this chapter; the Monte-Carlo standard error on the bias tables is ± 0.49 ($n = 200$) and ± 0.34 ($n = 600$) in units of $|\text{bias}| \cdot n$, which is *smaller* than the observed spread across dictionary sizes — that residual spread is carried by $\kappa(\Lambda_E)$, see Remark 1.1, and is not sampling noise.

A The direction leg, in full

This appendix is the measurement ledger behind Remark 3.10; it is kept complete rather than summarised because every figure in it is what retired a claim of mine, and because the script identifiers are the pointers used by Section 9.

Remark A.1 (what **b51**–**b55** measured: the residual is truncational, not directional). Proposition 3.10 was written by **b53** to test a stronger conjecture of mine, which it refuted: that $\min_v \lambda_1(v) = \lambda_1(v_*) = \lambda_{\min}(\mathcal{G})$, *i.e.* that the $n = 1$ member already attains the certificate at the right direction. Scripts **b51_net_lambda2_direction_moment.py**, **b52_boundary_layer_reconciliation.py** and **b53_exact_direction_tightness.py** run on the same 320 dictionary draws (identical seed and call order, eight cells, $q = 2, 3, 4$) and report, in this order: (i) **b51** is a measurement of its own grid. A 64×64 polar net gives $\min_{\text{net}} \lambda_1 = 0.0000$ of the gap in 7 of 8 cells, contradicting the analytic ray of **b50**; the net result is not wrong (its own gates pass: 0 side violations in 320, $\max \|\|v\| - 1\| = 2.2 \cdot 10^{-16}$, 0 root-filter rejections, moment exactness $\leq 6.6 \cdot 10^{-15}$) — it is blind, for the reason in (ii). (ii) *The descent set is a boundary layer.* Reconciling the two on the same draws, **b52** brackets the directions along which λ_n descends from v_1 inside $t < t_{\text{hi}} \approx 2.5 \cdot 10^{-5} \dots 1.6 \cdot 10^{-3}$, with $t_{\text{hi}}(\lambda_2) \equiv t_{\text{hi}}(\lambda_1)$: the width is a property of the direction functional, not of n . Every grid whose angle list starts at 10^{-3} contains v_1 and nothing else in the descent set, so its minimum is constant in the grid density *by construction*; replacing the list by $\text{geomspace}(10^{-8}, \pi/2, 300)$ lifts λ_2 to ≥ 0.9994 of the gap in six of the seven cells that carry a ray, and the grid-free route (analytic ray, $n = 2$) to ≥ 0.9965 — the $q = 4$ cell being the single exception at 0.9164 (grid) and 0.9221 (ray). The descent is wide in the azimuth u : rotating it by 0.2 rad retains 0.916–0.960 of the descent — only the polar resolution failed. (iii) *At v_* the identity holds and the $n = 1$ member fails.* **b53** checks (23) as an identity (no root finding involved) to $\|K^\top (\Lambda_E - \lambda I)^{-1} Ky - (1 - \lambda)y\| / ((1 - \lambda)\|y\|) \leq 2.5 \cdot 10^{-15}$ on 320/320, together with the p -block row on the same vectors ($\leq 2.2 \cdot 10^{-15}$). It also measures the intercept at v_* : $0 < c^2 - B(v_*) \leq 4.6 \cdot 10^{-6}$ in the seven $\dim E_0 = 1$ cells, and $|c^2 - B(v_*)| \leq 3 \cdot 10^{-15}$ — machine zero — in the $\dim E_0 = 3$ cell, whose top eigenspace is degenerate. Against that, $\lambda_1(v_*)$ sits 0.50–1.01 of the gap *above* $\lambda_{\min}(\mathcal{G})$ (median, per cell) — not below it, and not erratically, so this is truncation and not branch error — while $\lambda_2(v_*)$ is within $3.3 \cdot 10^{-7}$, $8.7 \cdot 10^{-7}$, $\dots$, $5.0 \cdot 10^{-4}$ of it, with 0 side violations in 320/320. The one systematic exception is the degenerate cell, and (20) explains it rather than contradicting it: cf_1 is defined as the member with the largest intercept *and, among those, the largest slope* (18), and at fixed B the root of (20) decreases in C — now the proved $\partial r / \partial t < 0$ of Lemma 3.11, not a reading of the quadratic; when $\dim E_0 = 1$ saturating $B = c^2$ pins $v = v_1$ up to sign, but when $\dim E_0 = 3$ it leaves a two-sphere of directions, v_* among them without any reason to maximise C . Measured: $\lambda_1(v_*) > \text{cf}_1$ in 40/40 draws of that cell and in 0 of 280 elsewhere. That cell is the equal-angle one, and **b55** shows the degeneracy is not an accident of the draw: $K = \Lambda_E^{1/2} M$ with $M = Q_1 \text{diag}(c, \dots, c) Q_2^\top$ gives $A_0 = K^\top \Lambda_E^{-1} K = M^\top M = c^2 I_q$ *exactly* ($\max \|A_0 - c^2 I_q\| \leq 5.8 \cdot 10^{-15}$ over 12 draws at $q = 3$ and 12 at $q = 2$), so $\delta = 0$ identically and Corollary 3.12 applies — see (v). (iv) *The two knobs are not interchangeable.* The ray direction u of **b49** is within $5 \cdot 10^{-4}$ rad of v_* (two cells: identical to machine precision), yet $\lambda_1(\text{ray})$ removes only 0.0003–0.50 of the gap while $\lambda_2(\text{ray})$ removes 0.9965–1.0000 *as a per-cell median* (six of the seven ray cells; the $q = 4$ cell gives 0.9222). Those are medians, not guarantees: on the same draws the $q = 4$ cell’s per-draw minimum is 0.7804 with 13 of 40 below 0.90 (**b55**), and across a q -ladder of new tables (**b55**) the median falls 1.0000, 0.9995, 0.9715, 0.8932, 0.9034, 0.6547 at $q = 2, 3, 4, 5, 6, 8$ — 20/20 draws below 0.90 at $q = 8$. The ray leg is therefore a small- q phenomenon and retraction (q) retires the strength I had attached to it; on the very same tables $\lambda_2(v_*)$ still removes $1.00 \cdot 10^{+00}$ of the gap (median *and* max, 20/20 cells). Direction was already essentially right; what bought

the gap was the order of the rule. And $\text{angle}(v_*, v_1)$, computed independently of **b52** from the eigenvector, has median $1.3 \cdot 10^{-5} \dots 1.3 \cdot 10^{-3}$ — a second, unrelated computation of t_{hi} , which is the corroborating (not the proving) half of (ii): same draws, same scaling. (v) *The margin is exact, and it is one number.* Lemma 3.11 turns the trade-off of (iii)–(iv) into a criterion, and **b55** tests it on 550 draws in 20 cells (**b53**’s eight, plus p - and q -doubled tables on a new seed): the sign of $t - t_1 - \frac{\delta}{c^2} \frac{1-t_1\mu}{\mu}$ disagrees with the sign of $\lambda_1(v_*) - \text{cf}_1$ in 0 of them, each side condition of (25) holds on all 550, and the collapse (26) predicts the *size* to a relative $4 \cdot 10^{-4}$ (per-cell medians 0.9996–1.0003 for the ratio of $\lambda_1(v_*) - \text{cf}_1$ to its first-order value). That agreement is a statement about $|T|$, not about the law, so the same script drives $|T|$ up along the great circle $v(\vartheta) = \cos \vartheta v_* + \sin \vartheta w$, $w \perp v_*$: over 24,000 points with $|T|$ spanning 10^{-8} to $1.5 \cdot 10^3$ the sign never flips, while the first-order ratio drifts to 1.67, its departure from 1 being $\leq 5.5 \cdot 10^{-3}$ for $|T| \leq 1$. So the criterion carries, the quantitative form only within a measured regime, and $\min_{\|v\|=1} \lambda_1(v) = \text{cf}_1$ in the equal-angle cells by Corollary 3.12 (0 competitors below cf_1 — floor on a 20,000-point circle at $q = 2$ and on 60,000 points at $q = 3$, while $\lambda_1(v_*) > \text{cf}_1$ in 24 of 24 draws). The upshot for the open item it was meant to settle: retraction (n) established that cf_1 is a saddle, not a minimum, so a descent direction exists; (i)–(v) establish that chasing it cannot recover the residual at $n = 1$, because the $n = 1$ chord loses 0.5–1.0 of the gap *even at v_** , the direction that attains the exact fibre crossing of Proposition 3.10 — which is a different minimiser from $\min_v \lambda_1(v)$, and whether that infimum reaches $\lambda_{\min}(\mathcal{G})$ is now the open question (Section 7, open item 1). The residual of cf_1 is therefore ordered by the moment side, and the sentence of Remark A.2 that attributes it to “the fixed direction” is corrected here — see retraction (o). (vi) **b57 measures Lemma 3.13, and the near-degenerate cells refute the interpolation I hoped for.** Two independent routes to the boundary of $\{\lambda_1 \leq \text{cf}_1\}$ on a ray — the leading positive root of Φ and a 200-step bisection of $\lambda_1(v_\theta) - \text{cf}_1$ over a geometric grid of 3000 angles on $[10^{-10}, \pi/2]$ — are compared on 470 draws in thirteen cells, and the disagreement is reported in λ , multiplying by the measured $|\partial \lambda_1 / \partial \theta|$ at the boundary. Every cell satisfies the bookkeeping identity for that product to 0 digits, and the mismatch is at most $5.1 \cdot 10^{-16}$ (cell medians $5 \cdot 10^{-17}$ to $1.2 \cdot 10^{-16}$) except for two draws, in the 2–45–88 and $q = 4$ cells, whose bisected angle came out at $1.0 \cdot 10^{-10}$ — the grid’s own left endpoint, so the predicate fired on a round-off dip and bracketed nothing: the same left-endpoint artefact as retraction (p), in a grid I had chosen to avoid it. Read that way, the θ -level disagreement of $1.4 \cdot 10^{-6}$ in the shallowest cell is $2.4 \cdot 10^{-16}$ in λ , and no draw contradicts (25); the first-order root shift of the lemma is verifiable only where it exceeds that floor, and there it holds to a relative $2 \cdot 10^{-5}$ (four cells). The equal-angle falsifier fires nowhere: 0 of 120 random rays leave cf_1 , with deepest apparent dip $2 \cdot 10^{-16}$ against a gap of $4.8 \cdot 10^{-5}$. Driving the two top cosines together instead ($\cos(20 \pm d)$, $d = 10^{-3}$ to 5 degrees, 150 draws) does *not* collapse the layer: x_{hi} has log-log slope -0.28 and P slope $+0.71$ against my pre-registered 0 (both falsified; g and D pass at 0.02 and 1.01), $P \leq 0$ in 13–18 of 30 draws at $d \leq 0.1^\circ$ with no upper crossing on those rays, and $\lambda_1(v_*)$ then recovers 0.986–0.993 of the gap. The mechanism is the one already flagged at the definition of t_1 : as the top eigenvalue of $K^\top \Lambda_E^{-1} K$ becomes multiple the data stop selecting v_1 , so g , P and every statement referring to them become solver-chosen. A continuity statement in $\|K^\top \Lambda_E^{-1} K - c^2 I_q\|$ therefore cannot be written with v_1 in it; the parameter that controls this regime is the top-eigenvalue separation, not the norm defect.

(vii) **b59 tests (27) as a set statement and turns the ray into a grid-free inner problem; it narrows open item 1 without settling it.** On 264 rays through v_1 (eight cells, 12 draws each, rays = the steepest descent direction plus two random ones) and 2999 angles per ray — 791,736 evaluations — the equivalence (27) has 13,757 points where the two sides disagree in sign, all of them with $|\lambda_1(v_\theta) - \text{cf}_1| \leq 4.6 \cdot 10^{-16}$, none above 10^{-15} , and all at $\tan \theta \leq 2.2 \cdot 10^{-7}$, i. where the two quantities differ only by round-off. That bound is not impressive by itself, so it is measured against my own evaluator: the note’s root formula $((B + C) - \sqrt{\text{disc}})/(2C)$ subtracts two $O(1)$

numbers to produce $\lambda \approx 3.7 \cdot 10^{-3}$ and so loses about 2.5 digits, while the rationalised form $\lambda_- = 2B(1 - B)/((B + C) + \sqrt{\text{disc}})$ is an independent route through the product of the roots; those two routes disagree by up to $3.33 \cdot 10^{-16}$ on the same cloud. The worst exception is 1.38 times that noise floor, which is what “no set-level exception” means here — a statement *weaker* than (27), and written weaker on purpose. My first registered band for this test was 10^{-11} of the local gap ($\approx 1.8 \cdot 10^{-17}$), six orders below the evaluator’s own error, and it reported the iff as falsified; the fix was the second root route, not the band. With $x = \tan \theta$ the two moments are rational, $B(x) = (c^2 + ax^2)/(1 + x^2)$ and $C(x) = (t_1 c^2 + 2gx + bx^2)/(1 + x^2)$, so a ray minimiser of λ_1 is a root of the cleared stationary numerator $G(x, \lambda) = \hat{C}\lambda^2 - (\hat{B} + \hat{C})\lambda + \hat{B}(1 - 2B)$ with $\hat{B} = -2Dx$, $\hat{C} = 2(g(1 - x^2) + (b - t_1 c^2)x)$, and $\partial\lambda/\partial x = -F_x/F_\lambda$ on the λ_- branch ($F_\lambda = -\sqrt{\text{disc}}$). One caveat got me: differentiating C itself produces $b - t_1 c^2$, whereas the combination $C - t_1 B$ of the lemma carries $b - t_1 a$ — the two are not interchangeable, and I conflated them on paper first. G is validated exact-against-exact, against the derivative of the closed-form root rather than a finite difference, to a relative $9.6 \cdot 10^{-16}$ (scale $|\lambda|/x$) on all 264 rays with the sign matching in 5221 of 5221 places where the derivative clears the difference floor; the finite-difference route is reported as context only, because its own floor is $6 \cdot 10^{-11}$ to $5 \cdot 10^{-10}$ near a stationary point where it is the reference that is unusable, not G . Minimising over $\{\lambda_1 \leq \text{cf}_1\}$ then needs no grid: the candidates are the two endpoints $x = 0$ and $x = \infty$, the edges of the descent intervals (which are single intervals on 264/264 rays), and the bisected roots of G . The registered test — a grid may not undercut the exact ray minimum by more than ten times that ray’s own measured two-route floor — passes 0/264, the median grid excess being $8.6 \cdot 10^{-9}$ of the gap and the worst raw undercut $4.1 \cdot 10^{-6}$ of the gap in the cell whose gap is smallest, i. where the floor allows it; the earlier 103 apparent violations were my candidate list omitting $x = 0$, which is exactly the minimiser on a ray whose descent set is empty. For open item 1 this gives numbers inside the family it searches: every one of the 264 ray minima sits strictly *above* $\lambda_{\min}(\mathcal{G})$, the smallest deficit being 0.2357 of the gap (per-cell minima 0.2357–1.0000; the equal-angle cell attains 1.0000 because there the ray infimum is cf_1 itself), and 0 rays fall below the floor. 54 of 264 rays undercut $\lambda_1(v_*)$ strictly, but only 24 by more than 10^{-12} in λ , and the largest beat outside the equal-angle cell is $4.7 \cdot 10^{-6}$ of the gap ($2.8 \cdot 10^{-9}$ in λ); the equal-angle beats, up to $8.7 \cdot 10^{-3}$ of the gap, are the already-known $\lambda_1(v_*) > \text{cf}_1$ of retraction (o), not a new phenomenon. On one $q = 4$ draw, 120 random rays give 0 that beat v_* , the best ray sitting 0.3009 of the gap above $\lambda_{\min}(\mathcal{G})$ against $\lambda_1(v_*)$ ’s 0.2741. *What this does not buy, plus a wording I had to take back.* The first version of this sentence claimed that rays through v_1 sweep only a two-plane family rather than the sphere. That is wrong as stated: every unit v decomposes as $\cos \theta v_1 + \sin \theta u$ with $u \perp v_1$, so *the union over u of these rays is the sphere*, and $\inf_{u \perp v_1} \min_\theta \lambda_1(v_{\theta, u}) = \min_{\|v\|=1} \lambda_1(v)$ holds exactly. What b59 samples is three directions u per draw (the steepest plus two random), and 120 on one $q = 4$ table — a Monte-Carlo sample of a $(p-1)$ -dimensional set, not a two-plane restriction. The open item is therefore narrowed by evidence, not by geometry: within the sampled planes the descent set never reaches $\lambda_{\min}(\mathcal{G})$ and v_* is never beaten, while G makes the inner problem exact, which is what a genuine search over u needs in order to say anything. (viii) b60 runs that genuine search, and v_* loses. Since the union over $u \perp v_1$ of the rays above *is* the sphere, $\min_{\|v\|=1} \lambda_1(v)$ is a $(p-1)$ -dimensional problem, and G makes it grid-free pointwise: with $F_\lambda = -\sqrt{\text{disc}}$ on the λ_- branch, $\lambda_B = -(1 - \lambda - 2B)/F_\lambda$ and $\lambda_C = -(\lambda^2 - \lambda)/F_\lambda$, so the gradient of $\lambda_1(v) = \lambda_-(B(v), C(v))$ is $2\lambda_B A_0 v + 2\lambda_C B_2 v$ projected to $T_v S^{p-1}$ — no finite differences. That formula is checked *against* the closed-form ray derivative of (vii) ($6.8 \cdot 10^{-15}$ relative to the natural scale $|\lambda|/x$, gate 10^{-8}) and the root against **np.roots** on the same quadratic ($8.9 \cdot 10^{-15}$). Retraction with backtracking from 60 starts per draw, five cells $\times$ ten draws: *the sphere minimum is strictly below $\lambda_1(v_*)$ on all 50 draws.* The clause of (i)–(v) that

called v_* “a different minimiser from $\min_v \lambda_1(v)$ ” was until this script a logical distinction — the two minimisers could still have coincided — and it is now a measured one. The claim is restricted to the 38 of 50 whose beat exceeds ten times that draw’s own measured two-route floor, the largest being $2.1 \cdot 10^{-6}$ of the gap (2955 floors), and the winning direction sits within $4.4 \cdot 10^{-5}$ rad of v_* (median $< 5 \cdot 10^{-7}$). So v_* is near-optimal, not optimal, and the margin is three orders of magnitude smaller than the effect (vii) removed. *What the search does not reach*: 0 of 50 draws put the sphere minimum below $\lambda_{\min}(\mathcal{G})$ beyond its own floor, the smallest deficit over all 50 being 0.3378 of the gap (median 0.91938); 0 of 50 draws show two converged starts disagreeing by more than 10^{-6} of the gap (max $7.1 \cdot 10^{-8}$, i. single basin); a 200 000-point random direction grid never undercuts the optimiser by more than ten floors (0 of 50); and the sphere minimum never exceeds the 3-ray minimum of (vii) by more than 0.0625 floors (max $1.1 \cdot 10^{-13}$ of the gap), as monotonicity requires. Nineteen reported minima still had $\|\nabla_S \lambda_1\| > 10^{-6}|\lambda|$ at stopping, so the statements above compare *function values at concrete directions*, not stationarity. Open item 1 is thus answered in one part and sharpened in the other: the sphere infimum does not touch $\lambda_{\min}(\mathcal{G})$, and v_* is not its minimiser — what remains is not whether a better direction exists but *which printable quantity controls the angle between v_* and it*, since the per-cell beat spans three orders of magnitude ($1.3 \cdot 10^3$ to $1.7 \cdot 10^6$ floors) while the per-cell median angles stay in $[0, 1.05 \cdot 10^{-5}]$ rad. (ix) **b61** turns that residual into a law, and my own registered claim is what it refutes. At the sphere minimiser v_b let u be the geodesic direction of v_* and $\theta = \angle(v_b, v_*)$; then $\lambda_1(v_*) - \lambda_1(v_b) = d_1\theta + \frac{1}{2}\kappa\theta^2$ with *both* coefficients measured by two independent routes — κ by a Richardson-corrected second difference of values and by a centred difference of the gradient of (viii), whose ratio is 1.0001 (median, every cell) — and the expansion is then checked against the directly measured beat over the whole arc. Gates $1.7 \cdot 10^{-14}$ and $2.8 \cdot 10^{-17}$. The linear term never reaches 10% of the beat (0/50; median $1.7 \cdot 10^{-3}$), so the stopping-criterion caveat of (viii) does not contaminate it, and the curvature is positive at all 50 reported minima (0/50 negative beyond its own floor). Where the quadratic term stands ten times above its own floor — 10 of 50 draws, all in the near-equal-angle cell, $\theta \leq 7.3 \cdot 10^{-5}$ — the ratio $\text{beat}/(\frac{1}{2}\kappa\theta^2)$ is 1.0000 (median), 0.9914–1.0008, residual ≤ 0.31 of the floor. *So the fibre direction’s suboptimality for λ_1 is quadratic in the angle it is off by, with a curvature I can print* ($\kappa \in [0.150, 0.926]$ measured). The claim this refutes is mine: [B-CLAIM-62] of the previous round asserted that the beat and the angle are not controlled by one quantity, and one quantity controls them. The other 40 draws are reported as *no leverage* — the floor there is dominated by the linear coefficient’s own error times θ , which exceeds $\frac{1}{2}\kappa\theta^2$ — so the law is claimed on 10/50, not on 50/50. Two warnings travel with it. (a) *A pooled exponent is not the law’s exponent*: on those 10 draws $\log \text{beat}$ versus $\log \theta$ has slope 1.631, not 2, because within a cell κ and θ are anticorrelated ($\kappa \in [0.744, 0.915]$ falling as θ grows); the identity is per-draw and the regression is across-draws. (b) *θ is exactly zero on 10 of the 50 draws*, and the equal-angle cell’s median θ is 0 — the same draws whose **b60** beats sat inside the evaluator noise, which is the consistency one would want: there $v_b = v_*$ to machine precision, as (vii) already predicted from the ray infimum being cf_1 itself. What open item 1 asks now is narrower than either half of (viii): κ is measured and the descent is $\frac{1}{2}\kappa\theta^2$, so the whole question is *what controls $\theta = \angle(v_*, \arg \min_v \lambda_1(v))$* , which (viii) bounded above by $4.4 \cdot 10^{-5}$ rad on its own draws, and **b61** by $7.3 \cdot 10^{-5}$ on mine. A Newton reading of that θ was tested and stopped: formally $\theta = \|\nabla_S \lambda_1(v_*)\|/\kappa$ and $\text{beat} = \|\nabla_S \lambda_1(v_*)\|^2/2\kappa$, and both hold to 0.4% on the draws where the step clears its own floor (θ ratio 0.9992–1.0039, beat ratio ≤ 1.0037), but **b62** raised the number of measurable draws from 10/50 to 11/50 — ten of them the same cell **b61** could already see — so the recast is not offered as a result. Its useful by-product is a diagnosis: the measured two-route floor on $\|\nabla_S \lambda_1(v_*)\|$ (max $1.4 \cdot 10^{-6}$ in λ , largest route disagreement $3.1 \cdot 10^{-7}$) exceeds the quantity itself (med $2.5 \cdot 10^{-7}$), and relative to g^* that floor is $1.6 \cdot 10^{-3}$ in the cell where the step is resolvable but 129 in the equal-angle cell — so the bottleneck is measuring that gradient, not the law. One

registered falsifier did fire: on 24/50 draws the descent started at v_* stops short of the six-start best by more than ten copies of that draw’s evaluator floor ($\approx 4 \cdot 10^{-19}$), which is $4.3 \cdot 10^{-11}$ of the gap at the median and $7.5 \cdot 10^{-9}$ at worst — three to five orders below the 10^{-6} -of-gap criterion **b60** used for the same question. Read as registered, the single-basin statement therefore holds at the gap scale and not at the evaluator scale.

Open item 1 was then run as a *designed* sweep instead of a search (**b63**, **b63b**), and both registered directions were falsified. θ is not monotone in the ratio of the two smallest cosines (Spearman +0.548, three sign flips over eight swept values, with 5/5 leverage-bearing draws at every point), and it is instead *largest* where the two *largest* cosines coincide (med $\theta = 4.56 \cdot 10^{-3}$ at $d = 10^{-4}$ against $3.12 \cdot 10^{-4}$ at $d = 0.40$). Separating spectrum from subspace at fixed spectrum (40 draws per spectrum, 80/80 leverage-bearing, the first design in this thread that lost no draw to resolution) leaves a two-factor answer and no control variable: the cosine spectrum shifts the median θ by $23.5\times$ ($4.912 \cdot 10^{-3}$ against $2.090 \cdot 10^{-4}$) while the two distributions *overlap* (min = $2.685 \cdot 10^{-5}$ below max = $9.324 \cdot 10^{-4}$), and the subspace orientation alone spreads θ over 1.87–3.40 decades at fixed spectrum. The candidate mechanism $\rho_A = \lambda_1(A_0)/\lambda_2(A_0)$ is then inert *by construction*, not by measurement: with $M := \Lambda_E^{-1/2}K$ — the matrix whose singular values are the cosines of Prop. 3.5, so this factorisation is not a new lemma — $A_0 = K^\top \Lambda_E^{-1}K = M^\top M$ and $\text{spec}(A_0) = \{c_i^2\}$ (checked to $1.95 \cdot 10^{-14}$, and to $3.33 \cdot 10^{-15}$ as an identity in M), hence $\rho_A = (c_1/c_2)^2$ cannot vary at fixed cosines. My within-spectrum Spearman against ρ_A was therefore a correlation against rounding noise (within-group ranges $6.9 \cdot 10^{-15}$ and $1.8 \cdot 10^{-14}$) and is withdrawn as void rather than reported as a fired falsifier; what the same data do license is the invariance statement — $\text{spec}(A_0)$ fixed, θ spreading 1.87–3.40 decades, so no function of A_0 ’s spectrum explains the within-spectrum spread. The last number is the one that changes the question: v_* is *not* a critical point of λ_1 ($\|\nabla_S \lambda_1(v_*)\| = 8.0 \cdot 10^{-6}$ – $8.5 \cdot 10^{-4}$, at least $3.9 \cdot 10^8$ times its own rounding scale, on 80/80 draws). So θ is the angle between the exact eigenvector’s fibre block and *this functional’s own* minimiser — a model-discrepancy angle, not a convergence error — and it should not be read as evidence that the descent fails. In this design the only legitimate test of **b61**’s quadratic law remains its per-draw ratio beat/ $(\frac{1}{2}\kappa\theta^2)$ with κ measured by the two independent routes, and that has not been run; the within-group slopes (+1.984, residual geometric s.d. 0.091; +1.539, 0.618) are reported only to show that they mean nothing on their own, per warning (a) above.

Remark A.2 (what **b50** measured, and what the hierarchy does not buy). Script **b50_gauss_moment_hierarchy.py** runs this on 280 dictionary pairs in seven cells ($q = 2, 3, 4$, four cosine spectra, two near-isotropic families; 40 draws each, $\dim E_0$ listed per cell), after checking on every draw that the rules do integrate $1, y, \dots, y^{2n-1}$ exactly (max relative deviation $\leq 7 \cdot 10^{-15}$) — a rule that integrates no polynomial exactly is not a Gauss rule, and a side violation under it would be my bookkeeping rather than the theorem. Result: $\lambda_1 = \text{cf}_1$ to $\leq 7.2 \cdot 10^{-16}$; $\lambda_2, \lambda_3 \geq \lambda_{\min}(\mathcal{G})$ in 280/280 (per-draw floor $100\epsilon\|\mathcal{G}\|_2$); $\lambda_1 \geq \lambda_2$ in 280/280 and $\lambda_2 \geq \lambda_3$ in 239/239 finite pairs, and the pointwise ordering $G_1 \leq G_2 \leq G_3 \leq \phi_v$ in 240/240 test points — reported as measured, since the error formula certifies each rule separately and does not order successive n . The size is what makes it a proposition and not an exercise: $\lambda_2(v_1)$ removes 0.26–0.9998 of the residual gap $\text{cf}_1 - \lambda_{\min}(\mathcal{G})$ (cell medians 0.50, 0.63, 0.79, 0.81, 0.87, 0.998, 0.999), where the steepest-descent ray of **b49** removes 0.0004–0.72; one extra scalar moment buys an order of magnitude more than optimising the direction did. The cubic form (22) was checked against the eigenvalue-plus-bisection value on all 280 draws, max difference $2.5 \cdot 10^{-16}$, and against a two-atom measure computed by hand. Two limits, both stated by the numbers rather than assumed. (i) *The hierarchy terminates*: ρ_v carries at most p atoms, and where Λ_E has few distinct eigenvalues the rule stops early — in the two near-isotropic cells λ_2 is already *exact* for ϕ_v and 39 of 80 $n = 3$ rules break down ($h_2 = 0$).

(ii) *Higher n is not where the residual lives:* write $\lambda_{\text{ex}}(v_1)$ for the crossing of ϕ_{v_1} itself, which the hierarchy attains once n reaches the number of atoms of ρ_{v_1} . Over the same draws $|\lambda_2(v_1) - \lambda_{\text{ex}}(v_1)|$ is at most $2.0 \cdot 10^{-3}$ of the gap $\text{cf}_1 - \lambda_{\min}(\mathcal{G})$, while ρ_{v_1} carries 2–7 atoms, so the moment truncation is negligible and the rest of the gap is the *fixed direction* — *this last clause is superseded*: at the exact minimiser v_* of Proposition 3.10 the $n = 1$ chord still loses 0.50–1.01 of that gap, so what is left is the order of the rule rather than the direction; see Remark A.1 and retraction (o). Since (21) holds for every unit v , $\min_v \lambda_2(v)$ over any finite set of directions is again a certified bound, and $\lambda_{\min}(\mathcal{G}) \leq \min(\lambda_2(v_1), \min_v \lambda_2(v))$; what remains open is how fast that minimum descends, which is the one-dimensional fractional programme of Section 7 (1) with the moment side now essentially free. What the hierarchy does *not* do is rehabilitate cf: these draws include the cells where $\text{cf} < \lambda_{\min}(\mathcal{G})$ (195 of 280), the no-angle model for which Proposition 3.8 claims only $\lambda_{\min}(\mathcal{G}) \leq \text{cf}_1$.

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